

Diagonally symmetric alternating sign matrices

(OEIS A005163)

The full leading equivalent and its amplitude, the bounded logarithmic remainder, the fugacity pressure and limit laws, and elementary growth envelopes

Four research reports of one external research session (bundle Reports 128, 124, 121 and 120),
merged into one ProveIt report

Parts I–IV: 2 October 2026 merged: 5 October 2026

Abstract

Let a_n count the diagonally symmetric alternating sign matrices (DSASMs) of order n , the sequence A005163, and let $Z_n(s)$ weight each DSASM by s to the number of its nonzero diagonal entries, so that $a_n = Z_n(1)$. Put $\alpha = \frac{1}{2} \log(3\sqrt{3}/4)$, $\beta = \frac{1}{4} \log 3$ and $\kappa = 5/72$. This report prints four manuscripts as Parts I–IV. Part I (the base) proves the full leading equivalent $a_n \sim C_* e^{\alpha n^2 + \beta n} n^{-5/72}$ along all n , the leading term of Behrend, Fischer and Koutschan's Conjecture 11.1, and more generally $Z_n(s) \sim C(s) e^{\alpha n^2 + nB(s)} n^{-\kappa}$ locally uniformly on $\Re s > 0$; the positive amplitude is characterized by convergent Fredholm determinants and a convergent Fourier series, but is not evaluated. Part I also gives an amplitude-aware inverse threshold and the constant terms of every fixed diagonal cumulant. Its proof rests on Part II, which proves stability of the diagonal fugacity polynomials, cross-size interlacing, an exactly solvable calibration at $s = \sqrt{3}$, the bounded remainder $\log a_n = \alpha n^2 + \beta n - \kappa \log n + O(1)$, a holomorphic pressure with bounded error, bounded cumulant errors and an explicitly centred central limit theorem. Part III proves the fugacity pressure with $o(n)$ error by a second route and adds results found nowhere else here: the limiting root measure, half of whose mass escapes to infinity, Hoeffding concentration, a parity-lattice local limit theorem and a large-deviation principle with explicit rate. Part IV gives elementary growth envelopes by Jensen and chord inequalities on the roots and a projected contraction, with certified constants $p < \beta < q$. The numerical value of the amplitude, the finer corrections conjectured by Behrend, Fischer and Koutschan, any rate of convergence and effective constants remain open. The report is unrefereed, and nothing in it is formalized.

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Guide to this report

The four Parts are four manuscripts of the session bundle of Reports 1–243 (arrival commit 60f54ea06), placed in this directory by commit 612787fb4 (batch 104). All four are dated 2 October 2026 and form one programme on one object, written in the order 120, 121, 124, 128. They are printed in order of strength, the strongest first, because Report 128 implies every growth statement of the other three:

Part	Bundle report	Delivered title	Labels
I	128 (base)	The full leading equivalent for diagonally symmetric alternating sign matrices	<code>dsasm:</code>
II	124	Logarithmic asymptotics of diagonally symmetric alternating sign matrices	<code>dsasm:log:</code>
III	121	DSASM fugacity pressure and diagonal statistics	<code>dsasm:pr:</code>
IV	120	Diagonal fugacity roots and DSASM growth bounds	<code>dsasm:env:</code>

Part I is not self-contained. Report 128 takes four groups of results, (I1)–(I4) in Section 1, from Report 124, which it shipped unchanged as an “earlier input”. Report 124 is therefore printed in full as Part II, and Part I’s eight citations of it are internal references here. Read Part II’s Sections 11–21 and 23 before, or alongside, Part I’s Sections 2–7.

Each Part prints its manuscript’s text, with every result, proof, remark, open question and limitation; statements keep their delivered wording. Sections are numbered continuously and statements within sections, so a delivered statement number moves with its section only: Part I keeps Report 128’s Sections 1–8 and adds Section 9; Report 124’s Section k is Section $k + 9$ (Part II, Sections 10–24); Report 121’s Section k is Section $k + 24$ (Part III, Sections 25–40); Report 120’s Section k is Section $k + 40$ (Part IV, Sections 41–49). The four manuscripts number equations consecutively through the whole text; each Part keeps its manuscript’s equation numbers, prefixed by the Part in Parts II–IV: Report 124’s equation (k) is (II. k) here, Report 121’s is (III. k) and Report 120’s is (IV. k), and Part I’s equations keep their delivered numbers (k).

Reports 120, 121 and 124 share whole sections, almost word for word (8-gram overlap 0.89–1.00 by section; differences checked line by line with `diff`). Each is printed once. The stability section, the exact calibration and the logarithmic calibration are printed in Part II (Sections 11, 13 and 14); “Two consequences for the diagonal statistic”, shared by Reports 120 and 121 only, is printed in Part III (Section 27). The repeated sections of Parts III and IV keep their titles and numbers, a pointer, every sentence in which they differ from the printed text, and their equation numbers (an equation not reprinted there is cited at its printed place in Part II or III). Report 121’s interlacing section repeats Report 124’s except for an exact derivative identity and a mean bracket, which Section 33 prints.

Notes added when the manuscripts were merged are set in small type and tagged [**write**], with their date (5 October 2026). They give cross-references between the Parts, mark results proved twice (second routes), mark statements that a stronger Part supersedes, and record what the write checked. Superseded statements, scope sentences and questions are printed as delivered, with a dated note pointing to the Part that settles them. Each Part ends with a section of open questions: Part I with Section 9, added in the write, and Parts II–IV with their own closing sections, re-scoped by dated notes. Under Vladimir’s standing rule of 4 October 2026 these collect every claim of the Part stated without proof, with its source, its sketch and what is missing.

What is proved, and where

Here $c = \sqrt{3}$, $P(s) = (s+1)^2/(2(s+2))$ is the pressure, $D_n(t) = Z_n(\sqrt{t})Z_{n+1}(\sqrt{t})/\sqrt{t}$ is the companion polynomial, $f(t) = \text{Log } P(\sqrt{t})$, and S_n is the number of nonzero diagonal entries under the fugacity- s law $s^{S(A)}/Z_n(s)$.

Statement	Where
$a_n \sim C_* e^{\alpha n^2 + \beta n} n^{-5/72}$ with $C_* = C_{\text{cal}} 2^{-1/4} e^{E(1)/2} > 0$; $Z_n(s) \sim C(s) e^{\alpha n^2 + nB(s)} n^{-\kappa}$ locally uniformly on $\Re s > 0$; companion limit $E_n \rightarrow E$	Part I, Theorem 1.1 and (26)
Common calibration amplitude on both parities, $Z_n(c) \sim C_{\text{cal}} e^{\alpha n^2 + \beta n} n^{-\kappa}$; exact boundary formula; boundary-ratio limit $F_n(s) \rightarrow (s/c)/\sqrt{P(s)}$	Part I, (27), Proposition 3.1, (8)
Szegő-type constant for Jacobi ensembles with endpoint-vanishing Hölder symbols	Part I, Proposition 2.1
Inverse threshold with the amplitude in the centre; cumulants $n\mathcal{D}^{k-1}m(s) + \mathcal{D}^k L(s) + o(1)$	Part I, Corollaries 7.1 and 7.2
$\log a_n = \alpha n^2 + \beta n - \kappa \log n + O(1)$ for all $n \geq 1$; $\log(D_n(1)/D_n(3)) = n \log(2/3) + O(1)$; trace limit $D'_n(3)/D_n(3) = (n+1)(1 - \sqrt{3}/2) + o(1)$	Part II, Theorem 10.1 (bounds and determinant remainder superseded by Part I; the trace limit is not restated there)
Stability of Q_n on $\Re s_i > 0$, $Z_n(s) = s^{n \bmod 2} P_n(s^2)$ with strictly negative roots; weak cross-size interlacing; $ M_n(s) - M_{n-1}(s) \leq 1$; $Z_n(c)Z_{n+1}(c) = c2^n A_{n+1}$	Part II, Sections 11–14 (the same theorem in Reports 120 and 121)
Holomorphic pressure $\log(D_n(t)/D_n(3)) = nf(t) + O_K(1)$ on $\mathbb{C} \setminus (-\infty, 0]$, all positive fugacities; cumulants $n(s\partial_s)^{k-1}m(s) + O(1)$; $(S_n - nm(s))/\sqrt{nv(s)} \Rightarrow \mathcal{N}(0, 1)$	Part II, Theorem 23.1, (II.121), (II.122)
Pressure $\frac{1}{n} \log(D_n(t)/D_n(3)) \rightarrow f(t)$ with all derivatives on $\Re t > 0$ (second route); $\log a_n = \alpha n^2 + \beta n + o(n)$	Part III, Theorem 34.1, (III.84)
Root measure $\nu_n \Rightarrow \frac{1}{2}\delta_\infty + 3d\lambda/(\pi\sqrt{\lambda}(\lambda+1)(\lambda+4))$; Newton inequalities; Hoeffding bound; actual-mean CLT; parity-lattice local CLT; speed- n large deviations with explicit rate	Part III, (III.86), (III.17), (III.90), (III.91), (III.93), (III.97)
Envelopes $\alpha n^2 + pn - \kappa \log n - C \leq \log a_n \leq \alpha n^2 + qn - \kappa \log n + C$, $p = 0.26324\dots < q = 0.29604\dots$, certified by exact interval arithmetic; two-ceiling inverse	Part IV, Theorem 41.1, (IV.48)–(IV.50), Theorem 48.1

Open in all four Parts

The numerical value of C_* and a closed form for it; any rate in the full equivalent, and the finer corrections of Conjecture 11.1 (n^{-2} and parity-dependent $n^{-5/2}$ terms); an all-orders expansion; effective constants, onsets and a certified finite inverse; the scales at which the escaping roots move; local prefactors away from the central window; root simplicity; uniformity as $s \rightarrow 0$ or $s \rightarrow \infty$. See Sections 9, 24, 40 and 49.

Finite computations in all four Parts (exact rational and integer checks, interval certificates, adversarial input campaigns) check identities, normalizations and implementations; no asymptotic statement rests on them. Part IV's interval certificates for p and q are exact computations with explicit tails, and only certify those two constants.

The leading equivalent and Conjecture 11.1

The equivalent of Part I is the leading term of a conjecture of Behrend, Fischer and Koutschan [1]. In version 3 of their paper (arXiv:2309.08446v3, dated 1 October 2026; text checked by the write on 5 October 2026), Conjecture 11.1, equation (11.8), states that

$$|\text{DSASM}(n)| = C \left(\frac{3\sqrt{3}}{4} \right)^{n^2/2} 3^{n/4} n^{-5/72} \begin{cases} 1 - \frac{385}{31104n^2} + \frac{(-1)^{n/2}a}{n^{5/2}} + O(n^{-3}), & n \text{ even,} \\ 1 - \frac{385}{31104n^2} + \frac{(-1)^{(n+1)/2}b}{n^{5/2}} + O(n^{-3}), & n \text{ odd,} \end{cases}$$

with constants C , a , b that they estimate numerically from the exact counts for $n \leq 1000$ ($C \approx 0.72352852136732\dots$, $a \approx 0.0783$, $b \approx 0.1892$) and the further conjecture $b = (1 + \sqrt{2})a$; they report no closed expression for C . Their Theorem 11.2 proves the leading growth $|\text{DSASM}(n)|^{2/n^2} \rightarrow 3\sqrt{3}/4$. Since $(3\sqrt{3}/4)^{n^2/2} 3^{n/4} = e^{\alpha n^2 + \beta n}$, Part I, Theorem 1.1, proves the *leading equivalent* of (11.8): the limit $C = C_*$ exists, is positive, and is given by (7) and (26). It does *not* prove the finer corrections, the term $-385/(31104n^2)$, the parity-dependent $n^{-5/2}$ terms, the relation $b = (1 + \sqrt{2})a$, or any rate of convergence, and it does not evaluate C_* ; so it neither confirms nor uses the digits of C . The credit for the conjectured form, the exponent $-5/72$ and the numerical constant belongs to Behrend, Fischer and Koutschan; Part II's earlier theorem proves the conjectured shape up to a bounded factor, and Parts III and IV the linear coefficient and envelopes around it. The manuscripts cite this conjecture in the same version; their quotation of it (with an unnamed constant $C_{\text{DSASM}} > 0$ and “further parity-sensitive corrections”) is accurate. The numerical evidence about C and the correction terms is in Section 9, labelled uncertified.

Notation across the four Parts

Common to all Parts: $a_n = |\text{DSASM}(n)|$; $S(A)$ the number of nonzero diagonal entries of A ; $Q_n(s_1, \dots, s_n)$ the multivariate diagonal polynomial and $Z_n(s) = Q_n(s, \dots, s)$, with $Z_n(s) = X_n(1, s, 1)$ in the notation of [1]; $A_N = |\text{ASM}(N)|$; $\alpha = \frac{1}{2} \log(3\sqrt{3}/4) = g/2$, $\beta = \frac{1}{4} \log 3$, $\kappa = 5/72$, $b = \frac{3}{4} \log 3 - \frac{1}{2} \log 2$ (Parts II–IV also write $g = \log(3\sqrt{3}/4)$); $c = \sqrt{3}$, the solvable calibration; $\varepsilon = \varepsilon_n = n \bmod 2$; $N(T) = \min\{n \geq 1 : a_n \geq T\}$; $\mathcal{H}$ is a half-plane in Parts II–IV and, in Part I, $\mathcal{H}_+ = \{\Re s > 0\}$ while $\mathcal{H}$ (same macro) is the Hilbert space $L^2(w_y dy)$ of Section 3. The symbols below change meaning between the Parts or inside one. No symbol was renamed; each Part's own text fixes its meaning there, and this table and the reading conventions at the head of each Part are the convention.

The letter P . Read P_n by its argument. (a) With a polynomial argument, $P_n(t)$, $P_n(s^2)$, $P_n(0)$, $P_n(4)$, it is the parity-reduced polynomial of (II.10), $Z_n(s) = s^{\varepsilon_n} P_n(s^2)$, monic of degree $\lfloor n/2 \rfloor$ with strictly negative roots, and $D_n = P_n P_{n+1}$ (Parts II–IV; Part I does not use it). (b) As an operator factor, $P_n \mathcal{B} P_n$, $\text{Ran } P_n$, $X P_n Y^*$, it is a projection of rank n : onto polynomials of degree $< n$ in Part I, Section 3, and Part II, Section 15; onto the first n coordinates in Part I, Sections 2 and 6 (Part II writes this one P_n^0 , Section 20). (c) Without subscript, $P(s) = (s+1)^2/(2(s+2))$ is the pressure of Part I, (2); $P_{n-1}^{(a_J, b_J)}$ in Part II, Section 19, is a Jacobi polynomial. *Tempting false reading:* the identity $D_n = P_n P_{n+1}$ of Parts II–IV is a product of polynomials, not of projections, and Part I's determinant $\det(I + (t-3)P_n \mathcal{B} P_n)$ contains no polynomial P_n .

Symbol	Meanings
f	The pressure in t : $f(t) = \text{Log } P(\sqrt{t}) = 2\text{Log}(\sqrt{t} + 1) - \text{Log}(\sqrt{t} + 2) - \log 2$ (Parts I–III; (3), (II.110), (III.66)), $f(3) = 0$, $f(1) = \log(2/3)$. Also, in Part I: $f_j = \sqrt{w_y}\psi_j$ (Section 4) and $f(\theta) = \log \tan(\theta/2)$ (Section 5); the residual $L + (-1)^n d$ of Section 1 uses L , d as generic constants, not $L(s) = \text{Log } C(s)$. Part II uses f for a polynomial in Section 12 (as do Parts III and IV through it).
B, β	$B(s) = b + \frac{1}{2} \text{Log } P(s)$, the linear coefficient at fugacity s (Part I), which Part III writes $\beta(s)$ in (III.83); $B(1) = \beta(1) = \beta$. Also $\mathcal{B}$ the bounded operator (Parts I, II), B a matrix and $B_0(z)$ a symbol (Part I, Sections 3 and 6), B_n, B_e operators (Part I, Section 2), $B_{n,j}$ Bernoulli variables (Parts II, III), $B_{\pm}$ envelopes (Part IV).
E, E_n	Part I: $E_n(t) = \text{Log}(D_n(t)/D_n(3)) - nf(t)$ and its limit $E(t)$; $\mathcal{E}$ an operator. Part II, question 1 of Section 24: the number $E_n = \log[D_n(1)/D_n(3)] - n \log(2/3)$, which is Part I's $E_n(1)$. Part IV: $E_n = \log(P_n(3)/P_n(1))$ in (IV.35), unrelated. $\mathbb{E}$ is always an expectation.
D_n	The companion polynomial $D_n(t) = Z_n(\sqrt{t})Z_{n+1}(\sqrt{t})/\sqrt{t}$ in Parts I–III. Part IV: $D_n = \log(P_n(4)/P_n(1))$ in (IV.35), a number. <i>False reading</i> : Part IV's D_n is not a polynomial. Also $\mathcal{D} = s d/ds$ (Part I) and $\mathcal{D} = HM_w H + (I - H)M_w(I - H)$ (Part I, (12)).
m, v	$m(s) = \frac{1}{2} - 1/((s+1)(s+2))$ and $v(s) = s m'(s)$, the mean and variance slopes (Parts I–III); $m(1) = 1/3$, $v(1) = 5/36$. Also, in Part I: $m(y) = 1/(1 + e^{2ay})$ (Section 3), $v(F) = \sum k F_k^2$ (Section 2), the gamma phase v (Section 4); Part III: $m_n = \lfloor n/2 \rfloor$, $v_n = \text{Var}(T_n)$; Part IV: $m = \lfloor n/2 \rfloor$.
M_n	Parts II, III: the mean $M_n(s) = s Z'_n(s)/Z_n(s)$ of the diagonal count. Part I: M_n is multiplication by $e^{-in\theta}$ (a modulation) in Section 2, and $(M_n)_{ij} = \langle \psi_i, m\psi_j \rangle$ in Section 3.
C	$C(s)$ the individual amplitude, $C_* = C(1)$, C_{cal} the calibration amplitude, C_{ASM} , C_{odd} (Part I); C_{DSASM} the conjectured amplitude (Parts II–IV), which is Part I's C_* ; C_0 an operator (Parts I, II); C, C_K, C_η unspecified constants; C_n a matrix (Part III).
c	$c = \sqrt{3}$ (Parts I, III; Parts II and IV write $\sqrt{3}$); $c_0 = 11\pi/12$ a phase (Part I); c_n leading coefficients, $c_{n,k}$ coefficients (Part III, Section 27); $c_{\pm}$ the constants in (II.3); $c(T)$ the inverse centre in Part IV.
a, h	a_n the count. Part I: $a = \pi/(3h)$ with $h = \sqrt{3}$, a_j^{src} the source coefficients, $h_j = j!(3)_j$, H_j norms. Part IV: $a = (1 - \theta)/\theta$. a, b in “The leading equivalent and Conjecture 11.1” are Behrend–Fischer–Koutschan’s correction constants.
d	Part III: $d = 3 - 3\sqrt{3}/2 = m(\sqrt{3})$, the calibration mean slope; Part IV’s logarithmic calibration writes $d = 5/36$ where Parts II and III write d_{ASM} . Part I: $d_j = \sqrt{(j+1)(j+2)}$, $d(\theta) = \min(\theta, \pi - \theta)$. Part II: $d(r)$ in Section 22.
r	The inverse scale $r = \sqrt{\log T/\alpha}$ (Parts I, II), which Parts III and IV call n_0 ; $r(x)$ the tilted fugacity of the rate function (Part III, Section 38); $r(z) = 1 - z + z^2$ (Parts I, III); $r_{n,j}$ roots (Parts II, III); $r_{\pm}(T)$ roots (Part IV).
ν	Part III: ν_n, ν the root measures of (III.86). Part I: ν_t the source density of Section 3.
S	$S(A)$ and S_n the diagonal count; Part I: $S_n(\delta)$ a determinant, S the lower shift matrix, $\mathcal{S}$ a trace-class operator; Part III: S the random diagonal count.
T	T the count threshold (all Parts). Part I: $T_n(\delta)$ a determinant, T a convolution operator, $T_n(g) = g(S)$ a Toeplitz matrix, T_r a tilted compression. Part III: $T_n = (S - \varepsilon)/2$.
z, n_0	The real centre of the inverse: $z_C(T)$ in Part I, (81); $z(T)$ in Part II, (II.8) (without the amplitude term); $n_0 - \beta/(2\alpha)$ in Part III; $c(T)$ in Part IV.
Q_n	The multivariate polynomial (all Parts); Part I: $Q_n(\delta)$ a determinant in Section 2 and $Q_n = I - P_n$ in Section 6; Part III: Q_n^{mat} a matrix.
F	Part I: $F_n(s)$ the boundary ratio (8), F, F_k, F_δ angular functions, F_n^ϕ, F_n^ψ Gram matrices. Part II: $F(x)$ the inverse function of Section 22. Part IV: $F = \ell \circ u$.
κ, θ	$\kappa = 5/72$ throughout; Part IV: $\theta = \log 3/\log 4$; elsewhere θ is an angle.

Provenance and merge decisions

Four manuscripts of the session bundle of Reports 1–243, all dated 2 October 2026, each headed “REPORT N ”; none names an author or a tool, none carries “prepared for private review” wording, and none pins a ProveIt commit or names a repository path. They arrived in commit 60f54ea06 and were placed by 612787fb4 (batch 104).

- **Part I**, bundle Report 128, archive A005163_Full_Leading_Equivalent_and_Individual_Amplitude_Source.zip (report128.tex with eight files sections/01_main.tex–08_references.tex, 27 pages). The base: its text became this article.tex. It ships Report 124’s TeX and PDF byte for byte as earlier_input/ (checked at placement). Contributes the full equivalent at every fugacity, the companion limit, the common calibration amplitude, the boundary theorem, the Szegő-type proposition, the amplitude-aware inverse and the cumulant constants.
- **Part II**, bundle Report 124, archive A005163_Exact_Logarithmic_Scale_and_Inverse_Thresholds_Source.zip (report124.tex, 34 pages). Contributes the shared exact prerequisites, the operator theory that Part I uses, the bounded remainder, the trace limit, the inverse with the log log term, the holomorphic pressure, bounded cumulant errors and the explicitly centred CLT.
- **Part III**, bundle Report 121, archive A005163_Linear_Growth_Fugacity_Pressure_and_Limit_Laws_Source.zip (report121.tex, 27 pages). Contributes the Jacobi-matrix route to the pressure, the linear coefficient, the root measure, Newton inequalities, Hoeffding, the actual-mean and local CLTs and the large deviations.
- **Part IV**, bundle Report 120, archive A005163_Diagonal_Roots_Growth_Bounds_and_Inverse_Brackets_Source.zip (report120.tex, 15 pages). Contributes the Jensen and chord inequalities on the roots, the projected two-step contraction, the certified constants p, q and the two-ceiling inverse.

Where the merge had to choose:

1. *Order and base*. Strength order, with Report 128 as base (the placement decision): its theorem implies every growth statement of the others. Logical order would put Report 124 first, since Part I depends on it; the Guide says so, and Part I’s references to Report 124 are internal.
2. *Shared sections*. Printed once, as listed in the Guide: Report 124’s Sections 2, 4, 5 (Sections 11, 13, 14) stand for Reports 121’s and 120’s Sections 2, 4, 5; Report 121’s Section 3 (Section 27) for Report 120’s Section 3; Report 124’s Section 3 (Section 12) for Report 121’s Section 9 except its two additional equations. The differences, checked by diff of the delivered sources, are: in Reports 120 and 121 the stability proof ends “This proves the structural part of Theorem [main]” and writes ε for ε_n ; Report 120’s closing non-claim of that section is longer (printed in Section 42) and Report 121’s is Report 124’s; Report 120’s logarithmic calibration writes d for d_{ASM} ; Report 121’s interlacing section has the identities (III.63)–(III.64) and a different last sentence (printed in Section 33).
3. *Citations of Report 124 in Part I*. The eight \cite{r124} of Report 128 became references to Part II, with section numbers converted (Report 124’s Section k is Section $k + 9$); the item list (I1)–(I4) of Section 1 and the bibliography entry keep the delivered numbers beside the

converted ones. Wherever Part I says “the earlier article”, “the earlier proof” or “the supplied article”, it means Part II.

4. *Front matter and bibliography.* The four title blocks and abstracts are recorded at the head of their Parts; one table of contents replaces four. The four bibliographies are merged into one, keeping every detail of every entry (where two manuscripts annotate an entry differently, both annotations are kept, each with its Part). The write added no reference.
5. *Write additions.* This front matter, the reading conventions of each Part, Section 9, the pointer sections of Parts III and IV, and the dated notes. No theorem was added; the notes contain short deductions, each stated with its justification (listed in the README).

What was checked, and what was not

The report is unrefereed and not formalized. At intake (dossier of batch 104, 5 October 2026) the four manuscripts were read in full, and the following were checked independently of the delivered programs. Exact counts from an independent Pfaffian and leading-minor computation of the kernel of [1, (4.1)] reproduce the twenty OEIS terms in Report 120’s fixtures and extend A005163 exactly to $n = 177$; $Z_n(2) = a_{n+1}$ ($n \leq 39$), $Z_{2m}(\sqrt{3}) = 4^m O_m$ ($m \leq 20$) and $Z_n(c)Z_{n+1}(c) = c^{2^n} A_{n+1}$ ($n \leq 39$) hold exactly. The exact polynomials P_n , $n \leq 31$, are monic with $P_n(0) > 0$ and only negative real roots, and satisfy weak interlacing, the Newton inequalities and $\max |M_n - M_{n-1}| = 1$ at the fugacities tested. Quadratures confirm $\mu(\text{Log}(1 + \delta w)) = 2f(3 + \delta)$ to twenty digits (the input of (25)) and that the Stieltjes transform of Part III’s root density is $f'(t)$. The boundary limit (8) at $s = \frac{1}{2}, 1, 3$, the calibration constant $C_{\text{ASM}}/C_{\text{cal}}^2 = \sqrt{2/3}$ (30 digits), the constant $\sqrt{2}3^{-1/4}$ of (34), the expansion (83), the relations $m = \mathcal{D}B$, $v = \mathcal{D}m$, Part III’s Legendre transform and rate endpoints, and the mean bracket (III.64) were checked by hand or numerically. The delivered verification suites pass on copies (README). Nothing tested failed, and no claim of the four manuscripts was found to be wrong.

What was *not* done: **Part I’s operator-theoretic Sections 2 and 4–6 were read for structure and every displayed algebraic identity that could be checked, but were not re-derived line by line at intake**; the same holds for the analytic estimates of Part II’s Sections 16–20 and 23, on which Part I rests. The limit theorems are therefore the manuscripts’ proofs, corroborated numerically, and not an independent verification. The external inputs (Borcea–Brändén, Krattenthaler, Rosengren, Frenzen–Wong, Simon, Breuer–Duits, Casper–Zurrián, Alseidi–Margaliot–Garloff, Cao–Li–Lin, DLMF) were not re-checked against their sources, except Behrend–Fischer–Koutschan: on 5 October 2026 the write checked against the HTML text of arXiv:2309.08446v3 that version 3 is dated 1 October 2026 and that its equations (4.1), (4.11), (5.10) (Proposition 5.1), (11.3)–(11.6), (11.8) (Conjecture 11.1) and Theorem 11.2 are as the manuscripts use them. The finite checks of the delivered suites and of the dossier corroborate identities; none proves an asymptotic statement, and the numerical amplitude estimates of Section 9 are uncertified.

Relation to neighbouring reports and formal projects

No other report of the collection treats alternating sign matrices, DSASMs, A005163 or the paper [1]: a search of the repository on 5 October 2026 found only a bibliography item (Bressoud’s *Proofs and Confirmations*) in a q -Pochhammer monograph of the Fabius tree and a library-search log

line naming A005163 in `a271619-strict-twice-partitions`. Nothing here answers or continues another report, and no reciprocal note is needed. Placement in the collection confers no formal status: no Lean or Rocq file in the repository treats alternating sign matrices, and no statement of this report is formalized.

Part I

The full leading equivalent and individual amplitudes

Bundle Report 128, archive A005163_Full_Leading_Equivalent_and_Individual_Amplitude_Source.zip, delivered as `report128.tex` with the eight modules `sections/01_main.tex`–`sections/08_references.tex` (27 pages), dated 2 October 2026. Delivered title block: “REPORT 128 / The full leading equivalent for diagonally symmetric alternating sign matrices / Individual fugacity amplitudes and finite-section boundary limits / 2 October 2026 / A proof with an unchanged earlier-input appendix and offline replay”. Printed as delivered, with [write] notes; section, statement and equation numbers are those of the delivery. Section 9 is added in the write.

Reading conventions of Part I	Elsewhere in the report
$P(s)$ the pressure; P_n a projection (degree $< n$ in Section 3, first n coordinates in Sections 2 and 6)	Parts II–IV: $P_n(t)$ also the parity-reduced polynomial
$E_n(t)$, $E(t)$ companion remainder and its limit	Part II: the number $E_n = E_n(1)$; Part IV: $E_n = \log(P_n(3)/P_n(1))$
$B(s)$ linear coefficient; $\mathcal{B}$, B , B_n , B_e operators	Part III: $\beta(s) = B(s)$
M_n modulation by $e^{-in\theta}$ (Section 2)	Parts II, III: $M_n(s)$ the mean
$m(y)$, $v(F)$, gamma phase v , f_j , $f(\theta)$	Parts I–III: $m(s)$, $v(s)$ slopes; $f(t)$ pressure
“the earlier article / proof / supplied article”	Part II (Report 124)

Abstract of this Part (as delivered). For the number a_n of diagonally symmetric alternating sign matrices, A005163, we prove a full-sequence equivalent $a_n \sim Ce^{\alpha n^2 + \beta n} n^{-5/72}$, where $\alpha = \frac{1}{2} \log(3\sqrt{3}/4)$ and $\beta = \frac{1}{4} \log 3$. The positive constant is characterized by convergent Fredholm determinants and a convergent scalar Fourier series. More generally, the partition function at every positive diagonal fugacity has a common individual amplitude, with a locally holomorphic limit on the right half-plane. There are two distinct new steps: convergence of the companion determinant remainder, and a quantitative finite-section boundary theorem that removes the alternating ambiguity in adjacent products. The latter retains the leading Galerkin correction and uses an exact Christoffel norm identity for weighted continuous-Hahn tightness. We give the amplitude-aware inverse threshold with a vanishing remainder before integer ceilings, and the constant terms of all fixed diagonal cumulants. No numerical amplitude, explicit rate of convergence, inverse-power correction to the general count, or all-orders expansion is claimed.

[write] **Reading Part I against Part II, 5 October 2026.** Report 128 was delivered with Report 124 as an unchanged “earlier input” (`earlier_input/report124.tex` and `.pdf`, byte-identical to the delivered Report 124). That copy is not shipped here: Report 124 is printed in full as Part II. The eight citations `\cite{r124}` of the delivery are references to Part II below, and Report 124’s Section k is Section $k + 9$ here. Where Part I speaks of “the earlier article”, “the earlier proof”, “the supplied earlier proof”, “the earlier exact identities”, “the earlier interlacing argument” or “the earlier global kernel bound”, the result meant is in Part II; the list (I1)–(I4) in Section 1 locates them. Part I’s equivalent is the leading term of Behrend, Fischer and Koutschan’s Conjecture 11.1; see “The leading equivalent and Conjecture 11.1” in the front matter for exactly what is and is not proved.

1 Definitions, theorem and proof dependencies

An alternating sign matrix has entries in $\{-1, 0, 1\}$, alternating nonzero entries along each row and column, and all row and column sums one. It is diagonally symmetric when it equals its transpose. Write

$$a_n = |\text{DSASM}(n)|, \quad S(A) = |\{i : A_{ii} \neq 0\}|, \quad Z_n(s) = \sum_{A \in \text{DSASM}(n)} s^{S(A)}, \quad Z_0(s) = 1.$$

Thus $a_n = Z_n(1)$ and a_n is A005163. All logarithms of positive numbers are natural. Put

$$c = \sqrt{3}, \quad \alpha = \frac{1}{2} \log \frac{3\sqrt{3}}{4}, \quad \beta = \frac{1}{4} \log 3, \quad \kappa = \frac{5}{72}, \quad b = \frac{3}{4} \log 3 - \frac{1}{2} \log 2. \quad (1)$$

On $\mathcal{H}_+ = \{\Re s > 0\}$ define

$$P(s) = \frac{(s+1)^2}{2(s+2)}, \quad B(s) = b + \frac{1}{2} \text{Log } P(s). \quad (2)$$

Here $\text{Log } P(s) = 2 \text{Log}(s+1) - \text{Log}(s+2) - \log 2$, with the indicated right-half-plane logarithms. In particular $P(c) = 1$ and $B(1) = \beta$. The companion polynomial and its centered logarithm are

$$D_n(t) = \frac{Z_n(\sqrt{t})Z_{n+1}(\sqrt{t})}{\sqrt{t}}, \quad f(t) = \text{Log } P(\sqrt{t}), \quad E_n(t) = \text{Log } \frac{D_n(t)}{D_n(3)} - n f(t), \quad (3)$$

where $t \in \Omega_t = \mathbb{C} \setminus (-\infty, 0]$ and the square root has positive real part. The quotient defining D_n is interpreted polynomially. Its logarithm is normalized to vanish at $t = 3$.

Theorem 1.1 (Full leading equivalent). *There is a holomorphic function E on Ω_t , real on $(0, \infty)$, with $E(3) = 0$, such that $E_n \rightarrow E$ locally uniformly. Define*

$$C_{\text{cal}} = \frac{3^{11/72} \pi^{1/6} e^{\zeta'(-1)/6}}{2^{1/24} \Gamma(1/3)^{1/3}} > 0 \quad (4)$$

and, on $\mathcal{H}_+$,

$$C(s) = C_{\text{cal}} \exp\{E(s^2)/2\} (s/c)^{1/2} P(s)^{-1/4}. \quad (5)$$

The powers use the analytic logarithms normalized at $s = c$. Then C is nowhere zero and positive for $s > 0$, and

$$Z_n(s) = C(s) e^{\alpha n^2 + n B(s)} n^{-\kappa} (1 + o(1)) \quad (6)$$

locally uniformly on $\mathcal{H}_+$, along the full integer sequence. In particular,

$$a_n \sim C_{\text{cal}} 2^{-1/4} e^{E(1)/2} e^{\alpha n^2 + \beta n} n^{-5/72}. \quad (7)$$

A convergent operator/Fourier expression for $E(t)$ is given in (26); it defines the constant without a numerical fit.

[write] **Conjecture 11.1 and the earlier Parts, 5 October 2026.** Formula (7) is the leading term of Behrend, Fischer and Koutschan's Conjecture 11.1, equation (11.8) of [1], whose constant C is $C_* = C_{\text{cal}} 2^{-1/4} e^{E(1)/2}$ here; the conjecture's finer terms are not proved (see "The leading equivalent and Conjecture 11.1" in the front matter, and Section 9). The theorem supersedes the two-sided bounds (II.3) of Part II (Theorem 10.1): the normalized counts $a_n e^{-\alpha n^2 - \beta n} n^{\kappa}$ converge, to C_* . It also supersedes the free energy (III.83) with error $o(n)$ of Part III and the envelopes (IV.5) of Part IV; at $s = 1$, $B(1) = \beta$ lies strictly between Part IV's p and q . Theorem 1.1 does not restate Part II's calibration trace limit (II.7).

1.1 What is supplied in the reader bundle

The unchanged earlier article (Report 124, printed here as Part II), supplied both as a PDF and as its complete TeX source in the bundle's `earlier_input` folder, proves the bounded-remainder stage. Its scope statements are historical: its failure to assert a limit is strengthened here. All new convergence, calibration and boundary steps are proved below; no unpublished proof note is a required reader input.

For precision, these are the earlier results used, with their locations in that supplied article:

- (I1) Sections 2–4 (here Sections 11–13): imaginary-axis zeros of Z_n , negative-real zeros of D_n , cross-size interlacing, the exact adjacent determinant and its finite Pascal reduction, and $Z_n(c)Z_{n+1}(c) = c2^n A_{n+1}$, where $A_N = |\text{ASM}(N)|$.
- (I2) Sections 6–8 (here Sections 15–17): the exact degree-space compression $D_n(t)/D_n(3) = \det(I + (t-3)P_n \mathcal{B} P_n)$, the bounded operator $\mathcal{B}$, its unweighted decomposition $C_0 + \mathcal{E}$ with $0 \leq C_0 \leq I/3$, $\mathcal{E} \in \mathcal{S}_2$, $\mathcal{S} = (\mathcal{E} + \mathcal{E}^*)/2 \in \mathcal{S}_1$, and the locally uniform nonzero relative determinant $\mathcal{A}(t)$. The exact continuous-Hahn/Fourier/Jacobi transform and the gamma-phase comparison are included there.
- (I3) Sections 9–11 (here Sections 18–20): the normalized Jacobi expansion, its endpoint and derivative bounds, the mean formula $\text{tr}(J_n F J_n) = (n+1)\pi^{-1} \int_0^\pi F + o(1)$ for endpoint-vanishing F with $\int |F|/\min(\theta, \pi - \theta) < \infty$, the global angular kernel bound $|K_n(\theta, \phi)| \leq C/|\theta - \phi|$, the finite tilted-projection variance bound, and the Hilbert–Schmidt weighted Hardy wave comparisons.
- (I4) Sections 12 and 14 (here Sections 21 and 23): the convergent trace-class cross factor $\Xi(t)$, the compact-uniform scalar bound $\log T_n(t-3) - 2nf(t) = O_K(1)$, the analogous companion bound, and local uniform bounds on adjacent individual logarithmic differences.

These are theorems and exact identities proved in Part II, rather than assertions supplied by finite checks. We restate the definitions needed to apply them. The additional published ingredients are the ASM/OSASM product asymptotics in [1], the continuous-Hahn normalization and generating function [11, 8], and the *polynomial* Jacobi-statistic central limit theorem [10]. We state precisely where each is used and prove the necessary extensions.

1.2 The two logically separate limits

An adjacent-product limit does not alone determine the individual amplitudes: a residual $L + (-1)^n d$ has a constant adjacent sum. Section 2 first proves $E_n \rightarrow E$. Sections 3–6 then prove the distinct boundary-ratio limit

$$F_n(s) := \frac{sZ_n(s)/Z_{n+1}(s)}{cZ_n(c)/Z_{n+1}(c)} \longrightarrow \frac{s/c}{\sqrt{P(s)}}. \quad (8)$$

The finite-section proof is at the $1/n$ coefficient scale. In particular, it does not substitute an infinite resolvent for a finite one merely from strong convergence. Section 7 combines the two limits, proves the inverse threshold and derives constant-order cumulants.

2 Convergence of the companion remainder

We use $\delta = t - 3 \in \Omega = \mathbb{C} \setminus (-\infty, -3]$. All analytic logarithms in this section vanish at $\delta = 0$. Distinguish the original real-line weight w_y in Section 3 from the angular multiplier

$$w(\theta) = \frac{1}{1 + 2 \cosh(\xi(\theta)/3)}, \quad \xi(\theta) = 2 \log \cot(\theta/2), \quad s_0(\theta) = \sqrt{w(\theta)}. \quad (9)$$

Thus $0 \leq w \leq 1/3$ and $w = O(d^{2/3})$ for $d(\theta) = \min(\theta, \pi - \theta)$. Let $U : \ell^2(\mathbb{N}_0) \rightarrow L^2(0, \pi)$ synthesize the real normalized Jacobi angular basis with parameters $(4/3, 2/3)$, and let H be the orthogonal Hardy projection obtained from the negative real half-line under Fourier and angular transformation. Explicitly it is unitarily conjugate to the Fourier image of multiplication by $\mathbf{1}_{(-\infty, 0)}$. Write P_n for the first n coordinate projection and $J_n = UP_nU^*$. Set

$$X = M_{s_0}HU, \quad Y = M_{s_0}(I - H)U, \quad A_n = P_nX^*XP_n, \quad B_n = P_nY^*YP_n.$$

Here A_n, B_n act on $\text{Ran } P_n$. Define

$$Q_n(\delta) = \det(I + \delta A_n) = \det(I + \delta B_n), \quad S_n(\delta) = \det(I + \delta(A_n + B_n)),$$

$$T_n(\delta) = \det_{\text{Ran } J_n} (I + \delta J_n M_w J_n).$$

Equality of the two single determinants follows from complex conjugation in the real Jacobi basis, which interchanges H and $I - H$. The supplied earlier proof gives, locally uniformly on Ω ,

$$\frac{D_n(t)/D_n(3)}{Q_n(t-3)} \longrightarrow \mathcal{A}(t), \quad \frac{S_n(t-3)}{T_n(t-3)} \longrightarrow \Xi(t). \quad (10)$$

For clarity, their complete operator definitions are

$$\mathcal{A}(t) = \det_2 \left(I + \delta(I + \delta C_0)^{-1} \mathcal{E} \right) \exp \left\{ \delta \text{tr}[(I + \delta C_0)^{-1} \mathcal{S}] \right\}, \quad (11)$$

$$\Xi(t) = \det \left[(I + \delta \mathcal{D})(I + \delta M_w)^{-1} \right], \quad \mathcal{D} = HM_wH + (I - H)M_w(I - H). \quad (12)$$

In (11) the real-line operators are exactly those defined in (29) below. The difference $\mathcal{D} - M_w$ is trace class; the two factors are invertible on Ω . Both $\mathcal{A}$ and Ξ are nowhere zero, normalized to one at calibration, and positive on $t > 0$. The gamma-phase comparison has limiting relative determinant one, which is already included in the first limit in (10).

2.1 A moving-edge mixed determinant limit

The weighted-wave result of Part II, Section 20, gives bounded maps

$$X_0 e_j = \frac{s_0(\theta) e^{-i(j+3/2)\theta + ic_0}}{\sqrt{2\pi}}, \quad Y_0 e_j = \frac{s_0(\theta) e^{i(j+3/2)\theta - ic_0}}{\sqrt{2\pi}}, \quad c_0 = 11\pi/12,$$

with $X - X_0, Y - Y_0 \in \mathcal{S}_2$. Importantly, $XY^* = 0$ exactly, since $UU^* = I$ and $H(I - H) = 0$. Define M_n to be multiplication by $e^{-in\theta}$ and define the Hilbert-Schmidt operator K by its kernel

$$K(\theta, \phi) = \frac{s_0(\theta)s_0(\phi)}{2\pi} \frac{e^{2ic_0 - (3i/2)(\theta + \phi)}}{1 - e^{-i(\theta + \phi)}}. \quad (13)$$

At either singular corner its square is bounded by $C\theta^{2/3}\phi^{2/3}/(\theta + \phi)^2$ or its reflection, which is integrable after setting $\theta = ru$, $\phi = r(1 - u)$. Thus $K \in \mathcal{S}_2$.

The geometric sum gives the exact identity

$$X_0 P_n Y_0^* = K - M_n K M_n.$$

Since $M_n \rightarrow 0$ by the Riemann–Lebesgue lemma and K is compact, $M_n K M_n \rightarrow 0$ strongly. Taking the strong limit gives $X_0 Y_0^* = K$. Set $E_X = X - X_0$, $E_Y = Y - Y_0$. Finite-rank approximation in $\mathcal{S}_2$ gives

$$X P_n Y^* - X_0 P_n Y_0^* = E_X P_n Y^* + X_0 P_n E_Y^* \rightarrow E_X Y^* + X_0 E_Y^* = -K$$

in Hilbert–Schmidt norm. Consequently

$$K_n := X P_n Y^* = -M_n K M_n + o_{\mathcal{S}_2}(1). \quad (14)$$

In particular $\text{tr}(A_n B_n) = \|K_n\|_2^2 \rightarrow \|K\|_2^2$. The moving term in (14) is retained; it does not tend to zero in Hilbert–Schmidt norm.

Let E_I extend functions on $(0, \pi)$ by zero to the circle $(0, 2\pi)$. Let P_+° project onto circle modes $j \geq 1$ and P_-° onto $j \leq -1$, and set $G_\pm = M_{e \pm 3i\theta/2}$. Define

$$A_e = M_{s_0} G_- E_I^* P_+^\circ E_I G_+ M_{s_0}, \quad B_e = M_{s_0} G_+ E_I^* P_-^\circ E_I G_- M_{s_0}. \quad (15)$$

The strict mode cutoffs follow from $n - j = 1, \dots, n$ in the finite wave sums, and yield

$$M_n^* X P_n X^* M_n \rightarrow A_e, \quad M_n Y P_n Y^* M_n^* \rightarrow B_e \quad (16)$$

strongly. Here is why the fixed Hilbert–Schmidt errors do not affect these limits. For fixed h , $P_n X^* M_n h$ is bounded and weakly null: pairing with g reduces to pairing $M_n h$ with $X P_n g \rightarrow X g$ in norm. Compactness of E_X then gives $E_X P_n X^* M_n h \rightarrow 0$, and compactness also gives $E_X^* M_n h \rightarrow 0$. Expand $X P_n X^* - X_0 P_n X_0^* = E_X P_n X^* + X_0 P_n E_X^*$ to obtain the first claim. The other is identical with reversed modulation. Both limits are positive and bounded by $I/3$.

For $0 \leq T \leq I/3$, $(I + \delta T)^{-1}$ is uniformly bounded for δ in compact subsets of Ω . Strong convergence in (16) gives strong convergence of these resolvents, locally uniformly on each fixed vector. A finite parameter net and the resolvent identity justify this uniformity.

Apply Sylvester’s determinant identity to the block map with entries $X P_n$ and $Y P_n$, and take its upper-left Schur complement. With

$$R_{X,n} = (I + \delta X P_n X^*)^{-1}, \quad R_{Y,n} = (I + \delta Y P_n Y^*)^{-1},$$

one obtains, with the order of factors as written,

$$\frac{S_n(\delta)}{Q_n(\delta)^2} = \det[I - \delta^2 R_{Y,n} K_n^* R_{X,n} K_n]. \quad (17)$$

Conjugate the product on the right by M_n , and use (14). Products of two Hilbert–Schmidt factors are trace class, so all terms containing its $o_{\mathcal{S}_2}(1)$ error tend to zero in trace norm, uniformly on compact parameter sets. The main term is

$$\delta^2 (M_n R_{Y,n} M_n^*) K^* (M_n^* R_{X,n} M_n) K.$$

Strong resolvent convergence and finite-rank approximation of K now give locally uniform trace-norm convergence. Hence

$$\frac{S_n(\delta)}{Q_n(\delta)^2} \longrightarrow \Psi(\delta) := \det \left[I - \delta^2 (I + \delta B_e)^{-1} K^* (I + \delta A_e)^{-1} K \right]. \quad (18)$$

Every finite ratio is nonzero on Ω , and $\Psi(0) = 1$. Hurwitz's theorem makes Ψ nowhere zero. It is positive for real $\delta > -3$, as the nonzero limit of positive ratios. Thus

$$2 \log Q_n(\delta) - \log S_n(\delta) \longrightarrow -\text{Log } \Psi(\delta) \quad (19)$$

locally uniformly, with the common normalization at zero. Alternatively, the modulated positive block matrices have strong limit $\begin{pmatrix} A_e & -K \\ -K^* & B_e \end{pmatrix}$ bounded by $I/3$; its Schur complement directly proves invertibility of the limiting factor in (18).

2.2 The scalar constant: a polynomial theorem and its extension

For an angular function $F(\theta) = g(\cos \theta)$ put

$$\mu(F) = \frac{1}{\pi} \int_0^\pi F(\theta) d\theta, \quad F_k = \frac{1}{\pi} \int_0^\pi F(\theta) \cos(k\theta) d\theta, \quad v(F) = \sum_{k \geq 1} k F_k^2.$$

For real F , the squares in v are ordinary real squares.

Proposition 2.1. *If F is real and angular Hölder of exponent $1/2 < \gamma \leq 1$, with $F(0) = F(\pi) = 0$, then $v(F) < \infty$ and*

$$\log \det_{\text{Ran } J_n} (J_n e^g J_n) = (n+1)\mu(F) + \frac{1}{2}v(F) + o(1). \quad (20)$$

Here g denotes multiplication by the function whose angular form is F .

Proof. The rank- n Jacobi polynomial ensemble has density proportional to its squared Vandermonde times the Jacobi weight. Andreief's finite identity gives

$$\mathbb{E} e^{X_n(g)} = \det(J_n e^g J_n), \quad \mathbb{E} X_n(g) = \text{tr}(J_n g J_n), \quad \text{Var}_n(g) = \|(I - J_n)gJ_n\|_2^2.$$

The earlier global kernel bound gives for every real angular C^η function H , $\eta > 1/2$,

$$\text{Var}_n(H) = \frac{1}{2} \iint (H(\theta) - H(\phi))^2 |K_n(\theta, \phi)|^2 d\theta d\phi \leq C_\eta [H]_{C^\eta}^2, \quad (21)$$

independently of n . Endpoint vanishing is unnecessary for this bound.

The polynomial input is Corollary 2.2 of Breuer–Duits [10]: if the recurrence matrix has a Laurent right limit, every fixed real polynomial statistic, centered by its exact mean, converges to a centered Gaussian with variance $\sum_{k \geq 1} k \hat{p}_k \hat{p}_{-k}$, where the Fourier coefficients are those of the polynomial applied to the Laurent symbol. The fixed Jacobi recurrence here has off-diagonal limit $1/2$ and diagonal limit zero, hence the unique full-sequence symbol $(z + z^{-1})/2$. Its coefficients are bounded and the support is compact. Thus this theorem applies to every real polynomial p , with variance $v(p(\cos \theta))$. Only the polynomial theorem is used; its separate C^1 extension is not applied to the fractional endpoint symbol.

A weak CLT alone is insufficient for the determinant. To obtain exponential integrability, let $E_r = e^{rg/2}$ and $T_r = J_n e^{rg} J_n$ on $\text{Ran } J_n$. The isometry $E_r J_n T_r^{-1/2}$ has range projection $Q_r = E_r J_n T_r^{-1} J_n E_r$. For any bounded real multiplier a , commutation with E_r gives the exact identity

$$(I - Q_r) a E_r J_n T_r^{-1/2} = (I - Q_r) E_r (I - J_n) a J_n T_r^{-1/2}.$$

For $r \geq 0$, taking Hilbert–Schmidt norms gives $\text{Var}_{rg}(a) \leq e^{r \text{osc}(g)} \text{Var}_n(a)$. Taylor’s integral formula for the logarithmic moment generating function, also applied to $-g$, therefore yields for real u

$$\log \mathbb{E} e^{u(X_n(g) - \mathbb{E} X_n(g))} \leq \frac{1}{2} u^2 e^{|u| \text{osc}(g)} \text{Var}_n(g). \quad (22)$$

For fixed polynomial p , the bound at $u = 2$ and (21) prove uniform integrability of $e^{X_n(p) - \mathbb{E} X_n(p)}$. The polynomial CLT consequently implies

$$G_n(p) := \log \mathbb{E} e^{X_n(p) - \mathbb{E} X_n(p)} \longrightarrow v(p)/2. \quad (23)$$

We next justify extension uniformly in n . For real g, h put $a = g - h$ and $g_u = h + ua$, $0 \leq u \leq 1$. Differentiation of the finite determinant and then integration along its tilt path give

$$\frac{d}{du} G_n(g_u) = \mathbb{E}_{g_u} X_n(a) - \mathbb{E}_0 X_n(a) = \int_0^1 \text{Cov}_{rg_u}(X_n(a), X_n(g_u)) dr.$$

Tilted covariance Cauchy–Schwarz and the preceding ordered identity show

$$\left| \frac{d}{du} G_n(g_u) \right| \leq e^{\text{osc}(g_u)} \sqrt{\text{Var}_n(a) \text{Var}_n(g_u)}. \quad (24)$$

By (21), G_n are uniformly Lipschitz on bounded real C^η sets for $\eta > 1/2$.

Choose $1/2 < \eta < \gamma$. Extend F evenly to the circle. Its Fejér convolutions F_m are even trigonometric polynomials, converge uniformly, and have bounded C^γ seminorms. The elementary increment estimate

$$[H]_{C^\eta} \leq C \|H\|_\infty^{1-\eta/\gamma} [H]_{C^\gamma}^{\eta/\gamma}$$

proves convergence in C^η . Subtract $a_m + b_m \cos \theta$ with $a_m = (F_m(0) + F_m(\pi))/2$ and $b_m = (F_m(0) - F_m(\pi))/2$. The corrected functions Q_m vanish at both endpoints, are polynomials in $\cos \theta$, and converge to F in C^η , since $a_m, b_m \rightarrow 0$.

The circle $H^{1/2}$ difference-quotient identity, first verified by Fourier orthogonality for trigonometric polynomials, identifies $\sum |k| |\hat{H}_k|^2$ with a fixed positive multiple of

$$\iint \frac{|H(\theta) - H(\phi)|^2}{\sin^2((\theta - \phi)/2)} d\theta d\phi.$$

It is bounded by $C_\eta [H]_{C^\eta}^2$. Thus $v(F - Q_m) \rightarrow 0$ and $v(Q_m) \rightarrow v(F)$ by weighted Cauchy–Schwarz. Equations (23) and (24), first with $n \rightarrow \infty$ and then $m \rightarrow \infty$, give $G_n(g) \rightarrow v(F)/2$. Finally the earlier Jacobi mean theorem applies because $F = O(d^\gamma)$ at the endpoints. It gives $\mathbb{E} X_n(g) = (n+1)\mu(F) + o(1)$, proving (20). $\square$

Apply this proposition to

$$F_\delta(\theta) = \text{Log}(1 + \delta w(\theta)), \quad F_{\delta,k} = \frac{1}{\pi} \int_0^\pi F_\delta(\theta) \cos(k\theta) d\theta.$$

For real $\delta > -3$ it is real, endpoint-vanishing and $C^{2/3}$, and the exact pressure integral from Part II, Section 18, is $\mu(F_\delta) = 2f(3 + \delta)$. Therefore

$$\log T_n(\delta) - 2nf(3 + \delta) \longrightarrow 2f(3 + \delta) + \frac{1}{2} \sum_{k \geq 1} k F_{\delta,k}^2. \quad (25)$$

The earlier complex scalar bound makes the left side locally bounded and holomorphic on Ω . Vitali's theorem extends this convergence locally uniformly over Ω . The displayed series is its analytic expression there, with squares, not absolute squares. Indeed for $1/2 < \eta < 2/3$ the map $\delta \mapsto F_\delta$ is holomorphic as a C^η -valued map, by the uniform lower bound for $|1 + \delta u|$, $0 \leq u \leq 1/3$, on compact parameter sets. Its image on a compact set is compact in the weighted Fourier ℓ^2 space. Finite-coordinate projections then have uniformly vanishing tails on this image. The bilinear Cauchy–Schwarz inequality proves local uniform convergence and holomorphy of the square series. The real identity and analytic uniqueness prove the claimed expression.

2.3 The explicit limit and branch normalization

Combine (10), (19), and (25). Every constituent logarithm is normalized at $t = 3$, and their locally uniform limits give

$$E(t) = \text{Log } \mathcal{A}(t) + \frac{1}{2} \text{Log } \Xi(t) - \frac{1}{2} \text{Log } \Psi(t - 3) + f(t) + \frac{1}{4} \sum_{k \geq 1} k F_{t-3,k}^2. \quad (26)$$

This proves $E_n \rightarrow E$ locally uniformly on Ω_t . At $t = 3$ every term is zero. For positive t the three determinant factors are positive, so $E(t)$ is finite and real. Notice both the minus sign of the mixed factor and the scalar coefficient $1/4$: one first solves $S_n/Q_n^2 \rightarrow \Psi$ for $\log Q_n$, and then halves the scalar variance term. These fix the amplitude normalization. As a separate normalization check, for the polynomial $p(x) = x$ the only positive cosine coefficient is $1/2$, so $v(p) = 1/4$ and the centered logarithmic determinant tends to $1/8$. Directly the finite variance is the square of the single Jacobi recurrence coefficient crossing the degree- n boundary, which tends to $1/4$.

3 Calibration and the exact boundary resolvent

3.1 A common calibration amplitude on both parities

The earlier exact identities give

$$Z_{2m}(c) = 2^m \prod_{k=1}^m \frac{A_{2k}}{A_{2k-1}}, \quad Z_n(c) = c 2^{n-1} A_n / Z_{n-1}(c) \quad (n \text{ odd}).$$

Let O_m denote the number of off-diagonally symmetric ASMs of order $2m$. The classical products in [1] are

$$A_N = \prod_{i=0}^{N-1} \frac{(3i+1)!}{(N+i)!}, \quad O_m = \prod_{i=1}^m \frac{(6i-2)!}{(2m+2i)!}.$$

Successive ratios give

$$\frac{O_m}{O_{m-1}} = \frac{(6m-2)!(2m)!}{(4m)!(4m-2)!} = \frac{1}{2} \frac{A_{2m}}{A_{2m-1}}.$$

For example, the old denominator factors in the O_m quotient telescope to $(4m-2)!/(2m)!$. The initial products equal one, so $Z_{2m}(c) = 2^{2m}O_m$ exactly.

The published product asymptotics [1, Equations (11.3)–(11.6)], using the even-order OSASM branch, state

$$\begin{aligned} A_n &\sim C_{\text{ASM}} e^{2\alpha n^2} n^{-2\kappa}, \\ 2^n O_{n/2} &\sim C_{\text{cal}} e^{\alpha n^2 + bn} n^{-\kappa} \quad (n \text{ even}), \\ C_{\text{ASM}} &= \frac{2^{5/12} \pi^{1/3} e^{\zeta'(-1)/3}}{3^{7/36} \Gamma(1/3)^{2/3}}. \end{aligned}$$

These are known product asymptotics, not an assumption about DSASM amplitudes. Substitution in the odd exact identity gives the odd constant

$$C_{\text{odd}} = (c/2) e^{b-\alpha} C_{\text{ASM}} / C_{\text{cal}}.$$

Now $b - \alpha = \frac{1}{2} \log 2$ and $C_{\text{ASM}}/C_{\text{cal}}^2 = \sqrt{2/3}$ by the displayed constants. Therefore $C_{\text{odd}} = C_{\text{cal}}$, proving the full-sequence calibration

$$Z_n(c) \sim C_{\text{cal}} e^{\alpha n^2 + bn} n^{-\kappa}. \quad (27)$$

Only this equivalent is needed below; no finite-size correction is transferred to a noncalibrated fugacity.

[write] **A common calibration amplitude, 5 October 2026.** This answers the caveat of Parts II–IV that the calibration amplitudes of the two parities are not matched (Part II, Section 14, “without matching parity amplitudes”; Part IV, Section 41, “No equality of the two parity-dependent calibration amplitudes is asserted or used”): both parities have the amplitude C_{cal} . The input is the published product asymptotics; at intake $C_{\text{ASM}}/C_{\text{cal}}^2 = \sqrt{2/3}$ was confirmed to 30 digits and $\log Z_n(c) - \alpha n^2 - bn + \kappa \log n - \log C_{\text{cal}}$ was $-8.9 \cdot 10^{-4}$ and $-8.7 \cdot 10^{-4}$ at $n = 39, 40$, decaying like $1/n$ on both parities.

3.2 The fixed operator, basis and source

Put $h = \sqrt{3}$, $a = \pi/(3h)$, $p(y) = y^2 + 3/4$, and

$$w_y(y) = \frac{p(y)}{2(2 \cosh(2ay) - 1)}, \quad m(y) = \frac{1}{1 + e^{2ay}}, \quad \varphi(u) = \frac{1}{3(1 + 2 \cosh(2au))}. \quad (28)$$

Let ψ_j be the real orthonormal polynomials in $\mathcal{H} = L^2(w_y dy)$ whose leading coefficients are $(-1)^j/\sqrt{H_j}$. The degree- $< n$ projection is P_n . On polynomials put

$$(Tg)(y) = \int_{\mathbb{R}} e^{au} \varphi(u) g(y+u) du, \quad \mathcal{B} = mT.$$

The boundedness of $\mathcal{B}$ is part of the earlier proof. The unweighted transform of $\mathcal{B}$ has kernel

$$\tilde{\mathcal{B}}(y, x) = g_0(y) \varphi(x-y) h_0(x), \quad g_0 = \sqrt{w_y} m e^{-ay}, \quad h_0 = e^{ay} / \sqrt{w_y}, \quad C_0(y, x) = \mathbf{1}_{y < 0} \varphi(x-y) \mathbf{1}_{x < 0}. \quad (29)$$

Here $\mathcal{E} = \tilde{\mathcal{B}} - C_0$ is the operator in (11). The polynomial inverse of T is

$$(\mathcal{K}g)(y) = g(y) + e^{i\pi/3} g(y+ih) + e^{-i\pi/3} g(y-ih).$$

The exact degree-space determinant is $D_n(t)/D_n(3) = \det(I + (t-3)P_n\mathcal{B}P_n)$.

Define $d_j = \sqrt{(j+1)(j+2)}$, $h_j = j!(3)_j$, and

$$H_j = 2(27/16)^j \frac{j!(3)_j(5/3)_j(7/3)_j}{(3/2)_j(5/2)_j}. \quad (30)$$

The affine source $q_t \in \mathcal{H}$ is

$$q_t(y) = \frac{A_t e^{2ay} + (-y - t + 3/2)e^{ay/\sqrt{2}} + C_t + [(1-t)y/2 - t + 3/2]e^{-ay/\sqrt{2}}}{p(y) \cosh(ay)}, \quad (31)$$

where $A_t = \sqrt{3/2}(t+1)/4$ and $C_t = \sqrt{3/2}(t-2)$. Write $q_t = q_3 + (t-3)q^{(1)}$ and

$$\gamma_j(t) = \langle \psi_j, (I + (t-3)P_{j+1}\mathcal{B}^*P_{j+1})^{-1}P_{j+1}q_t \rangle, \quad a_j^{\text{src}} = \langle \psi_j, q_3 \rangle. \quad (32)$$

The inner product is linear in its second argument. The inverse in (32) acts on the finite-dimensional range, never on an unjustifiably substituted infinite section.

Proposition 3.1 (Exact all-parity boundary formula). *For every $n \geq 1$ and $s > 0$, with $t = s^2$,*

$$\frac{sZ_n(s)}{Z_{n+1}(s)} = (-1)^{n-1}d_{n-1}\sqrt{h_{n-1}/H_{n-1}}\gamma_{n-1}(t). \quad (33)$$

It follows that $F_n(s) = \gamma_{n-1}(s^2)/a_{n-1}^{\text{src}}$, and

$$a_j^{\text{src}} \sim \frac{(-1)^j \sqrt{2} 3^{-1/4}}{j+1}. \quad (34)$$

Proof. We supply the borders and source normalization explicitly. Let

$$K_t(u, v) = \frac{v-u}{1-uv} \left(t + \frac{(1+u)(1+v)}{1-u-v} \right), \quad M_{ij} = [u^i v^j] K_t(u, v).$$

The source Pfaffian formula [1, Equation (4.1)] and the earlier adjacent determinant give $H_n = (M_{i,j+1})_{0 \leq i,j < n}$ with $\det H_n = D_n(t)$. Set $b_i = M_{i0}$ and $v_t = te_0 - b$; directly $b_0 = 0$, $b_1 = -(t+1)$ and $b_i = -2$ for $i \geq 2$. For even n , replacing the last column of H_n by b and cycling it to the front gives $-\det(H_n)e_{n-1}^\top H_n^{-1}b = Z_n(s)^2$. The e_0 cofactor vanishes because its complementary skew matrix has odd size. For odd n the full skew determinant vanishes, while the e_0 complementary skew minor indexed $1, \dots, n-1$ has determinant $(Z_n(s)/s)^2$. Including the cofactor signs in either case proves

$$sZ_n(s)/Z_{n+1}(s) = e_{n-1}^\top H_n^{-1}v_t. \quad (35)$$

Use the finite matrices S (lower shift), $L_{ij} = \binom{i}{j}$, $E_{ii} = (-1)^i$, $\Pi_{ij} = \binom{i+j}{i}$, and $T_n(g) = g(S)$. The earlier finite Pascal factorization, or direct multiplication of the displayed generating functions, is

$$H_n = T_n(B_0)LEJ_n(t)EL^\top, \quad J_n(t) = I + T_n(G_t)\Pi,$$

where

$$B_0(z) = \frac{(z-2)(z+1)(2z-1)}{(1-z)(z^2-z+1)}, \quad G_t(z) = \frac{(t-4)z^2 + (4-t)z + t-1}{(2-z)(1-z)^2(1+z)}.$$

The last row of $L^{-\top}$ is $e_{n-1}^\top$. The generating function of v_t is $V_t(z) = t + (t+1)z + 2z^2/(1-z)$, so the binomial transform $(1-z)^{-1}[V_t/B_0](-z/(1-z))$ simplifies to

$$w_t(z) = \frac{(t-z)(z^2 - z + 1)}{(2-z)(1-z)^2(1+z)}.$$

Consequently (35) equals $(-1)^{n-1}e_{n-1}^\top J_n(t)^{-1}w_t^{(n)}$, where the vector contains its first n coefficients.

Put $D = \text{diag}(d_j)$, $A = D^{-1}SD$ and $C = D^{-1}VD$ with $V_{ij} = \binom{i+j+2}{i}$. For $r(z) = 1 - z + z^2$, $q(z) = 2 + z - z^2$, the exact finite identity is $D^{-1}J_n D = I + C + (t-3)[r(A)/q(A)]C$. Multiplying by $q(A)/r(A)$ gives the old row-action Gram matrix $F_n^\phi = \mathcal{K}_n^{\text{row}}(I + C) + (t-3)C$. The forcing is especially simple:

$$(q/r)w_t = (t-z)/(1-z)^2, \quad p_i(t) = [t(i+1) - i]/d_i.$$

Thus (35) becomes $(-1)^{n-1}d_{n-1}e_{n-1}^\top (F_n^\phi)^{-1}p(t)$.

Let $\phi_j = u_j/\sqrt{h_j}$ be the Meixner–Pollaczek basis from Part II, Section 15, with

$$\sum_{j \geq 0} u_j(y) z^j / j! = r(z)^{-3/2} e^{y\eta(z)}, \quad \eta'(z) = -1/r(z), \quad \eta(0) = 0.$$

The lower-triangular basis change $\psi = U_n \phi$ has last diagonal entry $\sqrt{h_{n-1}/H_{n-1}}$. Congruence gives $F_n^\psi = U_n F_n^\phi U_n^\top = \mathcal{K}_n^{\text{row}} + (t-3)M_n$, where $(M_n)_{ij} = \langle \psi_i, m\psi_j \rangle$. The matrix $(\mathcal{K}_n^{\text{row}})^{-1}M_n$ is exactly the matrix of $P_n \mathcal{B}^* P_n$ in column coordinates. If $\ell_t(\phi_i) = p_i(t)$, its transformed source has entries $\ell_t(T\psi_i)$. We now identify their genuine representing vector.

Define the exponentially decaying density

$$\nu_t(y) = \frac{(y+t-3/2)e^{ay} + [(1-t)y + (3-t)/2]e^{-ay}}{2\sqrt{2} \cosh(3ay)}.$$

The integral $\int e^{\eta y} / [2 \cosh(3ay)] dy = (h/2) \sec(h\eta/2)$ and its derivative yield, for $x = h\eta/2$,

$$L_t(\eta) := \int e^{\eta y} \nu_t(y) dy = \frac{3}{\sqrt{2}} \frac{3t \cos x + \sqrt{3}(2-t) \sin x}{[1 + 2 \cos(2x)]^2}, \quad |\Re \eta| < 2a. \quad (36)$$

The inverse change is $z = -2 \sin x / (\sqrt{3} \cos x - \sin x)$ and $r(z) = 3/(\sqrt{3} \cos x - \sin x)^2$. Substitution gives $r(z)^{-3/2} L_t(\eta(z)) = (t-z)/[\sqrt{2}(1-z)^2]$. Since $\sqrt{h_i} = i!d_i/\sqrt{2}$, coefficient comparison proves $\ell_t(g) = \int \nu_t g$ on polynomials.

Set $\rho_t(y) = \int e^{au} \varphi(u) \nu_t(y-u) du$. Exponential tails justify Fubini for each polynomial and show $\ell_t(T\psi_i) = \int \psi_i \rho_t$. Its Laplace transform on $-2a < \Re \eta < a$ is

$$\int e^{\eta y} \rho_t(y) dy = \frac{L_t(\eta)}{1 + 2 \cos(h\eta + \pi/3)}.$$

Applying the same sech integral and derivative at the shifts $2a, a, 0, -a$ shows that its inverse is

$$\rho_t(y) = \frac{A_t e^{2ay} + (-y-t+3/2)e^{ay}/\sqrt{2} + C_t + [(1-t)y/2 - t + 3/2]e^{-ay}/\sqrt{2}}{2 \cosh(3ay)}. \quad (37)$$

Fourier uniqueness identifies this expression with the convolution. The identity

$$w_y = p \cosh(ay) / (2 \cosh(3ay))$$

shows that its quotient by w_y is exactly (31). At positive infinity $\rho_t(y) = A_t e^{-ay} + O((1+y)e^{-2ay})$; at negative infinity $\rho_t(y) = O((1+|y|)e^{2ay})$. As $w_y(y) \sim (y^2/2)e^{-2a|y|}$, one has $|\rho_t|^2/w_y = O(y^{-2})$ on the positive tail and exponential decay on the negative one. Thus $q_t \in \mathcal{H}$. The basis change and finite inverse formula now prove (33). The norm (30) also follows by multiplying the exact recurrence $H_j/H_{j-1} = 3j(j+2)(3j+2)(3j+4)/[4(2j+1)(2j+3)]$, $H_0 = 2$.

Finally set $A_* = 3\sqrt{3}/4 = e^{2\alpha}$. Stirling in (30) gives $\sqrt{h_j/H_j} \sim [8/(9 \cdot 3^{1/4})]A_*^{-j}$. Equation (27) gives $cZ_n(c)/Z_{n+1}(c) \sim [c/(\sqrt{2}A_*)]A_*^{-n}$. Substitution in (33), with $d_{n-1} \sim n$, proves (34), including its alternating sign. $\square$

3.3 Stable inverses and the analytic coefficient reduction

For completeness, the finite nonnormal inverses are uniformly bounded on compact subsets of Ω . In the unweighted representation, $\tilde{\mathcal{B}} = C_0 + \mathcal{E}$, and the earlier proof gives

$$\delta(I + \delta P_n C_0 P_n)^{-1} P_n \mathcal{E} P_n \longrightarrow \delta(I + \delta C_0)^{-1} \mathcal{E}$$

in Hilbert–Schmidt norm, locally uniformly. Nonvanishing of $\mathcal{A}$ makes $I + \delta(I + \delta C_0)^{-1} \mathcal{E}$ invertible. The exact factorization of $I + \delta P_n \tilde{\mathcal{B}} P_n$ then bounds its inverse for all sufficiently large n ; each of the remaining finitely many sizes is controlled by the negative-root property. Taking adjoints gives the corresponding bound for $\mathcal{B}^*$. The resulting strong convergence on fixed vectors still does not control a coefficient after division by $a_{n-1}^{\text{src}} = O(1/n)$.

Every F_n is holomorphic and nowhere zero on $\mathcal{H}_+$, with $F_n(c) = 1$. The earlier interlacing argument gives, for $R_n(s) = \text{Log}[Z_n(s)/Z_n(c)]$,

$$|R_{n+1}(s) - R_n(s)| \leq \frac{\pi|s - c|}{\sqrt{c\Re s}}.$$

Thus F_n and $1/F_n$ are locally uniformly bounded. A locally convergent subsequence and all its derivatives converge by Cauchy’s formula. If every fixed Taylor coefficient at $t = 3$ of $\gamma_{n-1}(t)/a_{n-1}^{\text{src}}$ converges, every subsequential analytic limit has the same Taylor series at $s = c$. Analytic uniqueness and normal-family compactness then prove full-sequence locally uniform convergence on $\mathcal{H}_+$.

For $k \geq 1$ the relevant coefficient is exactly

$$\frac{(-1)^k}{a_{n-1}^{\text{src}}} \left\{ \langle \psi_{n-1}, (P_n \mathcal{B}^* P_n)^k P_n q_3 \rangle - \langle \psi_{n-1}, (P_n \mathcal{B}^* P_n)^{k-1} P_n q^{(1)} \rangle \right\}. \quad (38)$$

This is the finite Neumann derivative at zero; it involves no interchange of an infinite series and a size limit. We prove existence of each fixed-power limit next. The established companion limit will then evaluate the analytic boundary limit without any need to compute those powers separately.

4 Quantitative continuous-Hahn smoothing

The required estimate is a column rate, stronger than compactness. Let $f_j = \sqrt{w_y} \psi_j \in L^2(\mathbb{R})$ and $b_y = \sqrt{p(y)}$. Let v denote the mixed-gamma phase in the exact Jacobi transform: in the scaled variable $z = y/(3\sqrt{3})$,

$$G(z) = \Gamma(5/6 + iz)\Gamma(7/6 - iz), \quad v = G/|G|.$$

Theorem 4.1. *For the kernels in (29),*

$$\sqrt{j}\|\mathcal{E}f_j\|_2 \longrightarrow 0, \quad \sqrt{j}\|(M_v C_0 M_v^* - C_0)f_j\|_2 \longrightarrow 0. \quad (39)$$

The second assertion also holds with input $M_v f_j$ and with the conjugate phase difference $C_0 - M_v^ C_0 M_v$. All statements hold on the full degree sequence, irrespective of the consistent real leading-sign convention for ψ_j .*

4.1 An elementary fixed-variable asymptotic, including zero

We first derive a local asymptotic rather than invoke a large-variable formula excluding the origin. For positive A, B put $s = A + B$ and $W(z) = |\Gamma(A + iz)\Gamma(B + iz)|^2$. The standard continuous-Hahn polynomial $p_j(z; A, B, A, B)$ has positive leading coefficient and squared norm in $W(z) dz$

$$h_j^{A,B} = \frac{2\pi\Gamma(j+2A)\Gamma(j+s)^2\Gamma(j+2B)}{(2j+2s-1)\Gamma(j+2s-1)j!}. \quad (40)$$

The removable expression at $j = 0, s = 1/2$ is interpreted by $(2s-1)\Gamma(2s-1) = \Gamma(2s)$. Neither parameter pair below requires that convention. The classical generating function [8, DLMF 18.23.6] is

$${}_1F_1(A + iz; 2A; -it) {}_1F_1(B - iz; 2B; it) = \sum_{j \geq 0} \frac{p_j(z)t^j}{(2A)_j(2B)_j}.$$

Euler's beta integral for ordinary ${}_1F_1$ (DLMF 13.4.1 together with the normalization conversion 13.2.4) gives exactly

$$\begin{aligned} p_j(z) &= \frac{i^j \Gamma(j+2A)\Gamma(j+2B)}{j! W(z)} I_j(z), \\ I_j(z) &= \int_0^1 \int_0^1 (v-u)^j u^{A+iz-1} (1-u)^{A-iz-1} v^{B-iz-1} (1-v)^{B+iz-1} du dv. \end{aligned} \quad (41)$$

Absolute convergence and boundedness of the entire exponential on the integration square justify coefficient extraction. The norm and polynomial conventions agree with [11, Equations (1.1)–(1.4)].

Uniformly for real z in a compact set, the corners $(u, v) = (0, 1)$ and $(1, 0)$ yield

$$I_j(z) = \Gamma(A + iz)\Gamma(B + iz)j^{-s-2iz} + (-1)^j \Gamma(A - iz)\Gamma(B - iz)j^{-s+2iz} + O_K(j^{-s-1}). \quad (42)$$

To verify the error, at the first corner set $q = 1 - v$. The smooth remaining factor equals $1 + O_K(u + q)$. On the full triangle $u, q \geq 0, u + q \leq 1$ the leading Dirichlet integral is

$$\int u^{A+iz-1} q^{B+iz-1} (1-u-q)^j du dq = \frac{\Gamma(A + iz)\Gamma(B + iz)\Gamma(j+1)}{\Gamma(j+s+2iz+1)}.$$

The $O(u + q)$ term has absolute integral $O_K(j^{-s-1})$. Removing the portion outside a fixed corner neighborhood costs an exponentially small error. The second corner is the conjugate version with $(-1)^j$. Away from these corners $|v - u| \leq 1 - \varepsilon$; the beta powers have integrable absolute values independent of real z . This proves (42), uniformly through $z = 0$.

Let $F_j = \sqrt{W} p_j / \sqrt{h_j^{A,B}}$ and $\eta(z) = \Gamma(A + iz)\Gamma(B + iz) / |\Gamma(A + iz)\Gamma(B + iz)|$. The gamma-ratio estimate in (40) gives

$$F_j(z) = \frac{1}{\sqrt{\pi j}} \left\{ i^j \eta(z) j^{-2iz} + (-i)^j \overline{\eta(z)} j^{2iz} \right\} + O_K(j^{-3/2}). \quad (43)$$

This additive uniform error remains valid at zeros and for odd polynomials at zero. For compactly supported integrable H , squaring and using Riemann–Lebesgue gives

$$\int H(z) j F_j(z)^2 dz \longrightarrow \frac{2}{\pi} \int H(z) dz. \quad (44)$$

For compactly supported L^2 tests, (43) similarly implies $\int H \sqrt{j} F_j \rightarrow 0$. The oscillatory frequencies are $2 \log j$ and $4 \log j$, respectively.

4.2 A positive exact norm identity prevents escape of mass

For the desired pair $(A, B) = (5/6, 7/6)$ put $d_0 = 1/6$ and $V(z) = |\Gamma(1/6 + iz)\Gamma(5/6 + iz)|^2$. Gamma recurrence gives $W(z) = (z^2 + d_0^2)V(z)$ exactly. Let P_j^V, Q_j^W be the monic orthogonal polynomials for V, W , with squared norms H_j^V, H_j^W . Both have their degree parity. At id_0 , the terminating hypergeometric formula has upper parameter zero, so

$$P_j^V(id_0) = \frac{i^j (1/3)_j}{(2j)}, \quad r_j = -\frac{P_{j+2}^V(id_0)}{P_j^V(id_0)} = \frac{(j+1/3)(j+4/3)(j+1)(j+2)}{4(2j+1)(2j+3)} > 0.$$

The standard leading coefficient in this parameter pair is $\binom{2j}{j}$. Direct divisibility and orthogonality prove the quadratic Christoffel relations

$$Q_j^W(z) = \frac{P_{j+2}^V(z) + r_j P_j^V(z)}{z^2 + d_0^2}, \quad H_j^W = r_j H_j^V. \quad (45)$$

Indeed parity gives the two zeros in the numerator; multiplying by $z^2 + d_0^2$ proves orthogonality against all lower degrees, and the norm identity follows by pairing with the numerator.

Put $a_j^V = |P_j^V(id_0)|^2 / H_j^V$. For $k \leq j$ of the same parity, $P_k^V - [P_k^V(id_0)/P_j^V(id_0)]P_j^V$ is divisible by $z^2 + d_0^2$ with quotient of degree less than j . Orthogonality and monicity therefore give

$$\langle Q_j^W, P_k^V \rangle_V = H_j^V P_k^V(id_0) / P_j^V(id_0).$$

Opposite-parity pairings vanish. Expand Q_j^W in the V basis and use (45). For the same normalized function F_j as above this gives the positive exact identity

$$\int \frac{F_j(z)^2}{z^2 + 1/36} dz = \frac{1}{r_j a_j^V} \sum_{\substack{0 \leq k \leq j \\ k \equiv j \pmod{2}}} a_k^V. \quad (46)$$

The lowered norm formula (40) is $h_j^{1/6, 5/6} = 2\pi\Gamma(j+1/3)\Gamma(j+5/3)/(2j+1)$, so

$$a_j^V = \frac{(2j+1)\Gamma(j+1/3)}{2\pi\Gamma(1/3)^2\Gamma(j+5/3)} \sim \frac{j^{-1/3}}{\pi\Gamma(1/3)^2}.$$

For either parity, elementary power-sum comparison yields

$$\sum_{\substack{k \leq j \\ k \equiv j \pmod{2}}} a_k^V \sim \frac{3j^{2/3}}{4\pi\Gamma(1/3)^2}, \quad r_j \sim j^2/16.$$

Thus (46) proves

$$j \int \frac{F_j(z)^2}{z^2 + 1/36} dz \longrightarrow 12. \quad (47)$$

This uses a positive sum and no cancellation of asymptotic terms.

Let $k_0 = 3\sqrt{3}$, $y = k_0 z$. Up to the real leading sign, $f_j(y) = k_0^{-1/2} F_j(y/k_0)$ and $p(y) = k_0^2(z^2 + 1/36)$. Therefore

$$\|\sqrt{j} f_j/b_y\|_2^2 \longrightarrow 4/9. \quad (48)$$

On each bounded interval (44) gives the limiting weighted mass $(2/(\pi k_0)) \int dy/p(y)$, whose total over the real line is also 4/9. Subtracting the compact-interval limit from (48) and then enlarging the interval proves

$$\lim_{R \rightarrow \infty} \limsup_{j \rightarrow \infty} j \int_{|y| > R} \frac{|f_j(y)|^2}{p(y)} dy = 0. \quad (49)$$

The sequence $u_j = \sqrt{j} f_j/b_y$ is bounded by (48) and weakly null by the compactly supported test conclusion of (43), followed by density. Multiplication by a fixed bounded unimodular function preserves weak nullity and tightness.

4.3 A local compactness principle and the original kernel

If A is bounded and $A\mathbf{1}_{[-R,R]}$ is compact for every R , then a bounded, weakly null, L^2 -tight sequence u_j satisfies $\|Au_j\| \rightarrow 0$. Indeed the compact part tends to zero for fixed R , and the other part is at most $\|A\| \|\mathbf{1}_{|y| > R} u_j\|$. Taking limsup and then $R \rightarrow \infty$ proves the assertion.

Split off the rank-one positive-tail kernel

$$R_0(y, x) = \frac{1}{3} G_0(y) r_0(x), \quad G_0(y) = g_0(y) e^{2ay}, \quad r_0(x) = h_0(x) e^{-2ax} \mathbf{1}_{x \geq 0}.$$

Both factors lie in L^2 . We claim that $A_0 = (\mathcal{E} - R_0)M_{b_y}$, with the multiplier absorbed in the input kernel, is bounded and locally compact. For $x, y < 0$ set $r_x = e^{2ax}$ and $d_x = \sqrt{1 - r_x + r_x^2}$. The exact formulas in (29) are $g_0(y) = b_y/[\sqrt{2}(1 + r_y)d_y]$ and $h_0(x) = \sqrt{2}d_x/b_x$. The negative-quadrant kernel of A_0 is

$$\begin{aligned} & \mathbf{1}_{x,y < 0} (b_y - b_x) \varphi(x - y) \\ & + \mathbf{1}_{x,y < 0} b_y \varphi(x - y) \left[\frac{d_x}{(1 + r_y)d_y} - 1 \right]. \end{aligned}$$

The first is bounded by $|x - y| \varphi(x - y)$ since b_y is 1-Lipschitz, so Schur's test applies. Restricting its input to a bounded interval makes it square integrable. The second bracket is $O(e^{2ax} + e^{2ay})$; exponential off-diagonal decay and these half-line tails make that whole kernel square integrable.

For $x \geq 0$ the kernel is

$$g_0(y) \left[\varphi(x - y) - \frac{1}{3} e^{-2a(x-y)} \right] h_0(x) b_x.$$

Here $h_0(x) b_x = O(e^{2ax})$. The bracket is $O(e^{-4a(x-y)})$ for $x \geq y$ and $O(e^{2a(y-x)})$ for $x < y$. Use $g_0(y) = O(1 + |y|)$ at $-\infty$ and $O((1 + y)e^{-4ay})$ at $+\infty$. The resulting bounds on the three regions $y < 0 \leq x$, $0 \leq y \leq x$, and $0 \leq x < y$ are respectively

$$C(1 + |y|) e^{4ay} e^{-2ax}, \quad C(1 + y) e^{-2ax}, \quad C(1 + y) e^{-2ay}.$$

Each is square integrable, including the triangular regions. For $x < 0 \leq y$, the reference and rank-one kernels vanish; the remaining bound $C(1+y)e^{-6ay}e^{2ax}$ is square integrable. This proves the claim. The local compactness principle and (49) now show

$$\sqrt{j}\|(\mathcal{E} - R_0)f_j\|_2 = \|A_0u_j\|_2 \longrightarrow 0. \quad (50)$$

The exceptional rank-one term uses the calibrated coefficient, not a noncalibrated asymptotic. At $t = 3$, (31) simplifies to

$$q_3(y) = \frac{\sqrt{2}[\sqrt{3}e^{ay} - y - 3/2]}{p(y)}. \quad (51)$$

Put $F = \sqrt{w_y}q_3$. Then

$$b_y[r_0 - \sqrt{2/3}F] \in L^2(\mathbb{R}). \quad (52)$$

For $y \geq 0$, set $\rho = e^{-2ay}$, $d = \sqrt{1 - \rho + \rho^2}$. Exactly $r_0 = \sqrt{2}d/b_y$ and $F = [\sqrt{3} - (y+3/2)e^{-ay}]/(b_yd)$, so the left expression in (52) is $\sqrt{2}(d - 1/d) + \sqrt{2/3}(y+3/2)e^{-ay}/d$, which decays exponentially. On the negative tail $r_0 = 0$ and b_yF decays exponentially up to a polynomial; all factors are bounded near zero. Writing the function in (52) as h_0^{sub} , weak nullity and (34) give

$$\sqrt{j}\langle r_0, f_j \rangle = \sqrt{2/3}\sqrt{j}\langle F, f_j \rangle + \langle h_0^{\text{sub}}, u_j \rangle \longrightarrow 0.$$

The rank-one norm therefore tends to zero at the required scale. Together with (50) this proves the first assertion of (39).

4.4 The phase correction has a compact weighted kernel

The exact gamma phase in Part II, Section 17, satisfies, on $x, y < 0$,

$$|v(y)/v(x) - 1| \leq \frac{C|x - y|}{(1 + |x|)(1 + |y|)}.$$

Consequently for $D_v = M_v C_0 M_v^* - C_0$,

$$|D_v(y, x)b_x| \leq \frac{C\mathbf{1}_{x, y < 0}|x - y|\varphi(x - y)}{1 + |y|}.$$

The right side is square integrable, by integrating in the difference variable and then in y . Thus $D_v M_{b_y}$ is Hilbert–Schmidt. Compactness applied to $u_j \rightharpoonup 0$ proves $\sqrt{j}D_v f_j \rightarrow 0$. Apply the identical argument to $M_v u_j$ and to the conjugate phase difference to obtain all the stated variants. This completes the proof of Theorem 4.1.

5 A quantitative Jacobi–Hardy column comparison

Continue with the angular notation (9). Put $\gamma = 2/3$, $N_j = j + 3/2$, $c_0 = 11\pi/12$, and

$$v_j^\pm(\theta) = e^{\pm i(N_j\theta - c_0)}/\sqrt{2\pi}, \quad v_j = v_j^- + v_j^+, \quad Ue_j = u_j, \quad V_\pm e_j = v_j^\pm, \quad V = V_- + V_+.$$

These synthesis maps are bounded (the wave maps are restricted circle Fourier synthesis). Set $X = M_{s_0} H U$, $X_0 = M_{s_0} V_-$, $A = X^* X$, $T_0 = X_0^* X_0$.

Theorem 5.1. *One has $\sqrt{j}\|(A - T_0)e_j\|_2 \rightarrow 0$. After restoring the exact basis phase i^j , the Toeplitz model has real symmetric entries b_{i-k} where*

$$b_r = \frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} w(x + \pi/2) e^{irx} dx, \quad |b_r| \leq C(1 + |r|)^{-5/3}. \quad (53)$$

For the original ψ -coordinate matrix of $\mathcal{B}$, denoted B , and this Toeplitz matrix T_b , one therefore has

$$\|(B - T_b)e_j\| = o((j+1)^{-1/2}). \quad (54)$$

5.1 An endpoint Fourier lemma

Suppose $k(\theta, \phi)$ is locally absolutely continuous in ϕ , with envelope $a_0(\theta)$ such that

$$|k| \leq Ca_0(\theta)d(\phi)^{-1/2}, \quad |\partial_\phi k| \leq Ca_0(\theta)d(\phi)^{-3/2}.$$

Suppose that for each fixed interior θ , the two scaled expressions $d(\phi)^{1/2}k$ and $d(\phi)^{3/2}\partial_\phi k$ tend to zero at both endpoints. Then

$$\sqrt{N} \int_0^\pi k(\theta, \phi) e^{\pm iN\phi} d\phi \rightarrow 0 \quad (55)$$

pointwise, bounded in absolute value by $Ca_0(\theta)$. Split at $1/N$ and $\pi - 1/N$. The endpoint pieces are $O(a_0/\sqrt{N})$; one integration by parts on the middle gives the same order from its boundary and derivative terms. For little-o, first choose fixed small endpoint neighborhoods where the scaled bounds are arbitrarily small for this θ , and integrate by parts on the remaining compact interval, giving $O(1/N)$ there. This proves (55) for real $N \rightarrow \infty$. Dominated convergence gives its $L^2(q(\theta)d\theta)$ version whenever $\int qa_0^2 < \infty$.

5.2 Two explicit Hardy kernels

Let $f(\theta) = \log \tan(\theta/2)$, so $f' = 1/\sin \theta$. The difference between H and the compressed ordinary negative-frequency Hardy operator C_I has, apart from a fixed factor $-i/(2\pi)$, the kernel

$$d_H(\theta, \phi) = \frac{\sqrt{f'(\theta)f'(\phi)}}{f(\theta) - f(\phi)} - \frac{1}{\theta - \phi}.$$

The canceled diagonal is filled continuously. The explicit bounds are

$$|d_H| \leq \frac{C}{\sqrt{d(\theta)d(\phi)}}, \quad |\partial_\phi d_H| \leq \frac{C}{\sqrt{d(\theta)d(\phi)^{3/2}}}. \quad (56)$$

For completeness, in comparable left-endpoint scales set $\phi = \theta r$, $1/2 \leq r \leq 2$. After extracting the scale, the canceled quotient is smooth through $r = 1$, and its required derivatives are uniformly bounded; hence the bounds are C/θ and C/θ^2 . In noncomparable scales use $|f(\theta) - f(\phi)| \geq c|\log(\theta/\phi)|$, whose logarithm is bounded away from zero. The ordinary Cauchy terms obey (56) separately there. At opposite endpoints the denominator is bounded below by $c[1 + |\log d(\theta)| + |\log d(\phi)|]$; mixed compact/endpoint regions obey the same estimates. Direct differentiation uses

$$\partial_\phi \frac{\sqrt{f'(\theta)f'(\phi)}}{f(\theta) - f(\phi)} = \frac{\sqrt{f'(\theta)f'(\phi)}}{f(\theta) - f(\phi)} \left[-\frac{\cot \phi}{2} + \frac{\csc \phi}{f(\theta) - f(\phi)} \right].$$

For fixed interior θ , the logarithmic denominator gives the two endpoint little-o conditions. Since $\int w/d$ and $\int w^2/d$ are finite, the Fourier lemma proves

$$\sqrt{j}\|s_0(H - C_I)v_j^\pm\|_2 \rightarrow 0, \quad \sqrt{j}\|w(H - C_I)v_j^\pm\|_2 \rightarrow 0. \quad (57)$$

The commutator $[H, w]$ has, up to the same fixed factor, kernel

$$k_w(\theta, \phi) = (w(\phi) - w(\theta)) \frac{\sqrt{f'(\theta)f'(\phi)}}{f(\theta) - f(\phi)}.$$

It obeys the Fourier lemma with square-integrable envelope

$$a_0(\theta) = C \left\{ d(\theta)^{\gamma-1/2} + \frac{1}{\sqrt{d(\theta)}(1 + |\log d(\theta)|)} \right\}. \quad (58)$$

Here $|w^{(r)}| \leq C_r d^{\gamma-r}$ for $r = 0, 1, 2$. At comparable endpoint scales the canceled quotient is $O(\theta^{\gamma-1})$ and its derivative is $O(\theta^{\gamma-2})$. For $\phi \geq 2\theta$ near the same endpoint, use

$$\sup_{2\theta \leq \phi \leq \varepsilon} \frac{\phi^\gamma + \theta^\gamma}{\log(\phi/\theta)} \leq C \left\{ \theta^\gamma + (1 + |\log \theta|)^{-1} \right\}.$$

This follows by putting $v = \log(\phi/\theta)$; the term $\theta^\gamma e^{\gamma v}/v$ has no interior maximum. The case $\phi \leq \theta/2$ follows directly from $|w(\phi) - w(\theta)| \leq C\theta^\gamma$. Opposite endpoints use the logarithmic sum denominator from (56); mixed regions give the second summand in (58). Differentiation adds $d(\phi)^{-1}$, using the displayed derivative formula and the derivative bounds for w . For fixed θ the scaled expressions still tend to zero, so

$$\sqrt{j}\|[H, w]v_j^-\|_2 \rightarrow 0. \quad (59)$$

The borderline envelope square $d^{-1}/(1 + |\log d|)^2$ is integrable.

5.3 The direct wave defect

The normalized Jacobi expansion proved from the uniform Bessel formula in Part II, Section 18, implies, for $R_j = u_j - v_j$,

$$|R_j(\theta)| \leq C \min\{1, [(j+1)d(\theta)]^{-1}\}, \quad \|R_j\|_2 = O(j^{-1/2}), \quad \|s_0 R_j\|_2 = O(j^{-(1+\gamma)/2}). \quad (60)$$

On every compact interior interval $\sqrt{j}R_j$ tends uniformly to zero; its global L^2 norm is bounded. It is therefore weakly null. The earlier proof shows $[s_0, H]$ compact, so

$$\sqrt{j}\|s_0 H R_j\|_2 \rightarrow 0$$

by $s_0 H R_j = H(s_0 R_j) + [s_0, H]R_j$. This uses compactness on a bounded weakly null sequence with the scale already included. The ordinary compressed Hardy tail estimate is

$$|C_I v_j - v_j^-| \leq C\{L(N_j \theta) + L(N_j(\pi - \theta))\}, \quad L(r) = \begin{cases} 1 + |\log r|, & r \leq 1, \\ 1/r, & r \geq 1. \end{cases}$$

Integration against $w = O(d^\gamma)$ makes its weighted norm $O(j^{-(1+\gamma)/2})$. Together with (57) and (60), this proves

$$\sqrt{j}\|(X - X_0)e_j\|_2 \rightarrow 0. \quad (61)$$

5.4 The adjoint defect must be controlled separately

We prove

$$\sqrt{j} \|(U - V)^*(wv_j^-)\|_{\ell^2} \rightarrow 0. \quad (62)$$

First, uniformly in k ,

$$\text{Var}(wR_k) \leq C, \quad (wR_k)(0) = (wR_k)(\pi) = 0. \quad (63)$$

Here no uncontrolled remainder is differentiated. The exact angular Jacobi derivative identity for parameters α_J, β_J is

$$u'_k = q_J(\theta)u_k - \sqrt{k(k + \alpha_J + \beta_J + 1)} u_{k-1}^{(\alpha_J+1, \beta_J+1)},$$

where $q_J = [(2\alpha_J + 1)\cot(\theta/2) - (2\beta_J + 1)\tan(\theta/2)]/4$. Apply the separate uniform first-order expansions to the two families. The shifted family has the same N_k and phase $\Phi_k - \pi/2$; the leading derivative terms cancel, while $\sqrt{k(k+3)} - N_k = O(1/N_k)$. Consequently

$$|R'_k| \leq C[d^{-1} + (N_k d^2)^{-1}] \quad (N_k d \geq 1), \quad |R'_k| \leq CN_k \quad (N_k d < 1).$$

The latter uses the exact identity and endpoint powers $11/6, 7/6$, both greater than one. Integrating $w'R_k + wR'_k$ with (60) proves (63).

The first Jacobi correction is

$$A_J(\theta) = \frac{(1/4 - \alpha_J^2)\cot(\theta/2) - (1/4 - \beta_J^2)\tan(\theta/2)}{4}, \quad (\alpha_J, \beta_J) = (4/3, 2/3), \quad \Phi_k = N_k\theta - c_0.$$

The normalized expansion, with no $1/k$ amplitude term, gives

$$\|w[kR_k - \sqrt{2/\pi}A_J \sin \Phi_k]\|_1 \rightarrow 0. \quad (64)$$

Indeed $wA_J \in L^1$. On $d < 1/k$ both terms have integral $O(k^{-\gamma})$; on the complement the uniform $O((N_k d)^{-2})$ remainder, multiplied by $k w$, has the same bound. Replacing k/N_k by one adds $O(k^{-1})$.

Set $\beta_{kj} = \langle R_k, wv_j^- \rangle$. Integration by parts in (63) gives $|\beta_{kj}| \leq C/(j+1)$ uniformly in k . By (64), $k\beta_{kj}$ is, uniformly in j , a fixed linear combination of Fourier coefficients of wA_J at frequencies $k-j$ and $-(k+j+3)$, plus an error tending to zero as $k \rightarrow \infty$. Fix $\varepsilon > 0$ and M . The contribution to $j \sum_k |\beta_{kj}|^2$ from $k < \varepsilon j$ is at most $C\varepsilon$. The window $|k-j| \leq M$ contributes $O(M/j)$. On $k \geq \varepsilon j$ outside that window, Riemann–Lebesgue and the uniform error in (64), together with $\sum_{k \geq \varepsilon j} k^{-2} = O((\varepsilon j)^{-1})$, give a limsup bounded by $C\varepsilon^{-1} \sup_{|r| > M} |\widehat{wA_J}(r)|^2$. Take $j \rightarrow \infty$, then $M \rightarrow \infty$, then $\varepsilon \downarrow 0$. This proves (62).

Similarly w vanishes at both endpoints and $w' \in L^1$. Integration by parts and Riemann–Lebesgue give Fourier coefficients $o(1/|r|)$ uniformly in the tail. The entries of $V_+^*(wv_j^-)$ have frequencies $k+j+3$, so summing their squares gives

$$\sqrt{j} \|V_+^*(wv_j^-)\|_{\ell^2} \rightarrow 0. \quad (65)$$

The exact decomposition

$$(H - I)wv_j^- = [H, w]v_j^- + w(H - C_I)v_j^- + w(C_I - I)v_j^-$$

and (57)–(59) yield

$$\sqrt{j}\|(H - I)(wv_j^-)\|_2 \rightarrow 0. \quad (66)$$

The last ordinary Hardy term is $O(1/j)$ in L^2 because $2\gamma > 1$.

Now put $D = X - X_0$. Exactly

$$\begin{aligned} A - T_0 &= X^*D + D^*X_0, \\ D^*X_0e_j &= U^*(H - I)(wv_j^-) + (U - V)^*(wv_j^-) + V_+^*(wv_j^-). \end{aligned}$$

Equations (61), (62), (65), and (66) prove the first claim of Theorem 5.1.

5.5 Returning to the original operator and restoring phase

Let S be unitary scaling to $z = y/(3\sqrt{3})$, $\mathcal{F}$ the unitary Fourier transform with kernel $e^{-iz\xi}$, and J the Fourier-to-angular coordinate unitary. With the scaled interpretation of v , the exact transform $L = J\mathcal{F}SM_v$ satisfies

$$Lf_j = i^j u_j, \quad J\mathcal{F}SC_0S^*\mathcal{F}^*J^* = HM_wH.$$

The latter identity has no gamma phase. Therefore

$$L\tilde{\mathcal{B}}f_j - (HM_wH)Lf_j = L\mathcal{E}f_j + J\mathcal{F}S(M_vC_0 - C_0M_v)f_j.$$

The second term is the unitary image of $(M_vC_0M_v^* - C_0)M_vf_j$. The two precise smoothing variants in Theorem 4.1 prove an $o(j^{-1/2})$ error. Let $Q = \text{diag}(i^j)$. The original coordinate matrix is $Q^*U^*L\tilde{\mathcal{B}}L^*UQ$, and the Hardy model is Q^*AQ . Column norms are unchanged by the diagonal phases. The raw wave Gram entry is $(2\pi)^{-1} \int_0^\pi w(\theta)e^{i(i-k)\theta} d\theta$. Conjugating by Q multiplies it by i^{k-i} ; setting $x = \theta - \pi/2$ proves exactly (53) and (54).

The corresponding circle symbol is $w(x + \pi/2)$ on $|x| \leq \pi/2$ and zero elsewhere. It is endpoint $C^{2/3}$, with first and second derivative bounds $d^{-1/3}, d^{-4/3}$. Integrate once by parts, split at distance $1/|r|$ from each support endpoint, and integrate the interior part once more by parts. This proves $b_r = O(|r|^{-5/3})$. The symmetry $w(\pi - \theta) = w(\theta)$ makes b_r real and even. The absolutely convergent Fourier series vanishes at $x = \pi$, so

$$\sum_{r \geq 1} (-1)^r b_r = -b_0/2 = \sqrt{3}/4 - 1/2, \quad b_0 = 1 - \sqrt{3}/2. \quad (67)$$

The last value is the exact scalar mean already evaluated in Part II, Section 18. All parts of Theorem 5.1 are now proved.

6 Fixed-source closure and the finite-section edge theorem

6.1 An invariant density class

Choose R with $2h < R < 5h/2$. Let $\mathcal{C}$ consist of functions g whose density $\rho_g = w_y g$ is holomorphic on $|\Im y| < R$ and, on every closed narrower strip, satisfies for some integer $M \geq 0$

$$\begin{aligned} \rho_g(y) &= c_g e^{-ay} + O((1 + \Re y)^M e^{-2a\Re y}) & (\Re y \rightarrow +\infty), \\ \rho_g(y) &= O((1 + |\Re y|)^M e^{2a\Re y}) & (\Re y \rightarrow -\infty). \end{aligned} \quad (68)$$

Such functions lie in $\mathcal{H}$ by the same tail calculation used for (31). Cauchy's formula supplies the corresponding fixed-derivative estimates on narrower strips.

Both q_3 and $q^{(1)}$ belong to this class. In (37) the affine numerator vanishes identically in t at $y = \pm ih/2, \pm 3ih/2$, as direct substitution shows. These remove the first four zeros of $2 \cosh(3ay)$; the next possible poles are $\pm 5ih/2$. The tails give $c_{q_3} = \sqrt{3/2}$ and $c_{q^{(1)}} = c_{q_3}/4$.

The exact adjoint density formula is

$$\rho_{B^*g}(y) = e^{ay} \int_{\mathbb{R}} \varphi(y-x) e^{-ax} m(x) \rho_g(x) dx. \quad (69)$$

It follows directly from the unweighted kernel, first on the real axis by absolute convergence. Write $F(x) = e^{-ax} m(x) \rho_g(x)$. It is holomorphic on $|\Im x| < 3h/2$, the nearest poles of m , and its horizontal-strip tails are

$$F(x) = O(\text{poly}(x) e^{-4a\Re x}) \quad (+\infty), \quad F(x) = O(\text{poly}(|x|) e^{a\Re x}) \quad (-\infty).$$

The kernel is holomorphic for $|\Im u| < h$ and has tails $O(e^{-2a|\Re u|})$ on narrower strips. To extend (69) to $|\Im y| < 5h/2$, choose a contour height v with $|v| < 3h/2$ and $|\Im y - v| < h$. Overlaps agree by contour shifting; the vertical sides vanish by the displayed tails. A finite cover gives uniform estimates on each narrower closed strip.

The positive leading coefficient is exactly

$$c_{B^*g} = \frac{1}{3} \int_{\mathbb{R}} e^{ax} m(x) \rho_g(x) dx. \quad (70)$$

To see the error rigorously, subtract $e^{-2a(y-x)}/3$ from the kernel over the entire chosen contour. For $\Re x \leq \Re y$ its remainder is $O(e^{-4a(\Re y - \Re x)})$; for $\Re x > \Re y$ bound the kernel and subtracted exponential separately. Combining with the tails of F gives a polynomial times $e^{-4a\Re y}$ before multiplication by e^{ay} . The borderline integration over $0 < \Re x < \Re y$ only adds one polynomial power. The final remainder is $O(\text{poly}(y) e^{-3a\Re y})$, stronger than needed. The moment integrand in (70) is analytic in the contour strip and decays at both ends, so its value is contour independent. On the negative tail split at $\Re x = \Re y$ and $\Re x = 0$. The convolution is $O(\text{poly}(|y|) e^{a\Re y})$ before multiplication, giving the required $e^{2a\Re y}$ bound. Thus

$$B^*C \subset C. \quad (71)$$

Every member has the same leading coefficient shape:

$$\langle \psi_j, g \rangle = \frac{c_g}{c_{q_3}} a_j^{\text{src}} + o(1/(j+1)). \quad (72)$$

Here is a proof of the little-o claim, including the endpoint domain. Subtract $(c_g/c_{q_3})q_3$ and call the result d_g . Its density has two-sided $\text{poly}(y) e^{-2a|\Re y|}$ tails in the strip. In $z = y/(3h)$ put $G^\sharp(z) = \Gamma(5/6 - iz)\Gamma(7/6 + iz)$. The exactly normalized Fourier input in the Jacobi transform is $\sqrt{c_W} G(z) d_g(3hz)$, $c_W = 81\sqrt{3}/(8\pi^2)$. Since $w_y(3hz)$ is a constant times $G(z)G^\sharp(z)$, this input is a fixed constant times $\rho_{d_g}(3hz)/G^\sharp(z)$. The reciprocal gamma product is entire. Stirling on horizontal strips makes this quotient, and all its fixed polynomial multiples, exponentially decaying. Its holomorphy strip contains $\Im z = \pm 2/3$, because $R > 2h$. Contour shifting for its Fourier transform H_g yields

$$|H_g(\xi)| + |H'_g(\xi)| \leq C e^{-2|\xi|/3}.$$

The angular image $g_\theta(\theta) = \sqrt{2/\sin\theta} H_g(\xi(\theta))$ therefore satisfies $g_\theta = O(d^{5/6})$ and $g'_\theta = O(d^{-1/6})$. It belongs to $H_0^1(0, \pi)$ and $\int |g_\theta|^2/d^2 < \infty$. Hence it is in the quadratic-form domain of the Friedrichs Jacobi angular operator, with potential

$$\frac{\alpha_J^2 - 1/4}{4\sin^2(\theta/2)} + \frac{\beta_J^2 - 1/4}{4\cos^2(\theta/2)}, \quad (\alpha_J, \beta_J) = (4/3, 2/3).$$

This is the realization with eigenfunctions u_j and eigenvalues $(j + 3/2)^2$, also at the limit-circle endpoint. Its form expansion gives $\sum_j (j + 3/2)^2 |\langle \psi_j, d_g \rangle|^2 < \infty$; the unitary changes coefficients only by phases i^j . Thus $j \langle \psi_j, d_g \rangle \rightarrow 0$, proving (72). In particular these coefficients are $O((j + 1)^{-1})$ and their projection-tail norms are $O(n^{-1/2})$. By (71), all fixed iterates of both q_3 and $q^{(1)}$ have this shape. The constants in (70) retain the full operator, including its compact and rank-one pieces.

6.2 An abstract edge theorem with nonlocal tails retained

Theorem 6.1. *Let B be bounded on $\ell^2(\mathbb{N}_0)$ and $T_b = T_b^*$ a Toeplitz operator with entries b_{i-k} , where for some $\varepsilon > 0$*

$$|b_r| \leq C(1 + |r|)^{-1-\varepsilon}, \quad \|(B - T_b)e_i\| = o((i + 1)^{-1/2}).$$

Let q be fixed and $u_\ell = (B^)^\ell q$. Suppose for each fixed $\ell \geq 0$ that*

$$(u_\ell)_i = \kappa_\ell a_i + o((i + 1)^{-1}), \quad a_i \sim c_* (-1)^i / (i + 1), \quad c_* \neq 0.$$

Then for every fixed $\ell \geq 0$ the full-sequence limit

$$\frac{[(P_n B^* P_n)^\ell P_n q]_{n-1}}{a_{n-1}}$$

exists. Every fixed right-edge profile coordinate has a limit as well.

Proof. Put $Q_n = I - P_n$, $A_n = P_n B^* P_n$, and $e_{n,\ell} = A_n^\ell P_n q - P_n u_\ell$. Exactly

$$e_{n,0} = 0, \quad e_{n,\ell+1} = P_n B^* e_{n,\ell} - P_n B^* Q_n u_\ell. \quad (73)$$

The coefficient hypothesis gives $\|Q_n u_\ell\| = O(n^{-1/2})$, and boundedness of B then gives $\|e_{n,\ell}\| = O(n^{-1/2})$ for each fixed ℓ . Set $K = B - T_b$. If $\|v_n\| = O(n^{-1/2})$, the column hypothesis implies, uniformly for $i \geq \eta n$, $\eta > 0$ fixed,

$$|(K^* v_n)_i| \leq \|K e_i\| \|v_n\| = o(1/n). \quad (74)$$

Uniformity follows because $\sup_{i \geq \eta n} \sqrt{i+1} \|K e_i\| \rightarrow 0$. This uses columns of K to control coordinates of K^* , without assuming an adjoint-column estimate.

Induction in (73) gives the stronger pointwise bound

$$\sup_{\eta n \leq i < n} n |(e_{n,\ell})_i| \leq C_{\ell,\eta}. \quad (75)$$

For its Toeplitz part split the input at $\eta n/2$. On the upper part use the preceding inductive bound and $\sum |b_r| < \infty$. On the lower part the distance is at least $\eta n/2$; Cauchy-Schwarz bounds the Toeplitz row tail by $O(n^{-1/2-\varepsilon})$, which times $\|e_{n,\ell}\| = O(n^{-1/2})$ is $O(n^{-1-\varepsilon})$. The source tail $Q_n u_\ell$

has coordinates $O(1/n)$ for all $i \geq n$, so its Toeplitz image is $O(1/n)$. The two K^* terms satisfy (74). This proves (75), starting from zero.

For each fixed $r \geq 0$ let $E_\ell(r) = \lim_{n \rightarrow \infty} (e_{n,\ell})_{n-1-r}/a_{n-1}$ when it exists. It is zero for $\ell = 0$. Assuming it exists at level ℓ , the exact recurrence gives

$$E_{\ell+1}(r) = \sum_{s \geq 0} b_{s-r} E_\ell(s) - \kappa_\ell \sum_{m \geq 0} b_{-r-1-m} (-1)^{m+1}. \quad (76)$$

To justify all tails, on input indices $k \geq n/2$ the normalized defect coordinates are uniformly bounded by (75); absolute summability permits a fixed-distance cutoff followed by its removal. On $k < n/2$, the preceding row-tail estimate is $O(n^{-1-\varepsilon}) = o(a_{n-1})$. On the exterior tail $k = n + m$, the normalized source coordinates are uniformly bounded and converge, for each fixed m , to $\kappa_\ell (-1)^{m+1}$. Dominated summation therefore applies to the second series. The K^* terms are $o(a_{n-1})$ by (74). Each limiting profile is bounded by (75) with $\eta = 1/2$. This proves existence inductively and gives

$$\lim_{n \rightarrow \infty} \frac{[A_n^\ell P_n q]_{n-1}}{a_{n-1}} = \kappa_\ell + E_\ell(0).$$

There is no infinite-power interchange in this argument. $\square$

6.3 Closing the analytic boundary limit

Apply Theorem 6.1 to the original B in the ψ basis, using Theorem 5.1 with $\varepsilon = 2/3$ and the source shape (72). Apply it separately to q_3 and $q^{(1)}$. Every coefficient in (38) therefore converges along the full sequence. The normal-family argument following that formula proves $F_n \rightarrow F$ locally uniformly on $\mathcal{H}_+$. The companion identity gives exactly

$$F_n(s)F_{n+1}(s) = (s/c)^2 \frac{D_n(s^2)/D_n(3)}{D_{n+1}(s^2)/D_{n+1}(3)}.$$

Since $E_n \rightarrow E$ locally uniformly, the right side tends to $(s/c)^2/P(s)$. Thus $F(s)^2 = (s/c)^2/P(s)$. It is nonzero on $\mathcal{H}_+$, and $F(c) = 1$ selects the branch $F(s) = (s/c)/\sqrt{P(s)}$. This proves (8) without a residual parity ambiguity.

6.4 A nonzero leading Galerkin contribution as a check

There is a useful independent check on the sign and scale. Define

$$\tau_j = \frac{\langle B e_j, Q_{j+1} q_3 \rangle}{a_j^{\text{src}}},$$

with source and basis now in coordinate space. The $B - T_b$ part is $o(1)$ after division by a_j^{src} , by (54) and $\|Q_{j+1} q_3\| = O(j^{-1/2})$. For the Toeplitz part, the calibration asymptotic makes $a_{j+r}^{\text{src}}/a_j^{\text{src}}$ uniformly bounded for $r \geq 1$ and gives its fixed- r limit $(-1)^r$. Absolute summability yields

$$\tau_j \longrightarrow \sum_{r \geq 1} (-1)^r b_r = \sqrt{3}/4 - 1/2. \quad (77)$$

Thus the original finite-projection tail is genuinely leading order. It was retained in (76); compactness alone would not have justified discarding it.

7 Individual amplitudes, inverse thresholds and cumulants

7.1 Proof of the full equivalent

Write $R_n(s) = \text{Log}[Z_n(s)/Z_n(c)]$ on $\mathcal{H}_+$, normalized at c . The exact adjacent identity and the now-proved companion limit give

$$R_n(s) + R_{n+1}(s) = nf(s^2) + E_n(s^2) + \text{Log}(s/c). \quad (78)$$

The exact boundary definition gives $R_{n+1}(s) - R_n(s) = \text{Log}(s/c) - \text{Log} F_n(s)$. Because $F_n \rightarrow (s/c)e^{-f(s^2)/2}$ locally uniformly and all functions are nonzero with the same normalization, their analytic logarithms converge locally uniformly. Hence

$$R_{n+1}(s) - R_n(s) \rightarrow \frac{1}{2}f(s^2). \quad (79)$$

Subtract (79) from (78) and divide by two. This yields

$$R_n(s) - \frac{n}{2}f(s^2) \rightarrow \frac{1}{2}E(s^2) + \frac{1}{2}\text{Log}(s/c) - \frac{1}{4}f(s^2) \quad (80)$$

locally uniformly. Adding the real full-sequence calibration (27) proves Theorem 1.1, including its complex local uniformity. All factors in (5) are positive on the positive axis and nonzero on $\mathcal{H}_+$. There are no zeros or poles of P there; the definitions at c fix all roots uniquely. At $s = 1$, $P(1) = 2/3$, so $(1/c)^{1/2}P(1)^{-1/4} = 2^{-1/4}$. Also $b + \frac{1}{2}\log(2/3) = \beta$. This verifies (7).

7.2 Amplitude-aware inverse count thresholds

Let

$$C_* = C(1) = C_{\text{cal}}2^{-1/4}e^{E(1)/2}, \quad N(T) = \min\{n \geq 1 : a_n \geq T\}, \quad y = \log T, \quad r = \sqrt{y/\alpha}.$$

The minimum exists because (7) tends to infinity. The counts are nondecreasing: $A \mapsto \text{diag}(A, 1)$ is an injection. Only large T is considered below.

Corollary 7.1. *Define the real center*

$$z_C(T) = r - \frac{\beta}{2\alpha} + \frac{\kappa \log r + \beta^2/(4\alpha) - \log C_*}{2\alpha r}. \quad (81)$$

There exists a nonnegative function $\varepsilon(T) \rightarrow 0$ such that

$$\lceil z_C(T) - \varepsilon(T)/r \rceil \leq N(T) \leq \lceil z_C(T) + \varepsilon(T)/r \rceil. \quad (82)$$

Equivalently the displayed center is

$$\begin{aligned} z_C(T) = & \sqrt{\frac{\log T}{\alpha}} - \frac{\beta}{2\alpha} + \frac{\kappa \log \log T}{4\sqrt{\alpha \log T}} \\ & + \frac{\beta^2/(4\alpha) - \log C_* - (\kappa/2) \log \alpha}{2\sqrt{\alpha \log T}}. \end{aligned} \quad (83)$$

Thus the amplitude enters at the $1/\sqrt{\log T}$ scale, and the remaining error is little-o at that scale before ceilings.

Proof. Set $g(x) = \alpha x^2 + \beta x - \kappa \log x + \log C_*$ for $x > 0$. The full equivalent says $\log a_n = g(n) + \eta_n$ with $\eta_n \rightarrow 0$. On $[r/2, 2r]$, $g'(x) = 2\alpha x + \beta - \kappa/x \asymp r$. Substitution of (81), writing its last numerator as $L_r = \kappa \log r + \beta^2/(4\alpha) - \log C_*$, gives

$$g(z_C(T)) - y = O((\log r)/r) = o(1).$$

Indeed the quadratic and linear contributions are $y - \beta^2/(4\alpha) + L_r + O((\log r)^2/r^2)$, while $-\kappa \log z_C = -\kappa \log r + O(1/r)$. The leading terms cancel exactly.

The growth theorem first gives $N(T) \in [r/2, 2r]$ for all sufficiently large T (the endpoints have different fixed quadratic multiples of y). On this interval the tail supremum $\sup_{n \geq r/2} |\eta_n|$ tends to zero. Choose $\varepsilon(T) > 0$, tending to zero, larger by a fixed sufficient factor than this supremum plus $|g(z_C) - y|$ and $1/r$. The lower derivative bound then shows that every integer $n < z_C - \varepsilon/r$ has $\log a_n < y$, while the integer $n = \lceil z_C + \varepsilon/r \rceil$ has $\log a_n \geq y$. Integers outside $[r/2, 2r]$ are handled by the preliminary growth comparison and monotonicity. These two statements are exactly (82). Substituting $r = \sqrt{y/\alpha}$ and $\log r = (\log y - \log \alpha)/2$ proves (83). $\square$

The ceilings are essential: even a vanishing real error can decide which adjacent integer is the threshold when the center is near an integer. The theorem does not give a fixed explicit decay rate for $\varepsilon(T)$, nor does it justify discarding it inside a ceiling.

[write] **Superseded inverses, 5 October 2026.** Corollary 7.1 sharpens Part II's Corollary 10.2: the centre (81) is Part II's (II.8) with the amplitude term $-\log C_*/(2\alpha r)$ added, and the error C/r becomes $\varepsilon(T)/r$ with $\varepsilon(T) \rightarrow 0$. Part III's bracket (III.98) (error $\eta(T)$ without the $\log r/r$ term) and Part IV's Theorem 48.1 (limiting width $(q-p)/(2\alpha) \approx 0.1254$) are weaker. None of the four brackets is effective (Section 9, item 4).

7.3 Constant terms of every fixed diagonal cumulant

For fixed $s > 0$ assign probability $s^{S(A)}/Z_n(s)$ to a DSASM, and let S_n denote its number of nonzero diagonal entries. Put $\mathcal{D} = s d/ds$ and $L(s) = \text{Log } C(s)$, real on the positive axis. Define

$$m(s) = \frac{1}{2} - \frac{1}{(s+1)(s+2)}, \quad v(s) = \frac{s(2s+3)}{(s+1)^2(s+2)^2} > 0.$$

Then $\mathcal{D}B = m$ and $\mathcal{D}m = v$.

Corollary 7.2. *For each fixed $k \geq 1$, locally uniformly on positive fugacity intervals,*

$$\text{cum}_k(S_n) = n\mathcal{D}^{k-1}m(s) + \mathcal{D}^k L(s) + o(1). \quad (84)$$

In particular

$$\begin{aligned} \mathbb{E}_s S_n &= nm(s) + s^2 E'(s^2) + \frac{1}{2} - \frac{1}{2}m(s) + o(1), \\ \text{Var}_s(S_n) &= nv(s) + 2s^2 E'(s^2) + 2s^4 E''(s^2) - \frac{1}{2}v(s) + o(1). \end{aligned}$$

At $s = 1$ the linear coefficients are $m(1) = 1/3$ and $v(1) = 5/36$.

Proof. The locally holomorphic limit (80) implies, on a small complex disk about $z = 0$,

$$\log \mathbb{E}_s e^{zS_n} = n[B(se^z) - B(s)] + L(se^z) - L(s) + o(1)$$

locally uniformly. Cauchy's formula makes every fixed derivative of the error $o(1)$. Differentiating proves (84). Inserting $L(s) = \log C_{\text{cal}} + E(s^2)/2 + \frac{1}{2} \text{Log}(s/c) - \frac{1}{4} \text{Log } P(s)$ gives the displayed first two constants. $\square$

This strengthens the bounded cumulant errors in the earlier article. It relies on complex-local convergence, rather than differentiating pointwise real asymptotics.

[write] **Answered questions, 5 October 2026.** Corollary 7.2 strengthens Part II’s bounded errors (II.121) (that is the “earlier article” meant here) and Part III’s limits (III.88). It answers Part III’s question 2 (Section 40), already answered by Part II in the form $M_n(s) - nm(s) = O(1)$: the difference converges. The constants $E'(1)$ and $E''(1)$ are not evaluated; an uncertified estimate of $E'(1)$ is in Section 9, item 6.

8 Scope and reproducibility

The result is a full leading equivalent at each fixed positive fugacity, with a positive constant characterized by convergent operators and a Fourier series. Convergence is locally uniform on the right half-plane. It is not asserted uniformly as $s \rightarrow 0$ or $s \rightarrow \infty$, nor uniformly in an unbounded derivative order. No numerical digits for $C(1)$, explicit rate for the little-o terms, n^{-2} correction, or all-orders conjecture are established here.

[write] **Numerical amplitude, 5 October 2026.** Behrend, Fischer and Koutschan give a numerical estimate of $C = C(1)$ from the exact counts for $n \leq 1000$; an independent uncertified estimate from exact counts to $n = 177$, consistent with it, and the evidence about the conjectured n^{-2} and parity-dependent corrections are recorded in Section 9. Neither is part of Report 128’s claims.

The source bundle contains this article’s modular TeX, its PDF, the unchanged earlier-input article’s PDF and TeX, a closed SHA-256 inventory, deterministic clean-build and repack scripts, and exact finite checks with an adversarial input campaign. The finite checks corroborate identities and normalization; they are not proofs of any infinite-dimensional estimate, asymptotic convergence or numerical amplitude. The analytical arguments above establish those limits. The replay checks package immutability, normal and optimized Python behavior, independent exact data, two byte-identical clean PDF builds, and a closed deterministic ZIP round trip. Rebuilding needs only the locally installed TeX distribution and Python; the scripts make no network requests and install no dependencies.

[write] **The source bundle here, 5 October 2026.** The bundle described in this paragraph was delivered as the archive named in the Part’s heading. In this report the article’s TeX is `article.tex`; the PDFs, the closed SHA-256 inventory and the earlier-input copy of Report 124 are not shipped (Report 124 is Part II), and the scripts and data are shipped under prefixed names (README). The delivered replay cannot run in this directory.

9 Further questions and research

[write] **Added in the write, 5 October 2026.** This section is not part of Report 128. Under Vladimir’s standing rule of 4 October 2026 it collects the claims of Part I, and the questions of Parts II–IV that remain open after Part I, that are stated without proof, with their source, the sketch given and what is missing. Questions that Part I answers are recorded as answered at their places in Parts II–IV (Sections 24, 40 and 49).

1. **The value of the amplitude.** Theorem 1.1 proves that $C_* = C(1) = C_{\text{cal}} 2^{-1/4} e^{E(1)/2}$ exists and characterizes it by (26), “without a numerical fit”; Section 8 states that no digits of $C(1)$ are established. Open: a certified numerical value, and a closed form. Behrend, Fischer and Koutschan estimate $C \approx 0.72352852136732419487$ from the exact counts for

$n \leq 1000$ and report no conjectured closed expression [1, Conjecture 11.1]. *Uncertified evidence (intake dossier and write, 5 October 2026)*: from exact counts to $n = 177$ (independent Pfaffian and leading-minor computation, agreeing with the OEIS terms checked), the residual $r_n = \log a_n - \alpha n^2 - \beta n + \kappa \log n$ is -0.3236158 at $n = 100$ and -0.3236155 , -0.3236162 at $n = 176, 177$; averages over four consecutive n and Richardson extrapolation with $1/n$ or $1/n^2$ drift give $\log C_* = -0.323615 \pm 10^{-6}$, that is

$$C_* \approx 0.7235288 \pm 10^{-6} \quad (\text{uncertified}),$$

$$E(1) = 2 \log(C_*/(C_{\text{cal}} 2^{-1/4})) \approx -0.248070 \pm 2 \cdot 10^{-6},$$

with $C_{\text{cal}} 2^{-1/4} = 0.81907484878612193750 \dots$ from (4). This agrees with the estimate of Behrend, Fischer and Koutschan ($\log C = -0.32361531 \dots$) to within the stated error. What is missing: a certified evaluation of (26), or rigorous error bounds for an extrapolation, which would need an explicit rate (item 2).

2. **Rate and the finer corrections of Conjecture 11.1.** Part I gives $(1 + o(1))$ with no explicit rate (Section 8). Conjecture 11.1 of [1] predicts the factor $1 - 385/(31104n^2) + (-1)^{n/2} a n^{-5/2} + O(n^{-3})$ for even n and $1 - 385/(31104n^2) + (-1)^{(n+1)/2} b n^{-5/2} + O(n^{-3})$ for odd n , with $b = (1 + \sqrt{2})a$; none of this is proved here. *Uncertified evidence*: the increments $r_n - r_{n-1}$ show a decaying oscillation of period 4 (about $\pm 3 \cdot 10^{-7}$ and $\pm 7 \cdot 10^{-7}$ near $n = 175$, signs $+, +, -, -$), which is the period of the signs $(-1)^{n/2}$ (n even) and $(-1)^{(n+1)/2}$ (n odd) of the conjectured $n^{-5/2}$ terms; and the exact r_n for $170 \leq n \leq 177$ differ from the logarithm of the conjectured expansion, with the published estimates of C , a , b inserted, by at most $2 \cdot 10^{-10}$. What is missing: any quantitative form of the companion limit $E_n \rightarrow E$ and of the boundary limit (8); Part I's proofs (Vitali, normal families, strong resolvent convergence) are qualitative.
3. **An all-orders expansion.** Neither an n^{-2} correction nor an all-orders or transseries expansion is established (Section 8; Part II, question 3 of Section 24). Behrend, Fischer and Koutschan remark that, unlike the product-formula classes, the DSASM expansion appears to contain fractional exponents and irrational coefficients. Open.
4. **Effective constants and a certified inverse.** The function $\varepsilon(T)$ of Corollary 7.1 has no explicit decay and cannot be dropped inside the ceilings; Part II (question 4 of Section 24), Part III (question 5 of Section 40) and Part IV (question 5 of Section 49) ask for effective constants and onsets. Open: explicit Jacobi, kernel-ideal, overlap and Fredholm constants, a rate as in item 2, and hence a certified finite-target threshold algorithm. For a specific finite T the exact counts decide $N(T)$ directly.
5. **Uniformity.** Convergence is locally uniform on $\Re s > 0$ and for each fixed derivative order (Section 8); it is not asserted as $s \rightarrow 0$, $s \rightarrow \infty$, or for derivative orders growing with n . Open: joint regimes in (n, s) ; Part III, questions 3 and 4 of Section 40, ask related questions about root escape and prefactors away from the central window.
6. **Evaluating $E(t)$.** The operator/Fourier expression (26) has not been evaluated at any $t \neq 3$. *Uncertified evidence (dossier)*: the exact means give $M_n(1) - n/3 \rightarrow 0.4060$ approximately, on both parities; with Corollary 7.2 at $s = 1$, $\mathbb{E}_1 S_n - n/3 \rightarrow E'(1) + \frac{1}{2} - \frac{1}{6}$, this suggests $E'(1) \approx 0.0727$. The boundary ratios satisfy $F_n(s) - (s/c)/\sqrt{P(s)} = 1.2 \cdot 10^{-6}$, $4.3 \cdot 10^{-6}$, $-1.5 \cdot 10^{-5}$ at $s = \frac{1}{2}, 1, 3$ and $n = 100$, decreasing. Open: a convergent numerical scheme with error bounds for the determinants $\mathcal{A}$, Ξ , Ψ and the series in (26).

Part II

Logarithmic asymptotics: bounded determinant remainders and inverse thresholds

Bundle Report 124, archive A005163_Exact_Logarithmic_Scale_and_Inverse_Thresholds_Source.zip, delivered as `report124.tex` (34 pages), dated 2 October 2026; Report 128 ships the same TeX and PDF byte for byte as its earlier input. Delivered title block: “REPORT 124 / Logarithmic asymptotics of diagonally symmetric alternating sign matrices / Bounded determinant remainders and inverse thresholds / 2 October 2026 / Standalone proof and reproducibility companion”. Printed in full, with **[write]** notes. Report 124’s Section k is Section $k + 9$ here, its statement $k.j$ is $(k + 9).j$, and its equation (k) is (II. k).

Reading conventions of Part II	Elsewhere in the report
$P_n(t)$ the parity-reduced polynomial (Sections 10–12, 23); P_n the degree- $< n$ projection (Section 15); P_n^0 the coordinate projection (Section 20) $E_n = \log[D_n(1)/D_n(3)] - n \log(2/3)$ (Section 24) $z(T)$ inverse centre; $r = \sqrt{y/\alpha}$ C_{DSASM} conjectured amplitude	Part I: P_n only a projection, $P(s)$ the pressure Part I: $E_n(t)$, with $E_n = E_n(1)$ Part I: $z_C(T)$ adds $-\log C_*/(2\alpha r)$; Parts III, IV: $n_0 = r$ Part I: C_*

Abstract of this Part (as delivered). Let a_n count diagonally symmetric alternating sign matrices of order n , the sequence A005163. We prove

$$\log a_n = \frac{1}{2} \log \left(\frac{3\sqrt{3}}{4} \right) n^2 + \frac{1}{4} (\log 3) n - \frac{5}{72} \log n + O(1).$$

Thus the counts have positive finite upper and lower bounds at the full exponential and logarithmic shape in the conjecture of Behrend, Fischer and Koutschan. The proof retains the exact finite orthogonal-polynomial projections throughout. A regularized Fredholm comparison reduces the original determinant to a Jacobi projection of a Hardy-compressed scalar multiplier. Endpoint-controlled Jacobi means, a finite tilted-projection variance bound, and a weighted Hardy overlap estimate give the bounded remainder. Exact stability, cross-size interlacing, and product calibration transfer it to individual sizes. We also identify the constant correction to the first calibration trace and the logarithmic term in inverse count thresholds, with integer rounding retained. No convergence of the centered count or determinant, no limiting DSASM amplitude, and no full asymptotic equivalent are asserted. A complex-local extension gives bounded errors for every fixed diagonal cumulant and an explicitly centered central limit theorem at each positive fugacity.

[write] Status of Part II, 5 October 2026. Part I uses this Part’s results (II)–(I4) (Section 1) and strengthens its bounded statements to limits: the sentence “No convergence of the centered count or determinant, no limiting DSASM amplitude, and no full asymptotic equivalent are asserted” describes this Part, and is now historical, since Part I proves all three (Theorem 1.1). Sections 11, 13 and 14 are also Sections 2, 4 and 5 of Reports 121 and 120 (Parts III and IV), and Section 12 is also Report 121’s Section 9; they are printed only here, and the variant sentences of the other two manuscripts are printed at Sections 26, 33, 42 and 45.

10 Definitions and main conclusions

An alternating sign matrix is a square matrix with entries in $\{-1, 0, 1\}$ whose nonzero entries alternate in sign in every row and column, and whose row and column sums equal one. It is diagonally symmetric if it equals its transpose. Let

$$a_n = |\text{DSASM}(n)|, \quad S(A) = |\{i : A_{ii} \neq 0\}|, \\ Q_n(s_1, \dots, s_n) = \sum_{A \in \text{DSASM}(n)} \prod_{i: A_{ii} \neq 0} s_i, \quad Z_n(s) = Q_n(s, \dots, s). \quad (\text{II.1})$$

Use $a_0 = Z_0 = Q_0 = 1$ and natural logarithms. In the notation of Behrend–Fischer–Koutschan [1], $Z_n(s) = X_n(1, s, 1)$. The sequence a_n is OEIS A005163 [14]. Define

$$g = \log \frac{3\sqrt{3}}{4}, \quad \alpha = \frac{g}{2}, \quad \beta = \frac{1}{4} \log 3, \quad \kappa = \frac{5}{72}, \quad b = \frac{g + \log 2}{2} = \frac{3}{4} \log 3 - \frac{1}{2} \log 2. \quad (\text{II.2})$$

All asymptotic statements are along the full sequence of positive integers, unless specified otherwise.

Theorem 10.1. *There are constants $0 < c_- \leq c_+ < \infty$ such that for every $n \geq 1$,*

$$c_- e^{\alpha n^2 + \beta n} n^{-\kappa} \leq a_n \leq c_+ e^{\alpha n^2 + \beta n} n^{-\kappa}. \quad (\text{II.3})$$

Equivalently,

$$\log a_n = \alpha n^2 + \beta n - \kappa \log n + O(1). \quad (\text{II.4})$$

For the polynomial

$$D_n(t) = \frac{Z_n(\sqrt{t})Z_{n+1}(\sqrt{t})}{\sqrt{t}}, \quad (\text{II.5})$$

interpreted polynomially, the determinant remainder is bounded:

$$\log \frac{D_n(1)}{D_n(3)} = n \log(2/3) + O(1). \quad (\text{II.6})$$

Its first logarithmic derivative at the solvable calibration satisfies

$$\frac{D'_n(3)}{D_n(3)} = (n+1) \left(1 - \frac{\sqrt{3}}{2}\right) + o(1). \quad (\text{II.7})$$

The constants in (II.3) are not made numerically effective here.

[write] **Superseded bounds, 5 October 2026.** Part I, Theorem 1.1, proves that $a_n e^{-\alpha n^2 - \beta n} n^\kappa \rightarrow C_* > 0$, so (II.3) holds with every $c_- < C_* < c_+$ for all large n , and $\log(D_n(1)/D_n(3)) - n \log(2/3) \rightarrow E(1)$. The trace limit (II.7) is not restated in Part I and stands here; Part III, (III.8), proves the weaker $\ell_n/n \rightarrow 1 - \sqrt{3}/2$ by a different route.

Behrend–Fischer–Koutschan’s Conjecture 11.1 in the version dated 1 October 2026 predicts a full equivalent $a_n \sim C_{\text{DSASM}} e^{\alpha n^2 + \beta n} n^{-5/72}$ and further parity-sensitive corrections. Their Theorem 11.2 gives the leading quadratic growth. Theorem 10.1 proves two-sided bounds at the conjectured shape; it does not establish the limit of the normalized counts or determine C_{DSASM} .

[write] **Historical, 5 October 2026.** Part I proves the limit of the normalized counts; C_{DSASM} is Part I’s C_* , characterized by (26) but not evaluated (Section 9). The parity-sensitive corrections remain unproved.

Corollary 10.2. *Let $N(T) = \min\{n \geq 1 : a_n \geq T\}$, $y = \log T$ and $r = \sqrt{y/\alpha}$. Set*

$$z(T) = r - \frac{\beta}{2\alpha} + \frac{\kappa \log r + \beta^2/(4\alpha)}{2\alpha r}. \quad (\text{II.8})$$

For some $C, T_0 > 0$ and every $T \geq T_0$,

$$\lceil z(T) - C/r \rceil \leq N(T) \leq \lceil z(T) + C/r \rceil. \quad (\text{II.9})$$

In particular, the coefficient of $(\log \log T)/\sqrt{\log T}$ in this real-valued center is $\kappa/(4\sqrt{\alpha})$. The bracket, rather than an unrounded additive $o(1)$ assertion, is the integer inverse statement.

[write] **Sharpened, 5 October 2026.** Part I, Corollary 7.1, adds the amplitude term $-\log C_*/(2\alpha r)$ to the centre (II.8) and replaces C/r by $\varepsilon(T)/r$ with $\varepsilon(T) \rightarrow 0$.

10.1 Proof architecture and published inputs

Sections 11–14 establish the exact finite prerequisites. The negative-root property makes the regularized relative determinant nonvanishing, and cross-size interlacing controls the possible parity imbalance. The calibration is derived from a shifted determinant evaluation and the ordinary ASM product; no calibration amplitude is required. Sections 15–17 construct the fixed bounded-operator comparison, distinguish Hilbert–Schmidt from trace-class errors, and identify the exact Jacobi projection. The endpoint mean theorem in Section 18, the scalar determinant bound in Section 19, and the weighted Hardy overlap in Section 20 close the analytic argument. Section 21 transfers it to the count and its inverse.

The published combinatorial inputs are the triangular six-vertex bijection and determinant of [1]. Stability closure and the MOD operator are classical [3, 4]; the degree-two operator and the interlacing implication are proved explicitly below. The shifted determinant and Meixner–Pollaczek Gram formulas are classical inputs [5, 6]. The only orthogonal-polynomial asymptotic input is the fixed-parameter uniform Jacobi Bessel expansion of Frenzen–Wong [7], with its explicit DLMP normalization and remainder [8]. Standard Fredholm determinant continuity and regularization are used only after the relevant ideal convergence has been proved [9]. No general strong Szegő theorem is invoked, and no global bibliographic priority claim is made.

For later reference, stability and parity will give

$$Z_n(s) = s^{\varepsilon_n} P_n(s^2), \quad \varepsilon_n = n \bmod 2, \quad P_n(0) > 0, \quad (\text{II.10})$$

with P_n monic of degree $\lfloor n/2 \rfloor$ and all its roots strictly negative. Consequently $D_n = P_n P_{n+1}$ is monic of degree n and has only strictly negative roots, with multiplicities retained.

11 An orientation polynomial and diagonal stability

Let $\mathcal{H} = \{z \in \mathbb{C} : \Re z > 0\}$. A polynomial is $\mathcal{H}$ -stable if it is nonvanishing whenever all its variables belong to $\mathcal{H}$; the zero polynomial is not called stable here.

11.1 The exact triangular graph

The model from [1, Section 5.1, Equations (5.1)–(5.2)] has vertices

$$V = \{(i, j) : 1 \leq i \leq j \leq n\}, \quad B = \{(i, j) : i < j\}, \quad D = \{(i, i) : 1 \leq i \leq n\}.$$

Its internal undirected edges are

$$\begin{aligned} E_h &= \{(i, j), (i, j+1)\} : 1 \leq i \leq j < n\}, \\ E_v &= \{(i, j), (i+1, j)\} : 1 \leq i < j \leq n\}, \quad E_{\text{int}} = E_h \cup E_v. \end{aligned} \tag{II.11}$$

There are $n(n-1)$ internal edges. The n top boundary arrows point out of the triangle; the n right boundary arrows point into it. Each bulk vertex has degree four and must have indegree two. Each diagonal vertex has degree two and no indegree restriction.

In the local correspondence with matrix entries, the diagonal possibilities are as follows. “In” and “out” refer to the diagonal vertex, not to the whole triangle.

Diagonal entry	North arrow	East arrow	Indegree
+1	out	in	1
−1	in	out	1
0	in	in	2
0	out	out	0

Thus a nonzero diagonal entry is exactly a diagonal vertex of indegree one. The published correspondence is a bijection, so no extra multiplicity factor is needed.

Assign a variable x_v to every vertex $v \in V$ and form

$$H_n(\mathbf{x}) = \left(\prod_{i=1}^n x_{(i,n)} \right) \prod_{\{v,w\} \in E_{\text{int}}} (x_v + x_w). \tag{II.12}$$

Choosing x_v in the factor for $\{v, w\}$ directs that edge into v . The prefactor records the fixed incoming arrows on the right; fixed outgoing top arrows contribute one. A monomial exponent is therefore the corresponding indegree, including boundary arrows. Different orientations may give the same monomial, and the polynomial counts each of them with its proper multiplicity.

Each edge factor is nonzero when its variables lie in $\mathcal{H}$, and so is the prefactor. Hence H_n is $\mathcal{H}$ -stable. Enforce all the bulk ice rules by coefficient extraction:

$$F_n(\mathbf{x}_D) = \left[\prod_{v \in B} x_v^2 \right] H_n(\mathbf{x}). \tag{II.13}$$

Differentiation and specialization to a boundary point of $\mathcal{H}$ preserve stability, with a possible identically zero result; these standard closure properties are given in [3, Theorem 1.6 and Lemma 1.7] and [4, Remark 3.1]. For coefficient extraction, differentiate twice, set that variable to zero, and divide by $2!$. Boundary specialization also follows by taking positive real specializations tending to zero and applying Hurwitz’s theorem. A configuration corresponding to the identity matrix survives each extraction. All coefficients are nonnegative, so there is no cancellation and no zero intermediate result. Consequently F_n is stable.

The remaining variable $x_{(i,i)}$ has degree at most two. At the first corner it has degree at most one because the top arrow is fixed outward; at the last corner it has exponent at least one because the right arrow is fixed inward. When $n = 1$, (II.12) and (II.13) give $H_1 = F_1 = x_{(1,1)}$, as required. The polynomial H_n has total degree n^2 . Extraction removes $n(n-1)$, so F_n is homogeneous of degree n .

11.2 The boundary operator with all degree cases

For a variable x of degree at most two, define

$$\mathcal{T}_s(1) = 1, \quad \mathcal{T}_s(x) = s, \quad \mathcal{T}_s(x^2) = 1. \quad (\text{II.14})$$

This is the degree-two restriction of the MOD operator from [4, Section 8.4], which reduces monomial exponents modulo two and preserves weak Hurwitz stability. The following direct argument is included so that the exact case needed here is explicit.

Lemma 11.1. *Applying (II.14) to one variable of a stable multivariate polynomial whose degree in that variable is at most two produces a stable polynomial in the new variable s and all the other variables.*

Proof. Fix all the other variables at arbitrary points of $\mathcal{H}$. The resulting polynomial $f(x)$ is nonzero on $\mathcal{H}$, so it is not identically zero. Its actual degree is zero, one or two, even if a leading coefficient vanishes under the chosen specialization.

If $f(x) = c \neq 0$, the image is c . If $f(x) = c(x+a)$, then $\Re a \geq 0$ and the image $c(s+a)$ is nonzero for $s \in \mathcal{H}$. In degree two, write $f(x) = c(x+a)(x+d)$ with $c \neq 0$ and $\Re a, \Re d \geq 0$. Its image is

$$c(1+ad+s(a+d)).$$

If $a+d \neq 0$, its only possible zero is $s = -(1+ad)/(a+d)$, and

$$\Re \frac{1+ad}{a+d} = \frac{\Re(a)(1+|d|^2) + \Re(d)(1+|a|^2)}{|a+d|^2} \geq 0. \quad (\text{II.15})$$

That zero is outside $\mathcal{H}$. If $a+d = 0$, the nonnegative real parts imply $a = iy$, $d = -iy$ for some $y \in \mathbb{R}$; then $1+ad = 1+y^2 > 0$. The image is again nonzero. Since every allowed assignment of the other variables was covered, the full multivariate assertion follows. $\square$

Apply this operator to every diagonal variable in F_n . The local indegree table shows exactly that

$$Q_n(s_1, \dots, s_n) = \left(\prod_{i=1}^n \mathcal{T}_{s_i} \right) F_n(\mathbf{x}_D). \quad (\text{II.16})$$

Iterating Lemma 11.1 proves the joint stability of Q_n . Setting every $s_i = s$ proves that Z_n has no zero with $\Re s > 0$.

11.3 Parity and strictly negative roots

Each monomial of F_n has exponents $d_i \in \{0, 1, 2\}$ with $\sum_i d_i = n$. Replacing these exponents by their parities changes total degree by an even number. Thus

$$Z_n(-s) = (-1)^n Z_n(s), \quad (\text{II.17})$$

which gives (II.10) with nonnegative integer coefficients. One can also check parity directly from symmetry: $n = \sum_i A_{ii} + 2 \sum_{i < j} A_{ij}$, while each nonzero A_{ii} is congruent to one modulo two.

If a zero of Z_n had negative real part, its negative would have positive real part by (II.17), contradicting stability. Every zero of Z_n therefore lies on the imaginary axis. The antidiagonal permutation matrix is symmetric and has exactly ε_n nonzero diagonal entries; hence $P_n(0) > 0$. The identity matrix has n nonzero diagonal entries, giving degree $\lfloor n/2 \rfloor$. For any root t of P_n , choose s with $s^2 = t$. Then $s \neq 0$, $Z_n(s) = 0$, and s is purely imaginary. It follows that $t = s^2$ is strictly negative and real. This proves the asserted root structure. For $n = 1$, $P_1 = 1$ and the root statement is vacuous.

The root theorem by itself does not prove simplicity. Cross-size weak interlacing will be established separately in Section 12. Stability for arbitrary further six-vertex weights is not asserted.

12 Cross size interlacing and parity control

The joint stability established in Section 11 gives more than real-rootedness of each separate polynomial. We now derive a stable pencil and its consequences with multiplicities retained.

12.1 Corner deletion and the stable pencil

If $A_{11} \neq 0$ in an ASM, the topmost nonzero entry in its first column shows $A_{11} = 1$. Every entry of the first row is zero or one, so its row sum forces all other entries in that row to vanish. Symmetry gives the same conclusion in the first column. Deleting that row and column is a bijection to $\text{DSASM}(n-1)$, whose inverse adjoins an isolated $+1$ corner. Hence

$$[s_1]Q_n(s_1, \dots, s_n) = Q_{n-1}(s_2, \dots, s_n). \quad (\text{II.18})$$

This also proves inductively that the unique DSASM with all diagonal entries nonzero is the identity. Thus Z_n is monic of degree n , and P_n is monic as well.

Define

$$\mathcal{H}_n(\mathbf{x}) = i^{-n} Q_n(i\mathbf{x}), \quad q_n(x) = i^{-n} Z_n(ix).$$

Every monomial of Q_n has degree congruent to n modulo two, by the symmetry argument used in (II.17); thus $\mathcal{H}_n$ has real coefficients and $Q_n(-\mathbf{s}) = (-1)^n Q_n(\mathbf{s})$. If every $\Im x_j > 0$, then every $\Re(-ix_j) > 0$, and parity and right-half-plane stability give $Q_n(i\mathbf{x}) = (-1)^n Q_n(-i\mathbf{x}) \neq 0$. Therefore $\mathcal{H}_n$ is upper-half-plane stable. In particular q_n is monic, real-rooted, and has symmetric roots about zero.

Since Q_n is multiaffine, (II.18) gives exactly

$$\mathcal{H}_n(x+t, x, \dots, x) = q_n(x) + tq_{n-1}(x). \quad (\text{II.19})$$

For $\Im x > 0$ and $\Im t > 0$, all arguments on the left lie in the upper half-plane. This proves two-variable stability of the pencil. The general stable-pencil/interlacing criterion is classical [3, Definition 1.1 and Lemma 1.8]; the direct proof below fixes all residue signs and handles cancellation explicitly.

12.2 The rational function argument with common roots

Take the monic greatest common divisor $d_n = \gcd(q_n, q_{n-1})$ and write $f = q_n/d_n$, $g_0 = q_{n-1}/d_n$. These are coprime monic real-rooted polynomials of degrees differing by one. For $\Im z > 0$, stability in (II.19) means $-f(z)/g_0(z)$ cannot have positive imaginary part. Thus $R_0 = f/g_0$ maps the upper half-plane into its closure.

Every pole of R_0 is real. At a pole p of order r with real leading coefficient $a \neq 0$, the leading imaginary part at $z = p + \rho e^{i\theta}$ is $-a\rho^{-r} \sin(r\theta)$. If $r \geq 2$, this has both signs for $0 < \theta < \pi$, a contradiction for small ρ . Thus all actual poles are simple, and their residues are negative. Partial fractions and the monic degree difference give

$$R_0(z) = z + \beta - \sum_{j=1}^r \frac{c_j}{z - p_j}, \quad p_1 < \cdots < p_r, \quad c_j > 0, \quad \beta \in \mathbb{R}. \quad (\text{II.20})$$

On every real interval between consecutive poles, $R'_0(x) = 1 + \sum_j c_j/(x - p_j)^2 > 0$. The one-sided limits show that R_0 runs from $-\infty$ to $+\infty$ on each such interval and on both exterior intervals. It has exactly one simple zero in each, so f and g_0 strictly interlace. If $g_0 = 1$, the same conclusion is the elementary statement for $R_0(z) = z + \beta$.

Let $N_p(x)$ count the roots of a polynomial in $(-\infty, x]$ with multiplicity. Reduced interlacing gives $N_f(x) - N_{g_0}(x) \in \{0, 1\}$. Restoring the common roots leaves this difference unchanged. Therefore, if the original root lists are $a_1 \leq \cdots \leq a_n$ and $b_1 \leq \cdots \leq b_{n-1}$, then

$$a_j \leq b_j \leq a_{j+1} \quad (1 \leq j < n). \quad (\text{II.21})$$

The difference of counts in any interval, with a fixed convention at its endpoints, has absolute value at most one. This follows by subtracting the two half-line discrepancies, each in $\{0, 1\}$. No simplicity or strict interlacing of the original polynomials is asserted.

For the reverse ratio there is a useful positive-residue statement. If α_k are the simple roots of f , then

$$\frac{q_{n-1}(z)}{q_n(z)} = \sum_k \frac{w_k}{z - \alpha_k}, \quad w_k = \frac{1}{R'_0(\alpha_k)} > 0, \quad \sum_k w_k = 1. \quad (\text{II.22})$$

Only uncanceled poles appear. Writing a residue as $q_{n-1}(r)/q'_n(r)$ at an original multiple or common root could give $0/0$ and is not justified; (II.22) uses the reduced polynomials instead.

12.3 The adjacent mean bound

Let $r_{n,j}$ be all the real roots of q_n , with multiplicity. The roots of Z_n are $ir_{n,j}$. Pairing roots $r, -r$ in its logarithmic derivative cancels imaginary parts and gives

$$M_n(s) := sZ'_n(s)/Z_n(s) = \sum_{j=1}^n \frac{s^2}{s^2 + r_{n,j}^2} \quad (s > 0). \quad (\text{II.23})$$

Zero roots are included. For $0 < u < 1$, the superlevel set of the summand as a function of r is the interval $(-s\sqrt{u^{-1}-1}, s\sqrt{u^{-1}-1})$. Layer cake and (II.21) therefore prove

$$|M_n(s) - M_{n-1}(s)| \leq \int_0^1 1 \, du = 1.$$

This holds at every positive fugacity and includes $n = 1$ under the empty-size conventions. The constant cannot be improved uniformly, since $M_1 - M_0 = 1$.

Integrating the adjacent bound against ds/s gives

$$\left| \log \frac{Z_{n+1}(s)}{Z_{n+1}(c)} - \log \frac{Z_n(s)}{Z_n(c)} \right| \leq |\log(s/c)| \quad (s, c > 0). \quad (\text{II.24})$$

This is the finite-size estimate needed below. It avoids alternating sums of asymptotic errors, which need not be controlled.

13 The exact solvable calibration

Let $A_N = |\text{ASM}(N)|$ denote the ordinary ASM count, with $A_0 = 1$. We will prove

$$Z_n(\sqrt{3})Z_{n+1}(\sqrt{3}) = \sqrt{3}2^n A_{n+1} \quad (n \geq 1). \quad (\text{II.25})$$

This is derived from the published adjacent determinant by exact finite transformations, followed by a classical determinant evaluation.

13.1 From the source determinant to an adjacent minor

Put $t = s^2$ and interpret $D_n(t) = Z_n(s)Z_{n+1}(s)/s$ polynomially, including at $s = 0$. Define the skew-symmetric formal kernel

$$K_t(u, v) = \frac{v - u}{1 - uv} \left(t + \frac{(1 + u)(1 + v)}{1 - u - v} \right). \quad (\text{II.26})$$

Equation (4.11) of [1], specialized to $r = 1$, states

$$D_n(t) = \det_{0 \leq i, j \leq n} \begin{cases} [u^i v^j] \frac{K_t(u, v)}{(1 + u)(1 + v)}, & 0 \leq j < n, \\ (-1)^i, & j = n. \end{cases} \quad (\text{II.27})$$

All coefficient extractions here are at $u = v = 0$.

On the first n columns, multiply coefficient series by $1 + v$. On all $n + 1$ rows, multiply by $1 + u$. Both operations are finite unitriangular operations, so preserve the determinant. The last column becomes $e_0 = (1, 0, \dots, 0)^\top$. Expansion along it gives $(-1)^n$ times the minor with entries $[u^{i+1}v^j]K_t$, $0 \leq i, j < n$. Transposition and $K_t(v, u) = -K_t(u, v)$ contribute another $(-1)^n$. The signs cancel, and therefore

$$D_n(t) = \det_{0 \leq i, j < n} ([u^i v^{j+1}] K_t(u, v)). \quad (\text{II.28})$$

13.2 Finite Pascal and Toeplitz transformations

For an $n \times n$ matrix use indices $0, \dots, n - 1$. Let S_n be the lower shift and set

$$T_n(f) = f(S_n), \quad \Pi_{ij} = \binom{i+j}{i}, \quad L_{ij} = \binom{i}{j}, \quad E = \text{diag}((-1)^i),$$

where $\binom{i}{j} = 0$ for $j > i$. Because $S_n^n = 0$, only the first n coefficients of a formal series are used. In particular, $T_n(fh) = T_n(f)T_n(h)$ is an exact finite identity.

Let $Q(z) = z^2 - z + 1$ and write

$$\begin{aligned} H_t(u, v) &= \frac{K_t(u, v) - K_t(u, 0)}{v} = \frac{\mathcal{A}_t(u)}{1 - uv} + \frac{\mathcal{B}(u)}{1 - u - v}, \\ \mathcal{A}_t(u) &= (1 - u^2) \left(t - 1 - \frac{3u}{Q(u)} \right), \\ \mathcal{B}(u) &= \frac{(u - 2)(u + 1)(2u - 1)}{(1 - u)Q(u)}. \end{aligned} \quad (\text{II.29})$$

This identity follows by putting both sides over their common denominator. The coefficient matrix of H_t is $T_n(\mathcal{A}_t) + T_n(\mathcal{B})\Pi$. Since $\mathcal{B}(0) = 2$, factoring the second Toeplitz matrix gives

$$\begin{aligned} D_n(t) &= 2^n \det(\Pi + T_n(C_t)), \\ C_t(z) &= \frac{\mathcal{A}_t(z)}{\mathcal{B}(z)} = \frac{(1 - z)^2((t - 1)(z^2 - z + 1) - 3z)}{(z - 2)(2z - 1)}. \end{aligned} \quad (\text{II.30})$$

The finite binomial identities needed next are

$$\Pi = LL^\top, \quad L^{-1}T_n(f)L = T_n\left(f\left(\frac{z}{1 + z}\right)\right), \quad L^{-1}L^{-\top} = E\Pi E. \quad (\text{II.31})$$

For clarity, these do not involve an infinite-matrix determinant. Vandermonde's identity gives the first equality. The inverse of L has entries $(-1)^{i-j}\binom{i}{j}$, which gives the third equality by the first one. The second can be verified on the monomials $f(z) = z^k$, or from the finite truncation of the binomial transform $h(z) \mapsto (1 - z)^{-1}h(z/(1 - z))$. Every coefficient of degree less than n is unchanged by this truncation.

Multiply $\Pi + T_n(C_t)$ on the left by L^{-1} and on the right by $L^{-\top}$. Both determinants are one. Then conjugate by E . Using $ET_n(f)E = T_n(f(-z))$ yields

$$\begin{aligned} D_n(t) &= 2^n \det(I + T_n(G_t)\Pi), \\ G_t(z) &= C_t\left(\frac{-z}{1 - z}\right) = \frac{(t - 4)z^2 + (4 - t)z + t - 1}{(2 - z)(1 - z)^2(1 + z)}. \end{aligned} \quad (\text{II.32})$$

All transformations are finite, and preserve every boundary row and column of the indicated sections.

13.3 The shifted determinant and its indexing

At $t = 3$, the numerator of G_t is $(2 - z)(1 + z)$, and hence $G_3(z) = (1 - z)^{-2}$. The hockey-stick identity gives

$$(T_n((1 - z)^{-2})\Pi)_{ij} = \sum_{k=0}^i (i - k + 1) \binom{k + j}{k} = \binom{i + j + 2}{i}.$$

The $q \rightarrow 1$ specialization of the shifted determinant in [5, Theorem 34, Equation (3.24)], transposed if necessary, is

$$\det_{0 \leq i, j < n} \left(\delta_{ij} + \binom{i + j + 2}{i} \right) = \prod_{1 \leq i \leq j \leq n+1} \frac{n + i + j}{2i + j - 1} = A_{n+1}. \quad (\text{II.33})$$

Here the final equality has the indicated index $n + 1$, rather than n . To verify it directly, write the double product with $N = n + 1$ as R_N . Cancellation between successive sizes gives

$$\begin{aligned}\frac{R_N}{R_{N-1}} &= \prod_{j=1}^{N-1} \frac{N-1+2j}{N-1+j} \prod_{i=1}^N \frac{2N-1+i}{N-1+2i} \\ &= \frac{(N-1)!(3N-2)!}{(2N-1)!(2N-2)!}.\end{aligned}$$

The standard ASM product formula [1, Equation (1.1)],

$$A_N = \prod_{i=0}^{N-1} \frac{(3i+1)!}{(N+i)!}, \quad (\text{II.34})$$

has the same successive ratio, and $R_1 = A_1 = 1$. This proves the final equality in (II.33) without finite-term recognition. Substituting (II.33) into (II.32) proves (II.25).

14 Logarithmic calibration without matching parity amplitudes

We require the bounded-remainder estimate

$$W_n := \log Z_n(\sqrt{3}) = \alpha n^2 + bn - \kappa \log n + O(1). \quad (\text{II.35})$$

The same coefficients will hold on both parities. No common limiting constant in the $O(1)$ remainder is needed.

Put $d_{\text{ASM}} = 5/36 = 2\kappa$. The ratio from (II.34) is

$$\frac{A_N}{A_{N-1}} = \frac{(N-1)!(3N-2)!}{(2N-1)!(2N-2)!} = \frac{\Gamma(N)\Gamma(3N-1)}{\Gamma(2N)\Gamma(2N-1)}. \quad (\text{II.36})$$

Stirling's expansion, with a uniform remainder for these four fixed shifts, gives

$$\log \frac{A_N}{A_{N-1}} = g(2N-1) - \frac{d_{\text{ASM}}}{N} + O(N^{-2}). \quad (\text{II.37})$$

Here is an explicit check of the only inverse-power coefficient needed. For $a > 0$ and fixed c , the coefficient of N^{-1} in $\log \Gamma(aN + c)$ is $(c^2 - c + 1/6)/(2a)$. For the four factors of (II.36), their signed contributions are

$$\frac{1}{12} + \frac{13}{36} - \frac{1}{24} - \frac{13}{24} = -\frac{5}{36}.$$

The coefficient of N is $3 \log 3 - 4 \log 2 = 2g$, the constant term is $-g$, and the $\log N$ terms cancel. Summing (II.37), including its summable remainder, gives

$$\log A_N = gN^2 - d_{\text{ASM}} \log N + O(1). \quad (\text{II.38})$$

Taking ratios between neighboring instances of (II.25), and using $Z_1(\sqrt{3}) = \sqrt{3}$, gives the exact formula

$$Z_{2m}(\sqrt{3}) = 2^m \prod_{k=1}^m \frac{A_{2k}}{A_{2k-1}}. \quad (\text{II.39})$$

For example, $Z_2(\sqrt{3}) = 4 = 2A_2/A_1$ is the starting case. Substitute (II.37) with $N = 2k$ and sum:

$$\begin{aligned} W_{2m} &= m \log 2 + \sum_{k=1}^m \left(g(4k-1) - \frac{d_{\text{ASM}}}{2k} + O(k^{-2}) \right) \\ &= 2gm^2 + (g + \log 2)m - \frac{5}{72} \log m + O(1). \end{aligned}$$

Replacing $2m$ by n proves (II.35) on even n .

For odd $n \geq 3$, (II.25) at $n-1$ says

$$W_n = \frac{1}{2} \log 3 + (n-1) \log 2 + \log A_n - W_{n-1}.$$

Insert (II.38) and the even formula. Since $2\alpha = g$, $2b = g + \log 2$ and $d_{\text{ASM}} = 2\kappa$, the quadratic, linear and logarithmic terms are αn^2 , bn and $-\kappa \log n$ respectively. The remaining terms are bounded. This proves (II.35) uniformly over both parities. In particular, separate bounded parity constants are fully compatible with everything used below.

15 From the finite determinant to an exact polynomial operator

The next construction gives a fixed bounded operator whose finite compressions are the determinant ratio. The order of multiplication and the exact polynomial space are essential.

15.1 The finite matrix identity

In the notation of (II.32), let

$$V_n(i, j) = \binom{i+j+2}{i}, \quad d_i = \sqrt{(i+1)(i+2)}, \quad D = \text{diag}(d_i), \quad C_n = D^{-1}V_nD.$$

Thus

$$C_n(i, j) = \frac{(i+j+2)!}{\sqrt{i!(i+2)!j!(j+2)!}}.$$

Write $A_n = D^{-1}S_nD$ for the weighted lower shift; its entry in position $(i, i-1)$ is $\sqrt{i/(i+2)}$. In this subsection only, A_n denotes this shift, not the Hardy Gram matrix introduced later. Set

$$q(z) = 2 + z - z^2, \quad r(z) = 1 - z + z^2, \quad R(z) = r(z)/q(z).$$

Since $G_t = G_3[1 + (t-3)R]$, equation (II.32) gives the exact identity

$$\frac{D_n(t)}{D_n(3)} = \frac{\det\{q(A_n)(I + C_n) + (t-3)r(A_n)C_n\}}{2^n \det(I + C_n)}. \quad (\text{II.40})$$

All functions of the nilpotent shift are finite polynomial expressions. In particular, $r(A_n)$ is invertible with determinant one, and $\det q(A_n) = 2^n$. No infinite inverse is used.

15.2 The classical Gram forms and a degree lowering operator

Let $h = \sqrt{3}$, $a = \pi/(3h)$ and define monic polynomials by

$$u_0(y) = 1, \quad u_1(y) = \frac{3}{2} - y, \quad u_{k+1}(y) = (k + \frac{3}{2} - y)u_k(y) - k(k+2)u_{k-1}(y). \quad (\text{II.41})$$

Here “monic” refers to the variable $-y$; the leading coefficient in y is $(-1)^k$. Put $h_k = k!(3)_k$ and $\phi_k = u_k/\sqrt{h_k}$. The rescaled Meixner–Pollaczek Gram identities [6, Equations (2.4)–(2.6), (3.4a), (3.6)–(3.7)] are

$$w_{\pm}(y) = \frac{3}{4\pi} e^{\pm ay} |\Gamma(3/2 + iy/h)|^2, \quad (\text{II.42})$$

$$\int_{\mathbb{R}} u_i u_j w_+ dy = h_i \delta_{ij}, \quad \int_{\mathbb{R}} u_i u_j w_- dy = (3)_{i+j}.$$

They give $C_n(i, j) = \int \phi_i \phi_j w_-$. The stated constants can also be checked from the $k = 0$ integral and the recurrence (II.41); thus the Gram matrix for $w = w_+ + w_-$ is $I + C_n$.

On polynomials define $\mathcal{L}u_k = ku_{k-1}$. Its row-action matrix in the ϕ_k basis is A_n , because $k\sqrt{h_{k-1}/h_k} = \sqrt{k/(k+2)}$. Direct substitution in the recurrence proves the exponential generating function

$$\sum_{k \geq 0} \frac{u_k(y) z^k}{k!} = (1 - z + z^2)^{-3/2} \exp(y\eta(z)), \quad \eta(z) = - \int_0^z \frac{dv}{1 - v + v^2}.$$

In finite polynomial functional calculus, therefore,

$$\mathcal{L} = \tau(\partial_y), \quad \tau(w) = \frac{-2 \sin(hw/2)}{\sqrt{3} \cos(hw/2) - \sin(hw/2)}.$$

Indeed τ is the local inverse of η , which proves this identity on the generating function and then coefficientwise. The elementary trigonometric identity

$$\frac{q(\tau(w))}{r(\tau(w))} = 1 + 2 \cos(hw + \pi/3)$$

shows that $\mathcal{K} = q(\mathcal{L})/r(\mathcal{L})$ acts exactly as

$$(\mathcal{K}f)(y) = f(y) + e^{i\pi/3} f(y + ih) + e^{-i\pi/3} f(y - ih). \quad (\text{II.43})$$

It preserves every degree space, is real on real polynomials, and has diagonal coefficient two in any degree-ordered basis.

Let ψ_j be real orthonormal polynomials for $w(y)dy$. Multiplying the numerator in (II.40) on the left by $r(A_n)^{-1}$, then applying (II.42), gives the bilinear form

$$(f, g) \longmapsto \int (\mathcal{K}f)g w dy + (t - 3) \int fg w_- dy.$$

Changing both polynomial bases from ϕ_j to ψ_j divides its determinant by $\det(I + C_n)$. We obtain, with row-action indexing,

$$\frac{D_n(t)}{D_n(3)} = 2^{-n} \det_{0 \leq i, j < n} \left\{ \int (\mathcal{K}\psi_i)\psi_j w + (t - 3) \int \psi_i \psi_j w_- \right\}. \quad (\text{II.44})$$

Every integral here is absolutely convergent. The imaginary translations act only on polynomials; no contour shift or domain assertion for unbounded operators is hidden in this formula.

15.3 An exact positive convolution inverse

The elementary identity for $|\Gamma(3/2 + iy/h)|^2$ gives

$$w(y) = \frac{p(y)}{2(2 \cosh(2ay) - 1)}, \quad p(y) = y^2 + \frac{3}{4}, \quad m(y) = \frac{w_-(y)}{w(y)} = \frac{1}{1 + e^{2ay}}. \quad (\text{II.45})$$

Set

$$\varphi(u) = \frac{1}{3(1 + 2 \cosh(2au))}. \quad (\text{II.46})$$

The bilateral Laplace integral is

$$\int_{\mathbb{R}} e^{zu} \varphi(u) du = \frac{1}{1 + 2 \cos(hz)}, \quad |\Re z| < 2a. \quad (\text{II.47})$$

For completeness, $v = e^{2au}$ reduces the integral to $(6a)^{-1} \int_0^\infty v^{z/(2a)} / (v^2 + v + 1) dv$. Partial fractions at $e^{\pm 2\pi i/3}$ and the elementary Mellin integral $\int_0^\infty v^{\rho-1} / (v + c) dv = \pi c^{\rho-1} / \sin(\pi\rho)$, first in its strip of convergence and then by continuation within the convergent difference, give (II.47). In particular, $\int \varphi = 1/3$. Define on polynomials

$$(Tf)(y) = \int_{\mathbb{R}} e^{au} \varphi(u) f(y + u) du.$$

Its kernel is positive, has mass $1/2$, and has moment generating function $[1 + 2 \cos(hw + \pi/3)]^{-1}$. Differentiating (II.47) is justified on compact substrips. The Taylor series on a polynomial terminates, so

$$T = [1 + 2 \cos(h\partial_y + \pi/3)]^{-1} = \mathcal{K}^{-1} \quad (\text{II.48})$$

exactly on each finite degree space.

Write P_n for the orthogonal projection onto degree- $< n$ polynomials in $L^2(wdy)$. Transpose the row-action matrix in (II.44) before factoring. With $\delta = t - 3$ its determinant is $\det[I + \delta M_n(\mathcal{K}_n^{\text{col}})^{-1}]$. Since T preserves the degree space, this equals

$$\frac{D_n(t)}{D_n(3)} = \det_{\text{Ran } P_n} (I + \delta P_n \mathcal{B} P_n), \quad \mathcal{B} = mT. \quad (\text{II.49})$$

Boundedness of this fixed operator is proved next. This exact ordering is $m\mathcal{K}^{-1}$; interchanging these factors or compressing an unrelated infinite inverse would not prove (II.49).

16 A controlled relative Fredholm comparison

Conjugate $L^2(wdy)$ to $L^2(dy)$ by multiplication by $\sqrt{w}$. Let Π_n project onto the span of $\sqrt{w}\psi_j$, $j < n$. The transformed kernel of $\mathcal{B}$ is

$$\tilde{\mathcal{B}}(y, x) = g_0(y) \varphi(x - y) h_0(x), \quad g_0(y) = \sqrt{w(y)} m(y) e^{-ay}, \quad h_0(x) = e^{ax} / \sqrt{w(x)}. \quad (\text{II.50})$$

Our reference is

$$C(y, x) = \mathbf{1}_{y < 0} \varphi(x - y) \mathbf{1}_{x < 0}.$$

The Fourier multiplier of convolution by φ is $[1 + 2 \cosh(h\xi)]^{-1}$. Hence

$$0 \leq C \leq I/3. \quad (\text{II.51})$$

Lemma 16.1. *For $E = \tilde{\mathcal{B}} - C$ one has*

$$E \in \mathcal{S}_2, \quad S = (E + E^*)/2 \in \mathcal{S}_1, \quad \text{tr } S = 0. \quad (\text{II.52})$$

In particular $\mathcal{B}$ is bounded and (II.49) is a bounded-operator compression identity.

Proof. We give the kernel estimates, including a trace-class criterion. Partition $\mathbb{R}$ into $I_j = [j, j+1]$. If a kernel k is $C^{2,2}$ on each closed square $I_j \times I_l$, put

$$M_{jl} = \max_{0 \leq r, s \leq 2} \sup_{I_j \times I_l} |\partial_y^r \partial_x^s k(y, x)|.$$

If $\sum_{j,l} M_{jl} < \infty$, then the operator is trace class. Indeed, expanding each block in the orthonormal cosine bases, two integrations by parts in each variable, with the boundary derivative terms retained, bound the coefficient with indices r, s by $CM_{jl}/[(1+r^2)(1+s^2)]$. The absolute coefficient sum bounds its trace norm, and the block trace norms are summable. A jump at zero is permitted because zero is an interval boundary. The same proof justifies tracing a diagonal block by integrating its kernel on the diagonal, followed by the absolutely convergent block sum.

For $y < 0$ write $r_y = e^{2ay}$, $d_y = \sqrt{1 - r_y + r_y^2}$ and $b_y = \sqrt{p(y)}$. Directly from (II.45),

$$g_0(y) = \frac{b_y}{\sqrt{2}(1+r_y)d_y}, \quad h_0(y) = \frac{\sqrt{2}d_y}{b_y}.$$

On the negative quadrant this gives

$$\tilde{\mathcal{B}}(y, x) = \frac{b_y}{b_x} \varphi(x - y) + k_e(y, x),$$

where k_e and its derivatives through order two in each variable are bounded, for suitable fixed $M, c > 0$, by

$$C(1 + |x - y|)^M e^{-c|x-y|} (e^{ay} + e^{ax}). \quad (\text{II.53})$$

To verify it, note that $d_y \geq \sqrt{3}/2$, the smooth factors in r_y, r_x differ from one by $O(r_y + r_x)$, and $b_y/b_x \leq C(1 + |x - y|)$. Differentiation preserves these types of bounds. The unit-block sum in (II.53) is finite.

The remaining difference from C has kernel $(b_y/b_x - 1)\varphi(x - y)$. Since b is 1-Lipschitz and $b_x \geq c(1 + |x|)$, its modulus is at most $C|x - y|\varphi(x - y)/(1 + |x|)$. Its square is integrable. Its symmetric part is exactly

$$\frac{(b_y - b_x)^2}{2b_y b_x} \varphi(x - y). \quad (\text{II.54})$$

The same finite collection of derivatives is bounded by

$$\frac{C(1 + |x - y|)^M e^{-c|x-y|}}{(1 + |y|)(1 + |x|)}. \quad (\text{II.55})$$

This follows by differentiating (II.54), using bounded derivatives of b , inverse powers of b , and $|b_y - b_x| \leq |y - x|$. For fixed block offset the denominators sum like $\sum_{j \geq 1} j^{-2}$; the offset sum has exponential decay. The criterion proves trace class of the symmetric part.

For $x \geq 0$, first subtract the rank-one kernel

$$\frac{1}{3} [g_0(y) e^{2ay}] [h_0(x) e^{-2ax} \mathbf{1}_{x \geq 0}]. \quad (\text{II.56})$$

Both factors are in L^2 : the first decays exponentially at both ends, and the second is $O((1+x)^{-1})$ at $+\infty$ and bounded at zero. For $u \geq 0$, $\varphi(u) - e^{-2au}/3 = O(e^{-4au})$, with the same derivative bounds; for $u < 0$ the difference is $O(e^{-2au})$. The elementary tails needed to estimate the remainder are

$$\begin{aligned} g_0(y) &= O(1 + |y|), & y \rightarrow -\infty, \\ g_0(y) &= O((1+y)e^{-4ay}), & y \rightarrow +\infty, \\ h_0(x) &= O((1+|x|)^{-1}), & x \rightarrow -\infty, \\ h_0(x) &= O(e^{2ax}/(1+x)), & x \rightarrow +\infty. \end{aligned} \tag{II.57}$$

On $x \geq 0$, $x \geq y$, the remainder after (II.56) has summable differentiated block bounds: it decays exponentially in both tails when $y < 0$, and at least exponentially in x up to polynomial factors when $y \geq 0$. On $0 \leq x < y$ it is bounded by a polynomial times $e^{-2ay}/(1+x)$, with only $O(1+y)$ blocks for each y -block. The formula is smooth across $x = y$; the split is only for estimates. Finally, on $y \geq 0$, $x < 0$, (II.57) and $\varphi(x-y) = O(e^{2a(x-y)})$ give exponential decay in both tails, including derivatives. The criterion applies on every block, including those touching zero. This proves the two ideal assertions.

The diagonal formula is legitimate by the absolutely summable cosine expansions above and the usual rank-one formula. Since $g_0(y)h_0(y) = m(y)$ and $\varphi(0) = 1/9$,

$$\operatorname{tr} S = \varphi(0) \int_{\mathbb{R}} [m(y) - \mathbf{1}_{y < 0}] dy = 0,$$

because $m(y) + m(-y) = 1$ and the difference is integrable. The first-order negative-tail error was only Hilbert–Schmidt; it is its symmetric part that has the extra decay needed for trace class. These two classes cannot be conflated. $\square$

Proposition 16.2. *For $t > 0$, set*

$$Q_n^C(t) = \det_{\operatorname{Ran} \Pi_n} (I + (t-3)\Pi_n C \Pi_n).$$

There is a positive finite factor $\mathcal{A}(t)$ such that

$$\frac{D_n(t)/D_n(3)}{Q_n^C(t)} \longrightarrow \mathcal{A}(t). \tag{II.58}$$

The convergence and normalized analytic logarithms are locally uniform for $t \in \mathbb{C} \setminus (-\infty, 0]$.

Proof. The polynomial functions are complete in $L^2(wdy)$. In fact, w has exponential moments; if f is orthogonal to every polynomial, Cauchy–Schwarz shows that $fwdy$ has an exponentially integrable absolute value in a smaller strip. Its Fourier transform is analytic there with every derivative zero at the origin. Analytic uniqueness and Fourier uniqueness imply $f = 0$. Thus $\Pi_n \rightarrow I$ strongly.

Write $C_n = \Pi_n C \Pi_n$, $E_n = \Pi_n E \Pi_n$ on the full Hilbert space, extended by zero, and similarly $S_n = \Pi_n S \Pi_n$. Then $E_n \rightarrow E$ in $\mathcal{S}_2$, $S_n \rightarrow S$ in $\mathcal{S}_1$ and $C_n \rightarrow C$ strongly. For $\delta \in \Omega := \mathbb{C} \setminus (-\infty, -3]$,

$$R_n(\delta) = (I + \delta C_n)^{-1}, \quad R(\delta) = (I + \delta C)^{-1}$$

are uniformly bounded on compact subsets of Ω , and R_n and their adjoints converge strongly. This follows from $0 \leq C_n, C \leq I/3$ and continuous functional calculus on that fixed spectral interval. Therefore $R_n E_n \rightarrow R E$ in $\mathcal{S}_2$, locally uniformly. Strong convergence combined with compact ideal approximations justifies each of these ideal products.

The exact finite factorization is

$$\frac{\det[I + \delta(C_n + E_n)]}{\det(I + \delta C_n)} = \det_2(I + \delta R_n E_n) \exp\{\delta \operatorname{tr}(R_n E_n)\}. \quad (\text{II.59})$$

Here $\det_2(I + F) = \det(I + F)e^{-\operatorname{tr} F}$ in finite dimensions, and its continuous Hilbert–Schmidt extension is used for the limit. All kernels and projections are real. Write $E = S + K$ with K real and skew-adjoint. In a real basis $R_n(\delta)$ is transpose-symmetric, even for complex δ , so

$$\operatorname{tr}(R_n \Pi_n K \Pi_n) = 0$$

exactly. Hence $\operatorname{tr}(R_n E_n) \rightarrow \operatorname{tr}(RS)$ by trace-norm convergence of S_n . Standard regularized determinant continuity [9] gives the locally uniform limit

$$\mathcal{A}(t) = \det_2\left(I + (t-3)(I + (t-3)C)^{-1}E\right) \exp\left\{(t-3) \operatorname{tr}[(I + (t-3)C)^{-1}S]\right\}. \quad (\text{II.60})$$

Nonvanishing requires a separate argument. Each ratio in (II.59) is nonzero on Ω : the numerator equals $D_n(3+\delta)/D_n(3)$ and has only negative real t -roots; the denominator is nonzero by (II.51). The limit equals one at $\delta = 0$. Hurwitz’s theorem on the connected slit domain shows that it is nowhere zero. For real $t > 0$ the finite ratios are positive, so the limit is positive and finite. Normalized analytic logarithms converge as well. Finally $\mathcal{A}(3) = 1$ and $(\log \mathcal{A})'(3) = \operatorname{tr} S = 0$. $\square$

17 The exact Jacobi projection and the phase determinant

Rescale $y = 3\sqrt{3}z$ and put $a_0 = 5/6$, $b_0 = 7/6$. Let

$$G(z) = \Gamma(a_0 + iz)\Gamma(b_0 - iz), \quad v(z) = G(z)/|G(z)|.$$

Gamma reflection and recurrence give

$$w(3\sqrt{3}z) = \frac{27}{8\pi^2}|G(z)|^2, \quad G(z) = \frac{2\pi(1/6 - iz)}{\cosh(\pi z) - i\sqrt{3}\sinh(\pi z)}. \quad (\text{II.61})$$

Thus the polynomial subspace, after unitary rescaling, is $|G|$ times polynomials in z . Use the unitary Fourier transform

$$(\mathcal{F}f)(\xi) = (2\pi)^{-1/2} \int_{\mathbb{R}} e^{-iz\xi} f(z) dz.$$

The beta integral, since $a_0 + b_0 = 2$, gives

$$G(z) = \int_{\mathbb{R}} e^{iz\xi} \frac{e^{5\xi/6}}{(1 + e^\xi)^2} d\xi, \quad \mathcal{F}G = F_0(\xi) = \sqrt{2\pi} \frac{e^{5\xi/6}}{(1 + e^\xi)^2}.$$

Multiplication by a degree- k polynomial in z becomes a differential operator of degree k . For $x = \tanh(\xi/2)$,

$$\frac{d}{d\xi}[F_0(\xi)P(x)] = F_0(\xi) \left\{ \left(-\frac{1}{6} - x\right)P(x) + \frac{1-x^2}{2}P'(x) \right\}. \quad (\text{II.62})$$

The degree-raising coefficient on degree k is $-(k+2)/2 \neq 0$. Consequently the first n derivatives span exactly F_0 times polynomials of degree $< n$ in x , with no asymptotic substitution. Since

$$|F_0(\xi)|^2 d\xi = \frac{\pi}{4}(1-x)^{4/3}(1+x)^{2/3} dx, \quad (\text{II.63})$$

the rescaling, multiplication by v , and Fourier transform carry Π_n exactly to the Jacobi projection J_n with parameters $(\alpha_J, \beta_J) = (4/3, 2/3)$.

Under the rescaling the reference is

$$C_z = \mathbf{1}_{z < 0} L \mathbf{1}_{z < 0}, \quad Lf = \varphi_z * f, \quad \varphi_z(u) = \frac{\sqrt{3}}{1 + 2 \cosh(2\pi u)}.$$

Its multiplier is

$$\sigma(\xi) = \frac{1}{1 + 2 \cosh(\xi/3)}. \quad (\text{II.64})$$

Let $H = \mathcal{F} \mathbf{1}_{z < 0} \mathcal{F}^*$ and $A = H M_\sigma H$. Then $\mathcal{F} C_z \mathcal{F}^* = A$ and $0 \leq A \leq I/3$. The actual transformed reference includes v , so it is necessary to account for this phase rather than discard it.

Lemma 17.1. *For every $\delta \in \Omega$,*

$$C_v := M_v C_z M_v^*, \quad C_v - C_z \in \mathcal{S}_1, \quad \det[(I + \delta C_v)(I + \delta C_z)^{-1}] = 1.$$

Proof. A continuous phase from (II.61) is

$$\vartheta(z) = -\arctan(6z) + \arctan(\sqrt{3} \tanh \pi z).$$

Its limiting values are $\vartheta(\pm\infty) = \mp\pi/6$, and

$$\vartheta'(z) = \frac{-6}{1 + 36z^2} + \frac{\sqrt{3}\pi \operatorname{sech}^2(\pi z)}{1 + 3 \tanh^2(\pi z)}.$$

For each fixed $j \geq 1$, $|\vartheta^{(j)}(z)| \leq C_j(1 + |z|)^{-j-1}$. On the negative half-line, integration gives

$$|\vartheta(z) - \vartheta(s)| \leq \frac{C|z - s|}{(1 + |z|)(1 + |s|)}.$$

The same paired denominator, up to a polynomial in $|z - s|$, bounds the derivatives needed for the kernel $[e^{i(\vartheta(z) - \vartheta(s))} - 1]\varphi_z(z - s)$. The trace-class unit-square criterion of Lemma 16.1 applies. The bounds hold up to zero.

For each fixed $k \geq 1$, the kernel of C_z^k and its fixed-order derivatives are bounded by a polynomial in $|z - s|$ times $e^{-c|z-s|}$, by domination of the intermediate half-line integrals with full-line convolutions. Thus $C_v^k - C_z^k = M_v C_z^k M_v^* - C_z^k$ satisfies the same trace-class criterion and has identically zero diagonal. The justified trace formula yields $\operatorname{tr}(C_v^k - C_z^k) = 0$. For $|\delta| < 3$,

$$\log \det[(I + \delta C_v)(I + \delta C_z)^{-1}] = \sum_{k \geq 1} \frac{(-1)^{k+1} \delta^k}{k} \operatorname{tr}(C_v^k - C_z^k) = 0.$$

Absolute convergence follows from $\|C_v^k - C_z^k\|_1 \leq k 3^{1-k} \|C_v - C_z\|_1$. Analytic continuation on Ω proves the assertion. $\square$

Strong convergence $J_n \rightarrow I$, trace-norm convergence of the compressed difference, and uniform resolvent bounds give

$$\frac{\det[I + \delta J_n \mathcal{F} C_v \mathcal{F}^* J_n]}{\det[I + \delta J_n A J_n]} \longrightarrow 1.$$

All determinants are on $\text{Ran } J_n$. Combining with Proposition 16.2 proves the exact reference reduction

$$\frac{D_n(t)/D_n(3)}{Q_n^J(t)} \rightarrow \mathcal{A}(t) > 0, \quad Q_n^J(t) = \det(I + (t-3)J_n H M_\sigma H J_n). \quad (\text{II.65})$$

The phase has limiting relative determinant one, not just a bounded unspecified error. The remaining Hardy projections will be handled by a finite overlap bound; they cannot yet be replaced by a scalar multiplier.

18 Endpoint controlled Jacobi means

We now prove the mean estimate required for the determinant. The symbol has angular endpoint exponent $2/3$; a smooth-test theorem cannot be used without checking this endpoint behavior. Use temporarily general fixed Jacobi parameters a_J, b_J , set $s_J = a_J + b_J$, and write

$$h_k^J = \frac{2^{s_J+1} \Gamma(k + a_J + 1) \Gamma(k + b_J + 1)}{(2k + s_J + 1) \Gamma(k + 1) \Gamma(k + s_J + 1)}. \quad (\text{II.66})$$

The real orthonormal angular functions on $I = (0, \pi)$ are

$$u_k(\theta) = \sqrt{\frac{(1 - \cos \theta)^{a_J} (1 + \cos \theta)^{b_J} \sin \theta}{h_k^J}} P_k^{(a_J, b_J)}(\cos \theta), \quad \rho_n(\theta) = \sum_{k < n} u_k(\theta)^2. \quad (\text{II.67})$$

The constant $\pi/4$ in (II.63) cancels in the normalized projection density. In the application $(a_J, b_J) = (4/3, 2/3)$.

Theorem 18.1. *For these parameters, suppose $F(\theta) = f(\cos \theta)$ is real measurable and*

$$\int_0^\pi \frac{|F(\theta)|}{\delta(\theta)} d\theta < \infty, \quad \delta(\theta) = \min(\theta, \pi - \theta). \quad (\text{II.68})$$

Then

$$\text{tr}(J_n M_f J_n) = (n+1)\mu_f + o(1), \quad \mu_f = \frac{1}{\pi} \int_0^\pi F(\theta) d\theta. \quad (\text{II.69})$$

For an unbounded test the left side means the finite quadratic-form matrix with entries $\int F u_i u_j$; it does not assert that $M_f u_i$ lies in L^2 . The actual tests below are bounded.

18.1 The normalized uniform Bessel input

Set

$$N_k = k + (s_J + 1)/2, \quad \Phi_k = N_k \theta - \pi(a_J/2 + 1/4), \\ a_1(\theta) = \frac{(1/4 - a_J^2) \cot(\theta/2) - (1/4 - b_J^2) \tan(\theta/2)}{4}.$$

The uniform Frenzen–Wong Bessel expansion [7], in the form DLMF 18.15.6–18.15.9 [8], implies

$$u_k(\theta) = \sqrt{2/\pi} \left\{ \cos \Phi_k + \frac{a_1(\theta)}{N_k} \sin \Phi_k + O((N_k \delta(\theta))^{-2}) \right\} \quad (\text{II.70})$$

uniformly where $N_k \delta \geq 1$, with $|a_1(\theta)| \leq C/\delta$. Here and below finitely many small degrees can be absorbed in the constant. We give the normalization details because a spurious $1/k$ amplitude term would change the mean.

On $0 < \theta \leq \pi/2$, angular normalization of the source formula produces a coefficient $\varkappa_k$ satisfying

$$\varkappa_k^2 = \frac{\Gamma(k + a_J + 1)\Gamma(k + s_J + 1)}{N_k^{2a_J}\Gamma(k + 1)\Gamma(k + b_J + 1)}.$$

The first relative coefficient of the two gamma ratios is $a_J(s_J + 1)/k$, exactly that of $N_k^{2a_J}/k^{2a_J}$. DLMF 5.11.13 therefore gives

$$\varkappa_k = 1 + O((k + 1)^{-2}). \quad (\text{II.71})$$

There is no $1/k$ cosine-amplitude correction. With

$$g_J(\theta) = (1/4 - a_J^2)(\cot(\theta/2) - 2/\theta) - (1/4 - b_J^2)\tan(\theta/2), \quad B_0 = g_J/(4\theta),$$

the first normalized Bessel terms are

$$\begin{aligned} \varkappa_k \left[\sqrt{N_k \theta} J_{a_J}(N_k \theta) + \frac{\theta B_0(\theta)}{N_k} \sqrt{N_k \theta} J_{a_J+1}(N_k \theta) \right. \\ \left. + \frac{A_1(\theta)}{N_k^2} \sqrt{N_k \theta} J_{a_J}(N_k \theta) \right]. \end{aligned}$$

The source coefficients B_0, A_1 are bounded on this half interval. For truncation order $M = 1$, the normalized remainder on $N_k \theta \geq 1$ is $\theta O(N_k^{-3})$. The two-term fixed-order Bessel expansion (DLMF 10.17.3) gives the sine coefficient

$$-\frac{4a_J^2 - 1}{8\theta} + \theta B_0(\theta) = a_1(\theta),$$

and all remaining terms are $O(N_k^{-2}\theta^{-2})$. Equation (II.71) preserves that error. Reflection and interchange of parameters prove (II.70) on the other half interval.

The same source with $M = 0$ gives, uniformly in k, θ ,

$$|u_k(\theta)| \leq C, \quad |u_k(\theta)| \leq C((k + 1)\theta)^{a_J+1/2} \quad ((k + 1)\theta \leq 1), \quad (\text{II.72})$$

and the reflected bound with b_J at π . This uses the boundedness of $\sqrt{r}J_{a_J}(r)$ for $r \geq 1$ and its small-argument power. The normalized remainder in the small-argument region is $\theta^{a_J+2}O(N_k^{a_J-2M+1/2})$, which is dominated by the stated endpoint bound. The families required below are the base pair, its reflection, the simultaneous shift $(7/3, 5/3)$ and its reflection, and the single shifts $(7/3, 2/3)$ and $(5/3, 4/3)$. All parameters exceed $-1/2$, their sums exceed -1 , and their differences exceed -2 in each orientation needed. Thus the Frenzen–Wong Main Theorem condition $a_J - b_J > -2m$ holds already with truncation order $m = 1$. No shifted family is silently identified with the base family.

18.2 Global density bounds and the interior constant

We first show

$$\rho_n(\theta) \leq Cn, \quad |\rho_n(\theta) - n/\pi| \leq C/\delta(\theta). \quad (\text{II.73})$$

The first follows from (II.72). For the second, choose $m \asymp 1/\delta$ so that (II.70) applies to $k \geq m$. The first m terms contribute $O(1/\delta)$, or contain all terms. Squaring the expansion on the remaining terms gives

$$u_k^2 = \frac{1}{\pi} + \frac{\cos(2\Phi_k)}{\pi} + \frac{2a_1(\theta)}{\pi N_k} \sin(2\Phi_k) + O(N_k^{-2}\delta^{-2}).$$

The cosine sum is $O(1/|\sin \theta|) = O(1/\delta)$. Abel summation with the same geometric bound gives

$$\left| \sum_{k=m}^{n-1} \frac{e^{2ik\theta}}{N_k} \right| \leq \frac{C}{N_m |\sin \theta|}.$$

After multiplication by $a_1 = O(1/\delta)$ this is $O(1/\delta)$; the squared-error sum is also $O(1/(m\delta^2)) = O(1/\delta)$. If $\delta < 1/n$, the first inequality of (II.73) suffices. Summing the sine correction absolutely instead would lose its cancellation and introduce an unjustified logarithm.

For the constant term restrict now to a fixed $[\epsilon, \pi - \epsilon]$. The exact derivative identity is

$$u'_n(\theta) = q_J(\theta)u_n(\theta) - \sqrt{n(n+s_J+1)} u_{n-1}^{(a_J+1, b_J+1)}(\theta), \quad (\text{II.74})$$

where

$$q_J(\theta) = \frac{(2a_J+1)\cot(\theta/2) - (2b_J+1)\tan(\theta/2)}{4}.$$

This follows from DLMF 18.9.15 and the exact norm ratio in (II.66). The shifted function has frequency N_n , phase $\Phi_n - \pi/2$, and correction $a_1^{\text{shifted}} + q_J = a_1$. Applying (II.70) to the two families in (II.74) gives

$$u'_n = \sqrt{2/\pi} [-N_n \sin \Phi_n + a_1(\theta) \cos \Phi_n + O_\epsilon(n^{-1})]. \quad (\text{II.75})$$

We have used an exact derivative identity, rather than differentiated an uncontrolled asymptotic remainder. The orthonormal recurrence coefficient is

$$r_n^J = \frac{2}{2n+s_J} \sqrt{\frac{n(n+a_J)(n+b_J)(n+s_J)}{(2n+s_J-1)(2n+s_J+1)}} = \frac{1}{2} + O(n^{-2}).$$

The confluent Christoffel–Darboux formula is

$$\rho_n(\theta) = \frac{r_n^J}{\sin \theta} (u_n u'_{n-1} - u'_n u_{n-1}).$$

Write $N = N_n$, $\Phi = \Phi_n$. Substitution of (II.70) and (II.75) yields

$$u_n u'_{n-1} - u'_n u_{n-1} = \frac{2}{\pi} \left[(N - \tfrac{1}{2}) \sin \theta + \tfrac{1}{2} \sin(2\Phi - \theta) \right] + O_\epsilon(n^{-1}).$$

The cosine-cosine first corrections cancel; the sine-sine correction has factor $N/(N-1) - (N-1)/N = O(n^{-1})$. Thus

$$\rho_n(\theta) = \frac{n+s_J/2}{\pi} + \frac{\sin((2n+s_J)\theta - \pi(a_J+1/2))}{2\pi \sin \theta} + O_\epsilon(n^{-1}). \quad (\text{II.76})$$

Here $s_J = 2$, so the nonoscillatory constant is $(n+1)/\pi$.

Proof of Theorem 18.1. The trace is exactly $\int F \rho_n$. For fixed $\epsilon > 0$, (II.76) and the Riemann–Lebesgue lemma show

$$\int_{\epsilon}^{\pi-\epsilon} F(\theta) [\rho_n(\theta) - (n+1)/\pi] d\theta \longrightarrow 0.$$

The oscillatory multiplier $F/\sin \theta$ is integrable on that compact interval. By (II.73), the absolute contribution of the two omitted endpoints is at most

$$C \int_{\delta < \epsilon} |F(\theta)|/\delta(\theta) d\theta$$

uniformly in n . Condition (II.68) makes this tend to zero. First taking $n \rightarrow \infty$, then $\epsilon \rightarrow 0$, proves the result. The bounded angular basis also proves that the finite quadratic-form entries are well defined. $\square$

18.3 Exact scalar means

Under $x = \cos \theta = \tanh(\xi/2)$ one has

$$w(\theta) := \sigma(\xi(\theta)) = \frac{1}{1 + \cot^{2/3}(\theta/2) + \tan^{2/3}(\theta/2)}, \quad \xi(\theta) = 2 \log \cot(\theta/2). \quad (\text{II.77})$$

Thus $0 \leq w \leq 1/3$ and $w = O(\delta^{2/3})$ at both endpoints. The notation $w(\theta)$ here is the angular symbol, not the earlier real-line polynomial weight $w(y)$. Let

$$V(\theta) = \log(1 - 2w(\theta)), \quad -\log 3 \leq V \leq 0.$$

Since $|V| \leq 6w$, both tests satisfy (II.68). Their means are

$$\mu_w = 2 - \sqrt{3}, \quad \mu_V = 2 \log(2/3). \quad (\text{II.78})$$

Here is an elementary integral verification independent of any DSASM pressure theorem. Define, only as a scalar integral,

$$J(t) = \frac{1}{\pi} \int_0^\pi \log[1 + (t-3)w(\theta)] d\theta, \quad t > 0.$$

With $v = \tan^{1/3}(\theta/2)$, differentiation gives

$$J'(t) = \frac{6}{\pi} \int_0^\infty \frac{v^4 dv}{(1+v^6)(v^4 + (t-2)v^2 + 1)}.$$

For $t \neq 1, 4$ the rational integrand equals

$$\begin{aligned} & -\frac{1}{3(t-4)(v^2+1)} + \frac{v^2+1}{3(t-1)(v^4-v^2+1)} \\ & + \frac{v^2+1}{(t-1)(t-4)(v^4+(t-2)v^2+1)}. \end{aligned}$$

Using $\int_0^\infty (1+v^2)/(v^4+cv^2+1) dv = \pi/\sqrt{c+2}$ for $c > -2$, which follows by the change $v \rightarrow 1/v$ or by elementary partial fractions, we obtain

$$J'(t) = \frac{\sqrt{t}+3}{\sqrt{t}(\sqrt{t}+1)(\sqrt{t}+2)}.$$

The removable cases $t = 1, 4$ follow by continuity of the original integral. Since $J(3) = 0$, integration yields

$$J(t) = 2 \log \frac{(\sqrt{t} + 1)^2}{2(\sqrt{t} + 2)}. \quad (\text{II.79})$$

Evaluating $J'(3)$ and $J(1)$ proves (II.78). In particular,

$$\text{tr}(J_n M_\sigma J_n) = (n+1)(2 - \sqrt{3}) + o(1), \quad \text{tr}(J_n M_V J_n) = 2(n+1) \log(2/3) + o(1). \quad (\text{II.80})$$

19 The scalar determinant has a bounded remainder

We prove

$$\log T_n = 2n \log(2/3) + O(1), \quad T_n = \det_{\text{Ran } J_n} (J_n e^{M_V} J_n) = \det(I - 2J_n M_\sigma J_n). \quad (\text{II.81})$$

The proof combines a uniform endpoint kernel estimate with an exact finite-dimensional exponential-tilt identity. It does not require asymptotics for the tilted Jacobi weight.

19.1 Adjacent cancellation and a global kernel bound

For the base Jacobi pair, the reflected and reindexed contiguous identity DLMF 18.9.6 is

$$\begin{aligned} (2n + s_J)(1 - x)P_{n-1}^{(a_J+1, b_J)}(x) \\ = 2(n + a_J)P_{n-1}^{(a_J, b_J)}(x) - 2nP_n^{(a_J, b_J)}(x). \end{aligned}$$

After angular normalization,

$$u_n = a_n^* u_{n-1} - b_n^* \sqrt{1 - \cos \theta} u_{n-1}^{(a_J+1, b_J)},$$

where direct gamma cancellation gives

$$(a_n^*)^2 = \frac{(n + a_J)(n + s_J)(2n + s_J + 1)}{n(n + b_J)(2n + s_J - 1)}, \quad (b_n^*)^2 = \frac{(2n + s_J)(2n + s_J + 1)}{2n(n + b_J)}. \quad (\text{II.82})$$

Hence $a_n^* - 1 = O(1/n)$ and $b_n^* = O(1)$. Since $a_J + 1/2 \geq 1$, (II.72) gives $|u_{n-1}(\theta)|/n \leq C\theta$ on the left half interval: use boundedness for $\theta \geq 1/n$ and the endpoint power for $\theta < 1/n$. The shifted family is uniformly bounded. Thus

$$|u_n(\theta) - u_{n-1}(\theta)| \leq C\theta, \quad |u_n(\theta) + u_{n-1}(\theta)| \leq C(\pi - \theta). \quad (\text{II.83})$$

The second follows by Jacobi reflection and $b_J + 1/2 \geq 1$. Uniform boundedness extends both inequalities over all of I .

The angular Christoffel–Darboux kernel is

$$K_n(\theta, \phi) = \sum_{k < n} u_k(\theta) u_k(\phi) = r_n^J \frac{u_n(\theta) u_{n-1}(\phi) - u_{n-1}(\theta) u_n(\phi)}{\cos \theta - \cos \phi}.$$

Using first the difference, then the sum in (II.83), the numerator is bounded by both $C(\theta + \phi)$ and $C(2\pi - \theta - \phi)$. Meanwhile

$$|\cos \theta - \cos \phi| \geq c \min(\theta + \phi, 2\pi - \theta - \phi) |\theta - \phi|.$$

The recurrence coefficient is bounded, so

$$|K_n(\theta, \phi)| \leq C/|\theta - \phi| \quad (\theta \neq \phi), \quad (\text{II.84})$$

uniformly in n on the entire square. Endpoint cancellation, rather than extrapolation of a bulk estimate, proves this bound.

19.2 Uniform variance and the exact tilted projection

The symbol V is globally Hölder of exponent $\gamma = 2/3$. Indeed on the left half interval $w = r/(1+r+r^2)$ with $r = \tan^{2/3}(\theta/2)$; the derivative in $r \in [0, 1]$ is bounded and $r' = O(\theta^{-1/3})$. Reflection and $|d \log(1 - 2w)/dw| \leq 6$ give $|V(\theta) - V(\phi)| \leq C|\theta - \phi|^{2/3}$. For the angular projection $P = J_n$ the exact variance identity is

$$\begin{aligned} \mathcal{V}_n &= \|(I - P)VP\|_{\text{HS}}^2 = \text{tr}(PV^2P) - \text{tr}((PVP)^2) \\ &= \frac{1}{2} \int_I \int_I (V(\theta) - V(\phi))^2 |K_n(\theta, \phi)|^2 d\theta d\phi. \end{aligned} \quad (\text{II.85})$$

By (II.84), the integrand is at most $C|\theta - \phi|^{-2/3}$, which is integrable. Therefore

$$\sup_n \mathcal{V}_n < \infty. \quad (\text{II.86})$$

For $0 \leq u \leq 1$, set on $\text{Ran } P$

$$T(u) = Pe^{uV}P, \quad F(u) = \log \det T(u), \quad E_u = e^{uV/2}, \quad R_u = E_u P T(u)^{-1/2}, \quad Q_u = R_u R_u^*.$$

All inverses are finite-dimensional. Since $R_u^* R_u = I$, Q_u is the orthogonal projection onto $E_u \text{Ran } P$. Differentiating the finite determinant, and using cyclicity, gives exactly

$$F'(u) = \text{tr}(Q_u V), \quad F''(u) = \text{tr}(Q_u V^2) - \text{tr}(Q_u V Q_u V) = \|(I - Q_u)V R_u\|_{\text{HS}}^2 \geq 0. \quad (\text{II.87})$$

The crucial ordered identity is

$$(I - Q_u)V R_u = (I - Q_u)E_u(I - P)VP T(u)^{-1/2}.$$

It uses only $VE_u = E_u V$ and $(I - Q_u)E_u P = 0$. If $m \leq V \leq M$, the operator norms therefore yield

$$F''(u) \leq e^{u(M-m)} \mathcal{V}_n.$$

Here $m = -\log 3$, $M = 0$, so $F''(u) \leq 3^u \mathcal{V}_n$. Taylor's integral remainder, $F(0) = 0$, and $F'(0) = \text{tr}(PVP)$ give

$$0 \leq \log \det(Pe^V P) - \text{tr}(PVP) \leq \frac{2 - \log 3}{(\log 3)^2} \mathcal{V}_n = O(1). \quad (\text{II.88})$$

No commutation of P with the multiplier or its exponential has been assumed. Combining (II.88) with (II.80) proves (II.81).

20 A uniform weighted Hardy overlap bound

Return to the exact projection J_n in $L^2(d\xi)$ and put

$$A = HM_\sigma H, \quad B = (I - H)M_\sigma(I - H), \quad D = A + B, \quad (\text{II.89})$$

$$A_n = J_n A J_n, \quad B_n = J_n B J_n$$

on $\text{Ran } J_n$. These are now the Hardy Gram matrices; the weighted shift of Section 15 is no longer used. Although $AB = 0$ in the ambient space, the compressed matrices need not commute or have zero product. We prove the estimate which retains precisely this finite-projection interaction.

Theorem 20.1. *The exact compressed matrices satisfy*

$$\sup_{n \geq 1} \text{tr}(A_n B_n) < \infty. \quad (\text{II.90})$$

20.1 Angular Hardy coordinates and weighted wave comparison

The map

$$(U_{\text{ang}} f)(\theta) = \sqrt{2/\sin \theta} f(2 \log \cot(\theta/2))$$

is unitary from $L^2(d\xi)$ to $L^2(I, d\theta)$. Write $H_\theta = U_{\text{ang}} H U_{\text{ang}}^*$, $w(\theta) = \sigma(\xi(\theta))$, and $s(\theta) = \sqrt{w(\theta)}$. By (II.77),

$$0 \leq w \leq 1/3, \quad w(\theta) \leq C\delta(\theta)^\gamma, \quad \gamma = 2/3. \quad (\text{II.91})$$

Let $U : \ell^2(\mathbb{N}_0) \rightarrow L^2(I)$ synthesize the real Jacobi basis: $Ue_k = u_k$. Completeness makes it unitary. Define

$$\begin{aligned} N_k &= k + 3/2, \quad c = 11\pi/12, \quad v_k = \sqrt{2/\pi} \cos(N_k \theta - c), \\ v_k^- &= (2\pi)^{-1/2} e^{-iN_k \theta + ic}, \quad v_k^+ = (2\pi)^{-1/2} e^{iN_k \theta - ic}. \end{aligned} \quad (\text{II.92})$$

Let V_-, V_+ be the corresponding synthesis maps and $V = V_- + V_+$. Each one-sided family is the restriction of an orthogonal Fourier family on an interval of length 2π , multiplied by one fixed unimodular factor. Consequently $\|V_\pm\| \leq 1$ and $\|V\| \leq 2$. Orthogonality on I itself is neither asserted nor required.

Equations (II.70) and (II.72) give

$$|u_k - v_k| \leq C \min\{1, ((k+1)\delta)^{-1}\}.$$

With $\gamma = 2/3$, integration using (II.91) therefore gives

$$\begin{aligned} \|s(u_k - v_k)\|_2^2 &\leq C \left\{ \int_0^{1/(k+1)} r^\gamma dr + (k+1)^{-2} \int_{1/(k+1)}^{\pi/2} r^{\gamma-2} dr \right\} \\ &\leq C(k+1)^{-1-\gamma}. \end{aligned}$$

The column squares sum, so

$$M_s(U - V) \in \mathcal{S}_2. \quad (\text{II.93})$$

The unweighted map $U - V$ is separately bounded.

20.2 The coordinate defect has a weighted square integrable kernel

Let C_I be the compression to I of the negative-frequency Hardy projection on $L^2(\mathbb{R}, d\theta)$. It is a positive contraction, not generally a projection on $L^2(I)$. Set $f(\theta) = \log \tan(\theta/2)$, so $f' = 1/\sin \theta$ and $\xi = -2f$. The Fourier convention gives

$$\begin{aligned} C_I &= \frac{1}{2}I - \frac{i}{2\pi} \text{PV} \frac{1}{\theta - \phi}, \\ H_\theta &= \frac{1}{2}I - \frac{i}{2\pi} \text{PV} \frac{\sqrt{f'(\theta)f'(\phi)}}{f(\theta) - f(\phi)}. \end{aligned} \quad (\text{II.94})$$

These are kernel formulas with respect to $d\phi$. To check the sign, the original H is $\frac{1}{2}I + (i/2\pi) \text{PV}(\xi - \eta)^{-1}$; the decreasing coordinate map contributes the displayed minus sign and cancels its factor two against the square-root Jacobians. Locally H_θ thus selects the negative-frequency branch v_k^- . The transformed principal value is the ordinary angular principal value on compactly supported smooth inputs: the ratio of the two angular exclusion radii induced by a symmetric ξ exclusion tends to one. Its possible diagonal residual is proportional to the logarithm of that ratio, hence zero.

Define the diagonal-subtracted kernel

$$d(\theta, \phi) = \frac{\sqrt{f'(\theta)f'(\phi)}}{f(\theta) - f(\phi)} - \frac{1}{\theta - \phi}.$$

We claim the global row-square estimate

$$\int_I |d(\theta, \phi)|^2 d\phi \leq C/\delta(\theta). \quad (\text{II.95})$$

Here is a direct proof covering same and opposite endpoints at once. Fix $a = \delta(\theta)$ and split at $|\phi - \theta| < a/2$. On this inner interval, $f' \asymp 1/a$, $|f''| \leq C/a^2$, and Taylor comparison of the endpoint geometric mean of f' with its interval average bounds the difference of the two Cauchy kernels by C/a . Its row-square contribution is $O(1/a)$; the singularity on the diagonal cancels. Outside this interval, the ordinary Cauchy kernel has squared integral $O(1/a)$. For the transformed kernel, substitute $\eta = f(\phi)$: its squared row integral is

$$f'(\theta) \int_{\text{outside image}} \frac{d\eta}{|f(\theta) - \eta|^2}.$$

The excluded inner interval maps to an interval containing a fixed radius neighborhood of $f(\theta)$, because $f' \asymp 1/a$ there. The last integral is uniformly bounded, and $f'(\theta) \leq C/a$. This proves (II.95), including opposite endpoint interactions and all transition regions. Since $\int_I w/\delta < \infty$, it follows that

$$M_s(H_\theta - C_I) \in \mathcal{S}_2. \quad (\text{II.96})$$

The singular-integral identities first hold on smooth compactly supported inputs; the actual weighted Hilbert–Schmidt kernel and boundedness of the operators extend the identity to all L^2 inputs.

20.3 The noncommuting multiplier and the interval tails

We also need

$$[M_s, H_\theta] \in \mathcal{S}_2. \quad (\text{II.97})$$

Undoing angular coordinates, the multiplier is $s_0(\xi) = (1 + 2 \cosh(\xi/3))^{-1/2}$, with $s_0, s'_0 \in L^2(\mathbb{R})$. Its Hardy commutator has, up to a constant, kernel $(s_0(\xi) - s_0(\eta))/(\xi - \eta)$. For $|r| < 1$,

$$\|s_0(\cdot + r) - s_0\|_2^2 \leq r^2 \|s'_0\|_2^2,$$

and for $|r| \geq 1$ it is at most $4\|s_0\|_2^2$. Integrating after division by r^2 proves (II.97). Hence the ordered identity

$$M_s H_\theta(U - V) = H_\theta[M_s(U - V)] + [M_s, H_\theta](U - V) \in \mathcal{S}_2 \quad (\text{II.98})$$

is valid. A weighted bound on $U - V$ alone would not permit the weight to pass through the Hardy projection.

The full-line negative-frequency projection selects $e^{-iN\theta}$ and kills $e^{iN\theta}$ for $N > 0$. After compression to I , subtraction of the omitted tails gives

$$|C_I e^{-iN\theta} - e^{-iN\theta}| + |C_I e^{iN\theta}| \leq C\{L(N\theta) + L(N(\pi - \theta))\}, \quad (\text{II.99})$$

where $L(r) = 1 + |\log r|$ for $0 < r \leq 1$, and $L(r) = 1/r$ for $r \geq 1$. The tails are multiples of $\int_r^\infty e^{\pm iu} du/u$. Integration by parts bounds them by $2/r$ for $r \geq 1$; splitting at one gives $|\log r| + 2$ otherwise. The full-line wave identity can be justified by Abel damping before this subtraction. For $0 < \gamma < 1$, $\int_0^\infty r^\gamma L(r)^2 dr < \infty$; using (II.91) in (II.99) gives

$$\|s(C_I v_k - v_k^-)\|_2^2 \leq C(k+1)^{-1-\gamma}.$$

Therefore

$$M_s(C_I V - V_-) \in \mathcal{S}_2. \quad (\text{II.100})$$

20.4 Fixed Hilbert Schmidt errors and the finite cutoff

Define bounded maps from the coefficient space by

$$X = M_s H_\theta U, \quad Y = M_s(I - H_\theta)U, \quad X_0 = M_s V_-, \quad Y_0 = M_s V_+.$$

Combining (II.96), (II.98), and (II.100),

$$X - X_0 = M_s H_\theta(U - V) + M_s(H_\theta - C_I)V + M_s(C_I V - V_-) \in \mathcal{S}_2.$$

Also $Y - Y_0 = M_s(U - V) - (X - X_0) \in \mathcal{S}_2$. These are fixed operators on the full coefficient space, with finite norms independent of n . Let P_n^0 be the first- n coordinate projection in $\ell^2(\mathbb{N}_0)$. The exact finite matrices are

$$A_n = P_n^0 X^* X P_n^0, \quad B_n = P_n^0 Y^* Y P_n^0.$$

Finite-rank cyclicity gives

$$\text{tr}(A_n B_n) = \|X P_n^0 Y^*\|_{\text{HS}}^2. \quad (\text{II.101})$$

Every occurrence of the finite cutoff is retained.

The model operator $X_0 P_n^0 Y_0^*$ has kernel

$$\frac{s(\theta)s(\phi)}{2\pi} e^{2ic - (3i/2)(\theta + \phi)} \sum_{k=0}^{n-1} e^{-ik(\theta + \phi)}.$$

The geometric sum is at most $1/|\sin((\theta + \phi)/2)|$, so

$$\|X_0 P_n^0 Y_0^*\|_{\text{HS}}^2 \leq \frac{1}{4\pi^2} \int_I \int_I \frac{w(\theta)w(\phi)}{\sin^2((\theta + \phi)/2)} d\theta d\phi < \infty \quad (\text{II.102})$$

uniformly in n . The only singular corners are $(0, 0)$ and (π, π) . At the first the integrand is at most $C\theta^\gamma \phi^\gamma / (\theta + \phi)^2$. The change $\theta = ru$, $\phi = r(1 - u)$ leaves the factors $r^{2\gamma-1} dr$ and $u^\gamma (1 - u)^\gamma du$, both integrable. Reflection handles the second corner. Opposite corners are nonsingular because their sum is near π . Finally,

$$X P_n^0 Y^* = X_0 P_n^0 Y_0^* + (X - X_0) P_n^0 Y^* + X_0 P_n^0 (Y - Y_0)^*.$$

The last two terms have uniformly bounded Hilbert–Schmidt norms by the ideal property. Equation (II.102) then proves a uniform bound on (II.101), completing Theorem 20.1.

21 The determinant transfer and the first calibration trace

We now combine the estimates, keeping the finite matrices noncommuting throughout.

21.1 Complementary Hardy pieces and their cross term

Because the two summands of D in (II.89) have orthogonal ambient support,

$$0 \leq D \leq I/3, \quad A_n + B_n \leq I/3.$$

The difference $M_\sigma - D$ is trace class. Undoing the Fourier transform, it consists of the two opposite-half-line blocks of convolution by φ_z . There $|z - s| = |z| + |s|$, so its kernel and all fixed derivatives decay exponentially in both variables. The unit-square criterion applies including the squares touching zero. Put

$$S_n^J = \det(I - 2(A_n + B_n)), \quad T_n = \det(I - 2J_n M_\sigma J_n).$$

Strong convergence $J_n \rightarrow I$, trace-norm convergence of the compressed cross term and the uniform resolvent bound give

$$\frac{S_n^J}{T_n} \longrightarrow \Xi = \det[(I - 2D)(I - 2M_\sigma)^{-1}] > 0. \quad (\text{II.103})$$

The limiting relative operator is invertible, so its determinant is nonzero; positivity follows from the positive finite ratios. The cross kernel has zero diagonal, but that alone does not make the relative determinant equal to one.

Complex conjugation in the real Jacobi basis interchanges H and $I - H$, and commutes with J_n and M_σ . Thus A_n, B_n have the same eigenvalues and

$$\det(I - 2A_n) = \det(I - 2B_n) = Q_n^J(1).$$

For $0 \leq s, u \leq 1$ define

$$F(s, u) = \log \det[I - 2sA_n - 2uB_n], \quad R_{su} = (I - 2sA_n - 2uB_n)^{-1}.$$

The matrices are positive and $\|R_{su}\| \leq 3$. Direct finite matrix differentiation gives

$$\partial_s \partial_u F = -4 \operatorname{tr}(R_{su} A_n R_{su} B_n) = -4 \|A_n^{1/2} R_{su} B_n^{1/2}\|_{\text{HS}}^2.$$

Consequently

$$\Delta_n := 2 \log Q_n^J(1) - \log S_n^J = 4 \int_0^1 \int_0^1 \|A_n^{1/2} R_{su} B_n^{1/2}\|_{\text{HS}}^2 ds du \geq 0. \quad (\text{II.104})$$

To bound this without an invalid noncommutative trace estimate, put $C_s = I - 2sA_n$ and use the resolvent equation. It gives exactly

$$A_n^{1/2} R_{su} B_n^{1/2} = C_s^{-1} A_n^{1/2} B_n^{1/2} [I + 2u B_n^{1/2} R_{su} B_n^{1/2}]. \quad (\text{II.105})$$

The two outside factors have norm at most three. The squared Hilbert–Schmidt norm is thus at most $81 \operatorname{tr}(A_n B_n)$, and

$$0 \leq \Delta_n \leq 324 \operatorname{tr}(A_n B_n) = O(1). \quad (\text{II.106})$$

It would in general be unjustified simply to bound $\operatorname{tr}(R A R A')$ by $\|R\|^2 \operatorname{tr}(A A')$ for arbitrary positive noncommuting matrices; (II.105) supplies the needed ordered factorization.

By (II.81), (II.103), and (II.106),

$$\log Q_n^J(1) = n \log(2/3) + O(1).$$

The positive relative limit (II.65) now proves (II.6). Both the lower and upper bound have been established; no infinite-rank substitution was made in the overlap estimate.

21.2 The constant correction in the first trace

Let L be convolution by φ_z and $C_- = \mathbf{1}_{z < 0} L \mathbf{1}_{z < 0}$, $C_+ = \mathbf{1}_{z > 0} L \mathbf{1}_{z > 0}$. In the positive $|G|$ gauge the polynomial projection, denoted here $\widehat{\Pi}_n$, is invariant under $z \mapsto -z$, since its weight is even. The kernel of L is even, so

$$\operatorname{tr}(\widehat{\Pi}_n C_- \widehat{\Pi}_n) = \operatorname{tr}(\widehat{\Pi}_n C_+ \widehat{\Pi}_n).$$

The cross operator $L - C_- - C_+$ is trace class by the same opposite half-line argument and has trace zero. Strong projection convergence gives

$$\operatorname{tr}(\widehat{\Pi}_n C_- \widehat{\Pi}_n) - \frac{1}{2} \operatorname{tr}(\widehat{\Pi}_n L \widehat{\Pi}_n) \longrightarrow 0. \quad (\text{II.107})$$

The full-line phase difference $L_v - L = M_v L M_v^* - L$ is also trace class with trace zero. On the two same-sign quadrants this follows from the phase bounds of Lemma 17.1; on opposite-sign quadrants exponential decay controls both tails. Its kernel has zero diagonal, and the same criterion justifies the trace formula. The Fourier-phase transformation of the exact projection therefore gives

$$\operatorname{tr}(\widehat{\Pi}_n L \widehat{\Pi}_n) - \operatorname{tr}(J_n M_\sigma J_n) \longrightarrow 0.$$

Finally (II.49) implies $D'_n(3)/D_n(3) = \operatorname{tr}(\Pi_n \tilde{\mathcal{B}} \Pi_n)$. The skew part of E has zero finite compressed trace in the real basis, while $\operatorname{tr}(\Pi_n S \Pi_n) \rightarrow \operatorname{tr} S = 0$. Combining this with (II.107) yields

$$\frac{D'_n(3)}{D_n(3)} - \frac{1}{2} \operatorname{tr}(J_n M_\sigma J_n) \longrightarrow 0.$$

Equation (II.80) proves (II.7). This exact first-trace centering does not by itself control all trace powers or imply a determinant amplitude.

21.3 From adjacent products to individual counts

Let $c = \sqrt{3}$, $f_1 = \log(2/3)$ and $R_n = \log[Z_n(1)/Z_n(c)]$. The definition (II.5) and (II.6) give

$$R_n + R_{n+1} = nf_1 + O(1) - \frac{1}{2} \log 3.$$

By (II.24), $|R_{n+1} - R_n| \leq \frac{1}{2} \log 3$. Subtracting this bounded difference from the sum proves

$$R_n = \frac{n}{2} \log(2/3) + O(1).$$

Adding the all-size calibration (II.35) and using $b + \frac{1}{2} \log(2/3) = \beta$ proves (II.4). Exponentiation gives (II.3) for all sufficiently large n ; positivity of finitely many smaller counts permits enlargement of the constants to cover every $n \geq 1$. In particular,

$$0 < \liminf_{n \rightarrow \infty} \frac{a_n}{e^{\alpha n^2 + \beta n} n^{-5/72}} \leq \limsup_{n \rightarrow \infty} \frac{a_n}{e^{\alpha n^2 + \beta n} n^{-5/72}} < \infty.$$

22 Integer inverse thresholds and the logarithmic correction

The identity $Z_n(2) = a_{n+1}$ from [1, Proposition 5.1], with nonnegative fugacity coefficients, implies $a_{n+1} = Z_n(2) \geq Z_n(1) = a_n$. Alternatively, adjoining an isolated +1 corner gives a direct injection. Theorem 10.1 implies $a_n \rightarrow \infty$, so $N(T)$ is well defined.

Put

$$F(x) = \alpha x^2 + \beta x - \kappa \log x, \quad d(r) = \frac{\kappa \log r + \beta^2/(4\alpha)}{2\alpha}, \quad z = r - \frac{\beta}{2\alpha} + \frac{d(r)}{r}.$$

The polynomial terms cancel exactly:

$$\alpha z^2 + \beta z = y + \kappa \log r + \alpha d(r)^2/r^2.$$

Since $\log z = \log r - \beta/(2\alpha r) + O(\log r/r^2)$,

$$F(z) = y + O(1/r), \quad F'(z + O(1/r)) = 2\alpha r + O(\log r/r). \quad (\text{II.108})$$

Thus for fixed C ,

$$F(z \pm C/r) = y \pm 2\alpha C + o(1).$$

Choose M, n_0 such that $|\log a_n - F(n)| \leq M$ for $n \geq n_0$. Taking $C > M/(2\alpha) + 1$ and enlarging the onset gives

$$F(z - C/r) + M < y, \quad F(z + C/r) - M > y.$$

The function F is eventually strictly increasing. Every integer $n < z - C/r$ then satisfies $a_n < T$, while $n = \lceil z + C/r \rceil$ satisfies $a_n \geq T$. Finitely many $n < n_0$ are excluded by enlarging T_0 . This proves (II.9).

Expressed in y alone, the center is

$$\begin{aligned} z(T) &= \sqrt{y/\alpha} - \frac{\beta}{2\alpha} + \frac{\kappa}{4\sqrt{\alpha}} \frac{\log y}{\sqrt{y}} \\ &\quad + \left(\frac{\beta^2}{8\alpha^{3/2}} - \frac{\kappa \log \alpha}{4\sqrt{\alpha}} \right) \frac{1}{\sqrt{y}}. \end{aligned} \quad (\text{II.109})$$

The unrounded bracket has width $O(y^{-1/2})$ and eventually yields at most two adjacent integer candidates. When both ceilings agree, they determine $N(T)$. Thresholds arbitrarily close to a count are covered by retaining both ceilings. The displayed fixed $y^{-1/2}$ term is a choice of center: an unknown bounded remainder in (II.4) allows an additional unknown correction of this same order in a continuous inverse. A limiting amplitude would be needed to identify the full constant at that order. The constants C, T_0 are not effective here, so this is an asymptotic localization theorem, not a finite numerical certification procedure.

[write] **The amplitude at order $y^{-1/2}$, 5 October 2026.** The limiting amplitude that this paragraph says would be needed is supplied by Part I: the full constant at order $y^{-1/2}$ is that of (83). The constants remain non-effective.

23 Bounded holomorphic pressure and diagonal cumulants

The preceding real determinant proof admits a useful stronger form. Boundedness on a real interval alone would not justify differentiating an error term. We supply the extra complex estimates explicitly. Define on the slit plane $\Omega_t = \mathbb{C} \setminus (-\infty, 0]$

$$f(t) = 2 \operatorname{Log}(\sqrt{t} + 1) - \operatorname{Log}(\sqrt{t} + 2) - \log 2, \quad (\text{II.110})$$

with principal square root; $f(3) = 0$. Every determinant logarithm below is its analytic logarithm normalized to vanish at the calibration, not an arbitrary pointwise principal logarithm of a determinant.

Theorem 23.1. *Uniformly on every compact $K \Subset \Omega_t$,*

$$\log \frac{D_n(t)}{D_n(3)} = nf(t) + O_K(1). \quad (\text{II.111})$$

For each fixed integer $k \geq 1$, the k th derivatives have errors $O_K(1)$ locally uniformly. Uniformly on compact subsets of $\{\Re s > 0\}$,

$$\log \frac{Z_n(s)}{Z_n(\sqrt{3})} = \frac{n}{2} f(s^2) + O_K(1). \quad (\text{II.112})$$

Consequently, for every positive fugacity s ,

$$\log Z_n(s) = \alpha n^2 + \left[b + \frac{1}{2} f(s^2) \right] n - \kappa \log n + O(1), \quad (\text{II.113})$$

locally uniformly on compact positive intervals. No limit of any bounded remainder is asserted.

[write] **Second route and limit, 5 October 2026.** Part III, Theorem 34.1, proves the weaker statement $\frac{1}{n} \log(D_n(t)/D_n(3)) \rightarrow f(t)$ on $\Re t > 0$ by a different argument (finite Jacobi matrices and a pole-cleared determinant). Part I proves that the bounded remainder converges: $\log(D_n(t)/D_n(3)) - nf(t) = E_n(t) \rightarrow E(t)$ locally uniformly on $\mathbb{C} \setminus (-\infty, 0]$.

23.1 A scalar compression gap on the complex domain

For $\delta = t - 3 \in \Omega$, every self-adjoint T with $0 \leq T \leq I/3$ has

$$\|(I + \delta T)^{-1}\| \leq \max_{0 \leq u \leq 1/3} |1 + \delta u|^{-1}.$$

This bounds all the relevant resolvents uniformly on compact subsets, independently of n . If $K \Subset \Omega$, its segment hull $\{v\delta : 0 \leq v \leq 1, \delta \in K\}$ is also a compact subset of Ω : nonreal points retain the sign of their imaginary part, and real endpoints and their segments lie to the right of -3 . Thus integration from zero along these segments is legitimate.

Put $T = M_\sigma$, $P = J_n$, $Q = I - P$. The kernel and Hölder estimates used for (II.86) apply to σ itself, giving

$$\|PTQ\|_{\text{HS}}^2 \leq C \quad (\text{II.114})$$

uniformly. Define

$$g_n(\delta) = \text{tr} \log(I_P + \delta PTP) - \text{tr}[P \text{Log}(I + \delta T)P].$$

For $\delta \neq 0$, let $R = (I + \delta T)^{-1}$, $R_P = (I_P + \delta PTP)^{-1}$ and $R_Q = (I_Q + \delta QTP)^{-1}$. Differentiation gives

$$g'_n(\delta) = \delta^{-1} \text{tr}(PRP - R_P). \quad (\text{II.115})$$

The two block equations of $(I + \delta T)R = I$ give, with the order of all factors retained,

$$PRP - R_P = \delta^2 R_P P T Q R_Q Q T P (PRP). \quad (\text{II.116})$$

Indeed the lower block says $QRP = -\delta R_Q QTP(PRP)$; substitution in the upper block proves (II.116). The two off-diagonal factors are Hilbert–Schmidt, so their product with the bounded resolvents is trace class. Equations (II.114)–(II.116) imply $|g'_n(\delta)| \leq C_K |\delta|$ on a compact segment hull. The value at zero is removable, with $g_n(0) = g'_n(0) = 0$. Integration proves $g_n(\delta) = O_K(1)$.

The global density bound (II.73) also gives the scalar mean directly and uniformly. Since

$$\text{Log}(1 + \delta u) = \int_0^u \frac{\delta}{1 + \delta v} dv, \quad |\text{Log}(1 + \delta u)| \leq C_K u \quad (0 \leq u \leq 1/3),$$

the integrable endpoint envelope $w(\theta)/\delta(\theta)$ proves

$$\text{tr}[P \text{Log}(I + \delta T)P] = \frac{n}{\pi} \int_I \text{Log}(1 + \delta w(\theta)) d\theta + O_K(1).$$

The integral equals $2f(3 + \delta)$ on the positive real axis by (II.79); analyticity and the identity theorem extend it over Ω . Therefore

$$\log \det(I_P + \delta PTP) = 2nf(3 + \delta) + O_K(1). \quad (\text{II.117})$$

23.2 The complex mixed Hardy gap

For $0 \leq s, u \leq 1$ put $R_{su} = [I + \delta(sA_n + uB_n)]^{-1}$. The mixed derivative of the normalized logarithm is $-\delta^2 \text{tr}(R_{su} A_n R_{su} B_n)$. For complex δ this trace is not claimed positive. The exact resolvent factorization is

$$\begin{aligned} A_n^{1/2} R_{su} B_n^{1/2} &= (I + \delta s A_n)^{-1} A_n^{1/2} B_n^{1/2} \\ &\quad \cdot [I - \delta u B_n^{1/2} R_{su} B_n^{1/2}]. \end{aligned}$$

The outside factors have compact-domain uniform bounds. Exchanging A_n, s with B_n, u gives the reverse-order estimate as well. Hilbert–Schmidt Cauchy–Schwarz yields

$$\begin{aligned} |\text{tr}(R_{su} A_n R_{su} B_n)| &\leq \|B_n^{1/2} R_{su} A_n^{1/2}\|_{\text{HS}} \|A_n^{1/2} R_{su} B_n^{1/2}\|_{\text{HS}} \\ &\leq C_K \text{tr}(A_n B_n) \leq C_K. \end{aligned}$$

The two factors need not be adjoints. Integration on the unit square, and equality of the spectra of A_n, B_n , give

$$2 \log \det(I + \delta A_n) - \log \det(I + \delta(A_n + B_n)) = O_K(1). \quad (\text{II.118})$$

The trace-class cross-term comparison (II.103) works locally uniformly on Ω , by the same resolvent bound and trace-norm convergence; the limiting relative operator is invertible there. Its normalized logarithm is locally convergent. The earlier DSASM relative factor has this property by Proposition 16.2 and Lemma 17.1. Combining (II.117) and (II.118) with these factors proves (II.111). Cauchy's estimate on a slightly larger compact neighborhood proves every fixed-order derivative assertion. This is a complex proof of bounded errors, not differentiation of real-axis boundedness.

23.3 Interlacing transfers the complex estimate to individual sizes

Let $r_{n,j}$ be the full real root list of $q_n(x) = i^{-n} Z_n(ix)$, including zero roots and multiplicities. Interlacing in (II.21) gives

$$d_n(r) = \#\{j : r_{n+1,j} \leq r\} - \#\{j : r_{n,j} \leq r\} \in \{0, 1\}.$$

Set $c = \sqrt{3}$ and, for $\Re s > 0$, use right-half-plane logarithms in the function

$$h_s(r) = \text{Log}(s - ir) - \text{Log}(c - ir).$$

Monicity gives $R_n(s) := \log[Z_n(s)/Z_n(c)] = \sum_j h_s(r_{n,j})$. The function h_s tends to zero at both real infinities, and

$$h'_s(r) = -i \left[(s - ir)^{-1} - (c - ir)^{-1} \right].$$

An explicit uniform integrability bound is

$$\int_{\mathbb{R}} |h'_s(r)| dr \leq \frac{\pi |s - c|}{\sqrt{c} \Re s}, \quad (\text{II.119})$$

by Cauchy-Schwarz and $\int |s - ir|^{-2} dr = \pi / \Re s$. Stieltjes integration by parts therefore proves

$$R_{n+1}(s) - R_n(s) = - \int_{\mathbb{R}} d_n(r) h'_s(r) dr = O_K(1).$$

The boundary terms vanish even though $d_n(+\infty) = 1$, because $h_s(+\infty) = 0$. This full-root argument includes the parity root and common roots without an extra correction.

With normalized analytic branches the exact adjacent identity is

$$R_n(s) + R_{n+1}(s) = \log[D_n(s^2)/D_n(3)] + \text{Log}(s/c).$$

It holds without a multiple of $2\pi i$ because both sides exponentiate to the same function and vanish at $s = c$. The map $s \mapsto s^2$ sends the right half-plane into Ω_t . Combining the bounded adjacent difference with (II.111) proves (II.112). Adding (II.35) on the positive real axis proves (II.113).

23.4 All fixed cumulants and an explicitly centered central limit theorem

For a fixed $s > 0$, give a DSASM A probability $s^{S(A)}/Z_n(s)$, and denote its nonzero-diagonal count by S_n . Define

$$m(s) = \frac{1}{2} - \frac{1}{(s+1)(s+2)}, \quad v(s) = \frac{s(2s+3)}{(s+1)^2(s+2)^2} > 0. \quad (\text{II.120})$$

The logarithmic moment generating function, in a small complex disk about zero, is

$$\log \mathbb{E}_s e^{zS_n} = \log Z_n(se^z) - \log Z_n(s) = \frac{n}{2}[f(s^2 e^{2z}) - f(s^2)] + O(1)$$

locally uniformly and analytically. Cauchy's estimates thus give, for each fixed $k \geq 1$,

$$\text{cum}_k(S_n) = n(s\partial_s)^{k-1}m(s) + O(1). \quad (\text{II.121})$$

Indeed $s^2 f'(s^2) = m(s)$ and $sm'(s) = v(s)$, so in particular

$$\mathbb{E}_s S_n = nm(s) + O(1), \quad \text{Var}_s S_n = nv(s) + O(1).$$

At $s = 1$ these coefficients are $m(1) = 1/3$ and $v(1) = 5/36$. All constants may depend on the fixed derivative order and fugacity; no uniform assertion as $s \rightarrow 0$ or $s \rightarrow \infty$ is intended.

The negative-root factorization supplies a short independent CLT proof. If $P_n(t) = \prod_{j=1}^{\lfloor n/2 \rfloor} (t + \lambda_{n,j})$ with $\lambda_{n,j} > 0$, then

$$\frac{P_n(s^2 z)}{P_n(s^2)} = \prod_j \left[\frac{\lambda_{n,j}}{s^2 + \lambda_{n,j}} + \frac{s^2}{s^2 + \lambda_{n,j}} z \right].$$

Thus S_n has the same law as $\varepsilon_n + 2 \sum_j B_{n,j}$ for independent Bernoulli variables with parameters $s^2/(s^2 + \lambda_{n,j})$. This is a representation of the total statistic, not independence of actual matrix diagonal indicators. The centered summands are bounded by two and the total variance is $nv(s) + O(1) \rightarrow \infty$. Lindeberg's condition follows immediately, giving the actual-mean CLT. The bounded mean and variance errors then give

$$\frac{S_n - nm(s)}{\sqrt{nv(s)}} \implies \mathcal{N}(0, 1). \quad (\text{II.122})$$

These distributional conclusions do not imply convergence of the bounded enumeration remainder.

[write] **Answered and strengthened, 5 October 2026.** The explicit centring answers Part III's question 2 (Section 40): $M_n(s) - nm(s) = O(1) = o(\sqrt{n})$ by (II.121). Part I proves the convergence of the bounded enumeration remainder (Theorem 1.1) and the constant terms of the cumulants (Corollary 7.2).

24 Further research and reproducibility

The analytic estimates prove boundedness at logarithmic scale. Four questions remain beyond that conclusion.

1. **A limiting amplitude.** Put $E_n = \log[D_n(1)/D_n(3)] - n \log(2/3)$. We have proved $E_n = O(1)$, not convergence. Some relative Fredholm factors do converge, but the scalar compression gap and mixed Hardy determinant gap have only been bounded here. In particular, the model wave kernel contains an n -dependent oscillatory term. Its uniform Hilbert–Schmidt bound cannot be replaced by a Hilbert–Schmidt convergence claim.
2. **Parity at constant order.** Even a limit of the adjacent centered determinant would not by itself prove a common count amplitude. If $x_n = \log a_n - \alpha n^2 - \beta n + \kappa \log n$, knowledge of the limit of $x_n + x_{n+1}$ permits opposite even and odd shifts. A sufficiently precise adjacent-ratio or principal-minor estimate, such as $x_{n+1} - x_n \rightarrow 0$, would remove that ambiguity.

3. **Corrections beyond the bounded term.** A full parity-sensitive expansion, including any fractional or oscillatory correction terms predicted by the published conjecture, requires control of endpoint and finite-cutoff effects beyond the present ideal-norm bounds. Neither an all-order expansion nor a correction transseries is established here.
4. **Effective constants.** The proof leaves the uniform Jacobi, kernel-ideal, overlap and Fredholm constants uncomputed. Quantifying them and a valid onset would turn the two-sided bounds and inverse bracket into finite numerical certificates. The present asymptotic localization alone is not such a certificate.

[write] **Status of the four questions, 5 October 2026.** *Question 1 is answered* by Part I: $E_n = E_n(1) \rightarrow E(1)$ (Theorem 1.1, (26)), and the limiting amplitude is $C_* = C_{\text{cal}} 2^{-1/4} e^{E(1)/2}$; Part I controls the scalar compression gap by Proposition 2.1 and retains the oscillatory wave term, (14), in the mixed limit (18). Its numerical value is open (Section 9, item 1). *Question 2 is answered* by Part I: the boundary limit (8) gives $R_{n+1}(s) - R_n(s) \rightarrow \frac{1}{2} f(s^2)$, (79), and $x_n \rightarrow \log C_*$, so in particular $x_{n+1} - x_n \rightarrow 0$. *Questions 3 and 4 remain open*; see Section 9, items 2–4.

The companion source package contains this complete TeX source, a deterministic two-clean-build PDF script, closed file inventories, and independent finite exact checks. Those checks exercise algebraic normalizations, retained finite cutoffs, noncommuting finite-matrix identities and inverse-center cancellation. They are supplementary: no finite run proves the uniform analytic estimates or substitutes for the all-size arguments in this report. The package README gives the exact verification commands, expected scope and environment.

Part III

Fugacity pressure and diagonal statistics

Bundle Report 121, archive A005163_Linear_Growth_Fugacity_Pressure_and_Limit_Laws_Source.zip, delivered as `report121.tex` (27 pages), dated 2 October 2026. Delivered title block: “REPORT 121 / DSASM fugacity pressure and diagonal statistics / Finite Jacobi matrices and cross size interlacing / 2 October 2026 / Standalone proofs and exact reproducibility companion”. Printed with **[write]** notes; Report 121’s Section k is Section $k + 24$ here, its statement $k.j$ is $(k + 24).j$, and its equation (k) is (III. k). Its Sections 2, 4 and 5 repeat Part II’s Sections 11, 13 and 14 and are printed there; Sections 26, 28 and 29 keep their titles, their equation numbers and a pointer. Its Section 9 repeats Part II’s Section 12 except for two identities, which Section 33 prints. A reference to an equation of a repeated section points to its printed place in Part II.

Reading conventions of Part III	Elsewhere in the report
$M_n(s)$, $V_n^{\text{stat}}(s)$ mean and variance of the diagonal count	Part I: M_n a modulation operator
$\beta(s)$ the linear coefficient (III.83)	Part I: $B(s)$
$d = 3 - 3\sqrt{3}/2$; $\ell_n = D'_n(3)/D_n(3)$	Parts II, III (logarithmic calibration): $d_{\text{ASM}} = 5/36$; Part IV: $d = 5/36$
ν_n , ν root measures	Part I: ν_t a source density
$r(x)$ tilted fugacity; $q(z)$, $r(z)$ polynomials	Parts I, II: $r = \sqrt{\log T/\alpha}$
$n_0 = \sqrt{\log T/\alpha}$	Parts I, II: r

Abstract of this Part (as delivered). Let a_n count diagonally symmetric alternating sign matrices and let $Z_n(s)$ weight their number of nonzero diagonal entries by s . We prove

$$\log a_n = \frac{n^2}{2} \log \frac{3\sqrt{3}}{4} + \frac{n}{4} \log 3 + o(n),$$

thereby identifying the conjectured linear exponential coefficient. The central result is an explicit fugacity pressure, locally uniform with all derivatives in a complex half-plane. Its proof uses exact finite Jacobi identities, total positivity, a determinant product with a controlled Stirling remainder, and a pole-cleared determinant whose small singular values have a uniformly integrable logarithmic tail. A multiplicity-safe stable-pencil argument gives cross-size interlacing and controls parity. We obtain a compactified limiting root measure, all fixed cumulants, actual-mean central and parity-lattice local central limit theorems, and an explicit macroscopic large-deviation rate. For uniform DSASMs, the limiting mean and variance per order are $1/3$ and $5/36$. The calibration trace limit is recovered as a derivative of the pressure. The logarithmic correction, amplitude and full conjectured asymptotic equivalent remain unresolved. All prerequisites are included from explicitly cited published inputs; finite exact checks supplement, rather than replace, the all-size proofs.

[write] Status of Part III, 5 October 2026. Every growth statement of this Part is implied by Part II (with $O(1)$ errors) or Part I (with limits): the pressure (Theorem 34.1) by Part II’s Theorem 23.1, the linear coefficient by Theorem 10.1 and Theorem 1.1. The proofs here are different (finite Jacobi matrices, a controlled Stirling product, a pole-cleared determinant and a logarithmic singular-value tail, against Part II’s Fredholm and Hardy comparison) and are kept as a *second route*. Found only in this Part: the limiting root measure (III.86) with its escaping half, Hoeffding’s bound (III.90), the parity-lattice local limit theorem (III.93) and the large-deviation principle (III.97). The sentence “The logarithmic correction, amplitude and full conjectured asymptotic equivalent remain unresolved” is historical: Part II proves the logarithmic correction and Part I the amplitude and the full equivalent.

25 Definitions and scope

An alternating sign matrix is a square matrix with entries in $\{-1, 0, 1\}$ whose nonzero entries alternate in sign in each row and column and whose row and column sums are one. A diagonally symmetric alternating sign matrix, or DSASM, satisfies $A = A^\top$. Write

$$a_n = |\text{DSASM}(n)|, \quad S(A) = |\{i : A_{ii} \neq 0\}|, \\ Q_n(s) = \sum_{A \in \text{DSASM}(n)} \prod_{i: A_{ii} \neq 0} s_i, \quad Z_n(s) = Q_n(s, \dots, s). \quad (\text{III.1})$$

Use $Q_0 = Z_0 = a_0 = 1$. For $s > 0$, the probability of A is $s^{S(A)}/Z_n(s)$; define

$$M_n(s) = s \frac{Z'_n(s)}{Z_n(s)}, \quad V_n^{\text{stat}}(s) = s \frac{d}{ds} M_n(s). \quad (\text{III.2})$$

Thus $M_n(s)$ and $V_n^{\text{stat}}(s)$ are the mean and variance of S . Uniform DSASMs correspond to $s = 1$, whereas the exactly solvable calibration in this report is $s = \sqrt{3}$. These are different measures. Matrix symbols V_n below are unrelated to the variance notation.

Set

$$g = \log \frac{3\sqrt{3}}{4}, \quad \alpha = \frac{g}{2}, \quad b = \frac{g + \log 2}{2}, \quad \kappa = \frac{5}{72}, \quad d = 3 - \frac{3\sqrt{3}}{2}. \quad (\text{III.3})$$

The count a_n is OEIS A005163. In the notation of Behrend–Fischer–Koutschan [1], $Z_n(s) = X_n(1, s, 1)$. Their Proposition 5.1 gives the exact identity

$$Z_n(2) = a_{n+1}. \quad (\text{III.4})$$

Their Theorem 11.2 already establishes the leading quadratic growth $\log a_n \sim \alpha n^2$. Their Conjecture 11.1 predicts

$$a_n \sim C_{\text{DSASM}} e^{\alpha n^2 + (\log 3)n/4} n^{-5/72}, \quad C_{\text{DSASM}} > 0, \quad (\text{III.5})$$

with further parity-sensitive corrections. An amplitude and the full equivalent in (III.5) are not proved here.

[write] **Historical, 5 October 2026.** Part I proves the full equivalent and the existence of the amplitude, $C_{\text{DSASM}} = C_*$ (Theorem 1.1); its numerical value and the parity-sensitive corrections remain open (Section 9).

Theorem 25.1. *Put $\varepsilon = n \bmod 2$. The polynomial Q_n is nonzero when every $\Re s_i > 0$, and*

$$Z_n(s) = s^\varepsilon P_n(s^2), \quad \deg P_n = \lfloor n/2 \rfloor, \quad P_n(0) > 0. \quad (\text{III.6})$$

All roots of P_n are negative real numbers. The monic real-rooted polynomials $q_n(x) = i^{-n} Z_n(ix)$ weakly interlace across adjacent sizes, counting multiplicities. For every $n \geq 1$ and $s > 0$,

$$|M_n(s) - M_{n-1}(s)| \leq 1. \quad (\text{III.7})$$

For $D_n(t) = Z_n(\sqrt{t})Z_{n+1}(\sqrt{t})/\sqrt{t}$ and $\ell_n = D'_n(3)/D_n(3)$,

$$\frac{\ell_n}{n} \longrightarrow 1 - \frac{\sqrt{3}}{2}, \quad 3\ell_n \leq M_n(\sqrt{3}) \leq 3\ell_n + 1, \quad \frac{M_n(\sqrt{3})}{n} \longrightarrow d. \quad (\text{III.8})$$

All limits hold along the full sequence, with no parity subsequence restriction and no assumed rate of convergence.

[write] **Where this theorem is proved, 5 October 2026.** The stability and parity statements are proved in Part II, Section 11 (Report 121’s Section 2, printed once); weak interlacing and (III.7) in Part II, Section 12, with the bracket of (III.8) proved in Section 33, (III.64). Part II, (II.7), strengthens the trace limit to $\ell_n = (n+1)(1 - \sqrt{3}/2) + o(1)$.

The stronger central conclusion is the pressure theorem in Section 34:

$$\frac{1}{n} \log \frac{D_n(t)}{D_n(3)} \longrightarrow \log \frac{(\sqrt{t} + 1)^2}{2(\sqrt{t} + 2)}.$$

It implies the growth formula (III.84) and the uniform-law limits $M_n(1)/n \rightarrow 1/3$ and $V_n^{\text{stat}}(1)/n \rightarrow 5/36$. Sections 36–38 state the root law and probability consequences; Section 39 treats integer threshold inversion. The sharper pressure route also proves the first calibration trace limit without relying on that limit as an input.

The published inputs are the triangular six-vertex correspondence and adjacent determinant of [1], classical stability closure, and explicit determinant and orthogonal-polynomial evaluations. The stability operator is the degree-two case of the known MOD operator [4, Section 8.4]; the stable-pencil implication is classical [3, Definition 1.1 and Lemma 1.8]. The shifted Jacobi identities are derived explicitly below, inspired by the unshifted commuting construction of Casper–Zurrián [12]. These attributions do not imply a bibliographic priority claim for any DSASM-specific consequence.

26 An orientation polynomial and diagonal stability

[write] **Printed in Part II.** Report 121’s Section 2 is Part II’s Section 11 word for word, with two exceptions: it writes ε for ε_n , and its proof of the root structure ends “This proves the structural part of Theorem 25.1. For $n = 1$, $P_1 = 1$ and the root statement is vacuous.” Its labels are those of Section 11, and its equations (III.9)–(III.15) are (II.11)–(II.17) there; Lemma 2.1 of Report 121 is Lemma 11.1.

27 Two consequences for the diagonal statistic

The root theorem allows a factorization

$$P_n(t) = c_n \prod_{j=1}^m (t + \gamma_j), \quad m = \lfloor n/2 \rfloor, \quad c_n > 0, \quad \gamma_j > 0. \quad (\text{III.16})$$

Multiplicity is included in this list. The proofs below do not need $c_n = 1$.

Write $P_n(t) = \sum_{k=0}^m c_{n,k} t^k$. Newton’s inequalities give

$$\left(\frac{c_{n,k}}{\binom{m}{k}} \right)^2 \geq \frac{c_{n,k-1}}{\binom{m}{k-1}} \frac{c_{n,k+1}}{\binom{m}{k+1}} \quad (1 \leq k < m). \quad (\text{III.17})$$

In particular, these parity-reduced coefficients are log-concave after binomial normalization. Strict positivity follows as well from (III.16).

For a fixed positive fugacity s , give a matrix A probability $s^{S(A)}/Z_n(s)$. The probability generating function of the integer-valued total $(S - \varepsilon)/2$ is

$$\frac{P_n(s^2 z)}{P_n(s^2)} = \prod_{j=1}^m \left(\frac{\gamma_j}{s^2 + \gamma_j} + \frac{s^2}{s^2 + \gamma_j} z \right).$$

Consequently this total has the same distribution as a sum of independent Bernoulli variables with parameters $s^2/(s^2 + \gamma_j)$. Equivalently,

$$S \stackrel{\text{law}}{=} \varepsilon + 2 \sum_{j=1}^m B_j.$$

This is an independent-Bernoulli representation of one total statistic. It does not assert independence of the actual diagonal entries or their nonzero indicators in a random matrix.

28 The exact solvable calibration

[write] **Printed in Part II.** Report 121's Section 4 is Part II's Section 13 word for word (checked by `diff`); its equations (III.18)–(III.27) are (II.25)–(II.34) there.

29 Logarithmic calibration without matching parity amplitudes

[write] **Printed in Part II.** Report 121's Section 5 is Part II's Section 14 word for word; its equations (III.28)–(III.32) are (II.35)–(II.39) there. Part I, (27), now matches the parity amplitudes.

30 The finite calibration trace and its Jacobi matrix

All matrices in this section have indices $0, \dots, n-1$. Put

$$V_n(i, j) = \binom{i+j+2}{i}, \quad Q_n^{\text{mat}} = (I + V_n)^{-1}.$$

The notation Q_n^{mat} is distinct from the generating polynomial $Q_n(s)$. Differentiating (II.32) at $t = 3$ is legitimate because $I + V_n$ is invertible. In the finite lower-Toeplitz convention,

$$R(z) = \frac{\partial_t G_t(z)|_{t=3}}{G_3(z)} = \frac{z^2 - z + 1}{(2-z)(1+z)} = -1 + \frac{1}{2-z} + \frac{1}{1+z}. \quad (\text{III.33})$$

The derivative matrix is $T_n(R)V_n$. Jacobi's determinant identity and $V_n(I + V_n)^{-1} = I - (I + V_n)^{-1}$ give

$$\ell_n = \text{tr}(T_n(R)V_n(I + V_n)^{-1}) = \frac{n}{2} - \text{tr}(T_n(R)Q_n^{\text{mat}}), \quad (\text{III.34})$$

since $R(0) = 1/2$. The order of the Toeplitz and binomial factors and the lower-shift convention in this identity will matter below.

30.1 Symmetrization and reciprocal pairs

Define

$$\begin{aligned} d_i &= \sqrt{(i+1)(i+2)}, & D &= \text{diag}(d_i), & C_n &= D^{-1}V_nD, \\ C_n(i, j) &= \frac{(i+j+2)!}{\sqrt{i!(i+2)!j!(j+2)!}}, & F_n &= (I + C_n)^{-1}. \end{aligned} \quad (\text{III.35})$$

Let $E = \text{diag}((-1)^i)$, and let L be the lower-triangular matrix

$$L_{ij} = \frac{\sqrt{i!(i+2)!/[j!(j+2)!]}}{(i-j)!} = \binom{i}{j} \frac{d_i}{d_j} \quad (i \geq j). \quad (\text{III.36})$$

Then $C_n = LL^\top$. Indeed, clearing the positive factorial factors reduces it to Vandermonde's identity

$$\sum_{k=0}^{\min(i,j)} \binom{i}{k} \binom{j+2}{k+2} = \binom{i+j+2}{i+2}.$$

Finite binomial inversion gives $L^{-1} = ELE$. Consequently $B = LE$ satisfies $B^2 = I$ and

$$BC_n B = EL^\top LE = L^{-\top} L^{-1} = C_n^{-1}. \quad (\text{III.37})$$

The matrix B need not be symmetric. Positive definiteness of C_n and (III.37) imply reciprocal pairing of its positive spectrum, with multiplicities. Thus

$$\text{tr } F_n = \frac{n}{2}. \quad (\text{III.38})$$

This includes a possible eigenvalue one, whose contribution is $1/2$.

30.2 Exact commutation and reflection

Put $N = n(n+2)$ and define the symmetric tridiagonal matrix J_n by

$$\begin{aligned} J_n(i, i) &= 2i^3 + 9i^2 + (12 - N)i, \\ J_n(k-1, k) &= J_n(k, k-1) = \sqrt{k(k+2)[N - k(k+2)]} \quad (1 \leq k < n). \end{aligned} \quad (\text{III.39})$$

Every off-diagonal entry is positive. The identities we need are

$$[J_n, C_n] = 0, \quad L^{-1} J_n L = -E J_n E + 2\tau_n I, \quad \tau_n = \frac{3}{2}(n-1)(n+3). \quad (\text{III.40})$$

They hold as finite matrix identities, including the last row. Here is an algebraic verification that exposes both boundaries.

For a parameter $a > -1$, define explicitly

$$\begin{aligned} C_a(i, j) &= \frac{\Gamma(i+j+a+1)}{\sqrt{i!\Gamma(i+a+1)j!\Gamma(j+a+1)}}, \\ L_a(i, j) &= \frac{\sqrt{i!\Gamma(i+a+1)/[j!\Gamma(j+a+1)]}}{(i-j)!} \quad (i \geq j), \end{aligned}$$

with entries zero above the diagonal. Only $a = 2$ is needed for DSASMs. Define

$$f(k) = n(n+a) - k(k+a), \quad b(i) = 2i^3 + 3(a+1)i^2 + [(a+1)(a+2) - n(n+a)]i.$$

Let J_a have diagonal $b(i)$ and off-diagonal $\sqrt{k(k+a)}f(k)$. Dividing the (i, j) commutator entry by its positive moment entry reduces it to

$$\begin{aligned} &(i+j+a+1)[f(i+1) - f(j+1)] \\ &+ \frac{i(i+a)f(i) - j(j+a)f(j)}{i+j+a} + b(i) - b(j) = 0. \end{aligned} \quad (\text{III.41})$$

Substitution of the two polynomials, multiplication by $i + j + a$, and cancellation prove the identity. Only $a = 2$ is needed here, so this denominator is strictly positive. The missing lower index has its factor $i(i + a) = 0$; the missing upper index has $f(n) = 0$. Thus (III.41) proves commutation for the finite section.

For the reflection identity set $c = (a + 1)[n(n + a) - a - 1]$. Divide the (i, j) entry of $J_a L_a - L_a(-E J_a E + cI)$ by $(L_a)_{ij}$ for $i \geq j$, and multiply by $i - j + 1$. The result is

$$(i - j + 1)\{(i - j)[f(i) - f(j + 1)] + b(i) + b(j) - c\} \\ + (i + 1)(i + a + 1)f(i + 1) - j(j + a)f(j) = 0. \quad (\text{III.42})$$

Again substitution gives the zero polynomial. For $i = j - 1$, both matrix entries are the same off-diagonal Jacobi entry; farther above the diagonal both are zero. The boundary terms vanish exactly as before. At $a = 2$, $c = 2\tau_n$, proving (III.40). These direct finite identities are the shifted counterpart of the unshifted construction in [12, Theorems 3.2–3.3 and (4.4)]; the shifted formula is not attributed verbatim to that paper.

30.3 Why the spectral orders coincide

For increasing row indices $i_1 < \dots < i_r$ and column indices $j_1 < \dots < j_r$, the corresponding unnormalized moment minor is

$$\det \left[\int_0^\infty x^{i_p + j_q} x^2 e^{-x} dx \right]_{p,q=1}^r = \int_{0 < x_1 < \dots < x_r} \det[x_q^{i_p}] \det[x_q^{j_p}] \prod_{q=1}^r x_q^2 e^{-x_q} d\mathbf{x}. \quad (\text{III.43})$$

This is the finite Andréief identity, obtained by expanding both determinants, applying Fubini, and grouping the $r!$ ordering regions. Both generalized Vandermonde determinants are positive on the ordered positive region: factoring an ordinary Vandermonde leaves a Schur polynomial with nonnegative coefficients and a positive value there. The density is positive and the region has positive measure. Therefore every minor is strictly positive. Positive row and column scalings preserve the signs, so C_n is strictly totally positive.

The oscillation theorem for a strictly totally positive matrix gives positive simple eigenvalues in descending order, with the k th corresponding eigenvector having exactly $k - 1$ sign changes; zeros are deleted when counting changes. A precise modern statement is [13, Section 3.3 and Theorem 2]. For a real symmetric Jacobi matrix with positive off-diagonals, the Sturm convention is the same: the k th *largest* eigenvalue has an eigenvector with $k - 1$ sign changes. To see the orientation, the eigenvector entries, after choosing its first entry positive, are the leading-principal monic characteristic polynomials evaluated at the eigenvalue, divided by successive positive off-diagonal products. The three-term recurrence is a Sturm chain. Its largest eigenvector has no sign change, and successive eigenvalues introduce one more; a zero entry is passed using the recurrence, which forces opposite signs on its neighbors. The Jacobi spectrum is simple because its first entry determines an entire eigenvector by that recurrence.

Commutation in (III.40) makes the simple J_n eigenvectors also eigenvectors of C_n . The two oscillation counts place them in the same descending order. Moreover, (III.37) and (III.40) give

$$C_n B = B C_n^{-1}, \quad J_n B = B(2\tau_n I - J_n).$$

Thus a simultaneous eigenpair (μ, λ) is paired with $(2\tau_n - \mu, \lambda^{-1})$. Their ranks are reversed. In odd size the unique central pair is $(\tau_n, 1)$; in even size neither threshold is an eigenvalue. We have

proved the exact identity

$$\Pi_n = \mathbf{1}_{J_n < \tau_n} + \frac{1}{2} \mathbf{1}_{J_n = \tau_n} = \mathbf{1}_{C_n < 1} + \frac{1}{2} \mathbf{1}_{C_n = 1}. \quad (\text{III.44})$$

In this shared basis the F_n eigenvalues decrease as the J_n eigenvalues increase. This ordering is stronger than commutation alone and is essential when a singular Toeplitz factor is present.

31 A determinant product controls spectral smoothing

Write $(a)_k = a(a+1) \cdots (a+k-1)$, with $(a)_0 = 1$, and define

$$\begin{aligned} J_m^{\text{prod}} &= \frac{m!(2)_m(3)_{3m}(4)_{3m}}{(2)_{2m}(3)_{2m}}, \\ H_{2m}^{\text{prod}} &= \frac{2m!(3/2)_m(3/2)_{3m}(4)_{3m}}{(3/2)_{2m}(5/2)_{2m}}, \\ H_{2m+1}^{\text{prod}} &= \frac{(21/2)m!(5/2)_m(9/2)_{3m}(4)_{3m}}{(5/2)_{2m}(7/2)_{2m}}. \end{aligned} \quad (\text{III.45})$$

The superscripts distinguish these scalar products from the Jacobi matrix and its later normalization. The determinant evaluation of Ciucu–Eisenkölbl–Krattenthaler–Zare in [6, Theorem 4.4], specialized at $b = 3$, and the plus-case norms [6, (3.5b,d) and Theorem 3.1], give

$$\begin{aligned} \det(I - V_{2m}) &= \frac{(-1)^m 3^{2m} \prod_{j=0}^{m-1} (J_j^{\text{prod}})^2}{\prod_{i=0}^{2m-1} i!(3)_i}, \\ \det(I + V_n) &= \frac{\prod_{k=0}^{n-1} H_k^{\text{prod}}}{\prod_{i=0}^{n-1} i!(3)_i}. \end{aligned} \quad (\text{III.46})$$

For odd size the minus determinant is zero. The source minus matrix has entries $i!(3)_i \delta_{ij} - (3)_{i+j}$. Multiplication on the left by $\text{diag}(i!)^{-1}$ and on the right by $\text{diag}((3)_i)^{-1}$ gives $I - V_n$ in exactly our orientation. The plus norms above are the source norms multiplied by $2 \cdot 27^k$; the triangular polynomial basis has diagonal signs, so its squared determinant is one. This explains both scaling factors in (III.46).

Let

$$\Lambda_m = \frac{|\det(I - C_{2m})|}{\det(I + C_{2m})}, \quad \Lambda_0 = 1.$$

By similarity and (III.46),

$$\begin{aligned} \frac{\Lambda_{m+1}}{\Lambda_m} &= \frac{9(J_m^{\text{prod}})^2}{H_{2m}^{\text{prod}} H_{2m+1}^{\text{prod}}} \\ &= \frac{3}{7} \frac{(2)_m^2}{(3/2)_m(5/2)_m} \frac{(3)_{3m}^2}{(3/2)_{3m}(9/2)_{3m}} \\ &\quad \cdot \frac{(3/2)_{2m}(5/2)_{2m}^2(7/2)_{2m}}{(2)_{2m}^2(3)_{2m}^2}. \end{aligned} \quad (\text{III.47})$$

No differentiation with respect to the discrete determinant parameter is used.

31.1 The controlled Stirling step

For $x > 0$, the usual Stirling remainder bound is

$$\log \Gamma(x) = (x - \tfrac{1}{2}) \log x - x + \tfrac{1}{2} \log(2\pi) + \frac{1}{12x} + r(x), \quad |r(x)| \leq \frac{1}{360x^3}.$$

For any of the finitely many positive a, c below, Taylor expansion at $x = am + c$ consequently yields, uniformly as $m \rightarrow \infty$,

$$\log \Gamma(am + c) = (am + c - \tfrac{1}{2}) \log(am) - am + \tfrac{1}{2} \log(2\pi) + \frac{B_2(c)}{2am} + O(m^{-2}), \quad (\text{III.48})$$

where $B_2(c) = c^2 - c + 1/6$. The finitely many fixed shifts make the remainder constant uniform over every term in (III.47). The gamma factors in that ratio have signed triples (a, c, e)

$$\begin{aligned} & (1, 2, 2), (1, 3/2, -1), (1, 5/2, -1), \\ & (3, 3, 2), (3, 3/2, -1), (3, 9/2, -1), \\ & (2, 3/2, 1), (2, 5/2, 2), (2, 7/2, 1), (2, 2, -2), (2, 3, -2). \end{aligned}$$

Substituting them in (III.48) cancels the coefficients of $m \log m$, m , $\log m$ and the constant term, including the fixed gamma normalizations in the rising factorials. The remaining coefficient is

$$\sum_{(a,c,e)} \frac{eB_2(c)}{2a} = -\frac{3}{4}.$$

Thus

$$\log \frac{\Lambda_{m+1}}{\Lambda_m} = -\frac{3}{4m} + O(m^{-2}), \quad \Lambda_m = c m^{-3/4} (1 + O(m^{-1})), \quad c > 0. \quad (\text{III.49})$$

Indeed, subtract $-(3/4) \log(1 + 1/m)$ from the logarithmic increment. The resulting series has a summable $O(m^{-2})$ tail, so $\log \Lambda_m + (3/4) \log m$ converges to a finite real constant with error $O(m^{-1})$. Exponentiation proves the second statement and positivity of c . The stronger optional expansion begins $-3/(4m) + 3/(4m^2) - 961/(1152m^3)$, but is not needed.

31.2 Both parities and the half threshold

For even $n = 2m$, let $\lambda_j > 0$ be the C_n eigenvalues. They avoid one by (III.46), and

$$\sum_j \log \frac{1 + \lambda_j}{|1 - \lambda_j|} = -\log \Lambda_m = \frac{3}{4} \log m + O(1).$$

If $u = \min(\lambda, \lambda^{-1}) < 1$, then

$$h_0(\lambda) := \frac{u}{1+u} \leq \frac{1}{2} \log \frac{1+u}{1-u}.$$

Set $h_0(1) = 1/2$. Off the threshold, $h_0(\lambda)$ is precisely the error between $(1 + \lambda)^{-1}$ and $\mathbf{1}_{\lambda < 1}$, while at the half-valued threshold of (III.44) that error is zero. Therefore

$$\|F_n - \Pi_n\|_1 = O(\log(n+1)) \quad (n \geq 1). \quad (\text{III.50})$$

For completeness, the extension to odd n does not use a zero minus determinant. The entries of C_n do not depend on n , so C_n is a principal section of C_{n+1} . Cauchy interlacing implies that

their eigenvalue counts in any interval differ by at most one. Each superlevel set of the continuous unimodal function h_0 is an interval, and its maximum is $1/2$. The layer-cake formula gives

$$|\operatorname{tr} h_0(C_n) - \operatorname{tr} h_0(C_{n+1})| \leq \frac{1}{2}.$$

An odd size is thereby bounded by the following even size, proving (III.50) along both parities.

32 Jacobi moments and mixed finite path limits

Normalize the Jacobi matrix by

$$\begin{aligned} H_n &= (J_n - \tau_n I)/n^3, \\ a(x) &= x(1 - x^2), \quad b(x) = 2x^3 - x, \quad h(x, \theta) = b(x) + 2a(x) \cos \theta. \end{aligned} \tag{III.51}$$

The letter H_n now denotes this matrix, not the orientation polynomial used in Section 26. The coefficients of H_n converge uniformly to this profile with error $O(1/n)$. For the off-diagonal coefficient, use

$$\frac{\sqrt{k(k+2)}}{n} \frac{n(n+2) - k(k+2)}{n^2} = a(k/n) + O(n^{-1}),$$

uniformly in $1 \leq k < n$; the diagonal follows directly from (III.39). Absolute row sums give a common bound $\|H_n\| \leq K$.

Lemma 32.1. *For every continuous function f on a common compact spectral interval,*

$$\frac{1}{n} \operatorname{tr} f(H_n) \longrightarrow \int_0^1 \int_0^{2\pi} f(h(x, \theta)) \frac{d\theta}{2\pi} dx. \tag{III.52}$$

The limiting measure has no atom at zero and assigns positive mass to each sufficiently small interval immediately on either side of zero.

Proof. For a fixed power r , a diagonal entry of H_n^r is a sum over nearest-neighbor paths of length r that start and end at that index. Away from the $O(r)$ boundary rows, freeze each coefficient at $x = i/n$. Uniform convergence of coefficients makes the resulting error $o(1)$ uniform over rows. The frozen sum is the constant Laurent coefficient of $(b(x) + a(x)(z + z^{-1}))^r$, equivalently its circle integral. All path terms are bounded uniformly in n , so the boundary rows contribute $o(1)$ after division by n . Riemann summation proves (III.52) for powers and hence for polynomials. Uniform polynomial approximation on $[-K, K]$ proves it for continuous f .

For each $0 < x < 1$, $a(x) > 0$ and the equation $h(x, \theta) = 0$ has at most two solutions in $[0, 2\pi]$. Fubini therefore gives zero mass at zero. At $x = 1/2$, zero is attained at $\cos \theta = 1/3$, with nonzero θ derivative. A small neighborhood of this point has positive area on each side of zero and in each sufficiently small adjacent interval. This proves the last assertion. $\square$

The following extension will allow products containing the shift as well as a discontinuous spectral cutoff. Let

$$A = D^{-1} S_n D, \quad A_{i,i-1} = w_i = \sqrt{\frac{i}{i+2}} \quad (1 \leq i < n). \tag{III.53}$$

Then $\|A\| \leq 1$. For every fixed noncommutative polynomial in A, A^*, H_n , the normalized trace converges to the same phase-space integral, with scalar substitutions

$$A \mapsto z = e^{i\theta}, \quad A^* \mapsto \bar{z}, \quad H_n \mapsto h(x, \theta). \quad (\text{III.54})$$

Indeed, each word has a fixed bandwidth and finitely many contributing paths. For $i/n \rightarrow x > 0$, the weights w_i tend to one. Discard the first ηn rows and the bounded endpoint rows, apply the uniform path argument on the remainder, and then let $\eta \downarrow 0$. The uniform norm bounds control the discarded rows. The convention z or $\bar{z}$ for the lower shift changes no circle integral here.

Put $U = I - F_n$ and $P_+ = \mathbf{1}_{H_n > 0}$. By (III.50),

$$\|U - P_+\|_{\text{HS}} = o(\sqrt{n}). \quad (\text{III.55})$$

The possible central rank-one half value contributes only $O(1)$ to the squared Hilbert–Schmidt norm. To replace P_+ by polynomials in H_n , choose a continuous cutoff that agrees with the step off $[-\eta, \eta]$ and lies in $[0, 1]$, then approximate it uniformly by a polynomial. Equation (III.52) and the absence of a zero atom make the normalized squared Hilbert–Schmidt error arbitrarily small. Norms of all approximants can be bounded, for example by two. In a fixed mixed word, telescope one changed factor at a time and use $|\text{tr } XY| \leq \|X\|_{\text{HS}} \|Y\|_{\text{HS}}$ together with the bounded norms of the remaining factors. Consequently the mixed-trace limit (III.54) remains valid with

$$U \mapsto \mathbf{1}_{h(x, \theta) > 0}. \quad (\text{III.56})$$

This is a trace statement about finite matrices; it neither assumes that A commutes with H_n nor asserts operator-norm convergence to a discontinuous function.

33 Cross size interlacing and parity control

[write] **Printed in Part II, except the identities below.** Report 121’s Section 9 is Part II’s Section 12, with these differences only: the root formula (II.23) is written $M_n(s) = \sum_{j=1}^n s^2 / (s^2 + r_{n,j}^2)$, since M_n was defined in (III.2); the following two identities and their paragraph are inserted after the proof of the adjacent mean bound; and the last sentence differs. Its equations (III.57)–(III.62) and (III.65) are (II.18)–(II.23) and (II.24) there. The inserted text and the last sentence are printed here as delivered.

The determinant definition gives the exact identity

$$2t \frac{D'_n(t)}{D_n(t)} = M_n(\sqrt{t}) + M_{n+1}(\sqrt{t}) - 1. \quad (\text{III.63})$$

At $t = 3$, write $\delta_n = M_{n+1}(\sqrt{3}) - M_n(\sqrt{3}) \in [-1, 1]$. Then

$$M_n(\sqrt{3}) = 3\ell_n + \frac{1 - \delta_n}{2} \in [3\ell_n, 3\ell_n + 1]. \quad (\text{III.64})$$

Thus a trace limit transfers to the individual mean with a bounded error; an alternating sum of uncontrolled $o(n)$ errors is never used. More generally, integrating the adjacent bound against ds/s gives (II.24). This version will control parity for the individual free energy.

34 The fugacity pressure

The pole in R at -1 prevents a direct bounded-symbol treatment of (III.34). A finite multiplication clears that pole before any asymptotic passage. We first prove the pressure on $3 < t < 4$, where an accretivity estimate controls the entire lower singular-value tail, and then use negative real roots to continue the result analytically.

Theorem 34.1. *For $\Re t > 0$, define $p_n(t)$ as the analytic logarithm satisfying $p_n(3) = 0$ of $D_n(t)/D_n(3)$, divided by n . Then, locally uniformly with every complex derivative,*

$$p_n(t) \longrightarrow f(t) = \log \frac{(\sqrt{t} + 1)^2}{2(\sqrt{t} + 2)}. \quad (\text{III.66})$$

The square root and logarithms use their branches real on the positive axis. In particular,

$$\begin{aligned} f(1) &= \log(2/3), \quad f(3) = 0, \quad f(4) = \log(9/8), \\ f'(t) &= \frac{\sqrt{t} + 3}{2\sqrt{t}(\sqrt{t} + 1)(\sqrt{t} + 2)}, \\ f'(3) &= 1 - \frac{\sqrt{3}}{2}, \quad f''(3) = \frac{1}{4} - \frac{\sqrt{3}}{6}. \end{aligned} \quad (\text{III.67})$$

[write] **Second route, 5 October 2026.** Part II, Theorem 23.1, proves $\log(D_n(t)/D_n(3)) = nf(t) + O_K(1)$ on the larger domain $\mathbb{C} \setminus (-\infty, 0]$, and Part I proves that the remainder converges. The proof below is a different, independent route to the $o(n)$ statement and is kept as such.

34.1 A bounded finite determinant after clearing the pole

Keep the weighted shift A from (III.53), and put

$$q(z) = 2 + z - z^2 = (2 - z)(1 + z), \quad r(z) = 1 - z + z^2 = 3 - q(z), \quad U = I - F_n.$$

Here the polynomial q has no size subscript and is unrelated to the interlacing polynomials q_n . Since G_t is affine in t , (II.32) gives the exact finite identity, for $\delta = t - 3$,

$$\begin{aligned} \frac{D_n(3 + \delta)}{D_n(3)} &= \det[I + \delta R(A)U] = \frac{\det \mathcal{M}_n(\delta)}{2^n}, \\ \mathcal{M}_n(\delta) &= q(A) + \delta r(A)U. \end{aligned} \quad (\text{III.68})$$

Indeed, after the diagonal similarity the base matrix is $I + C_n$ and the perturbation is $\delta R(A)C_n$. Right multiplication by $(I + C_n)^{-1}$ proves the first equality. Left multiplication by $q(A)$ proves the second, since $\det q(A) = q(0)^n = 2^n$. The inverse $q(A)^{-1}$ is used only in this exact algebra; it is not estimated. Both A and U are contractions, so $\mathcal{M}_n(\delta)$ has uniformly bounded norm for bounded δ .

34.2 Accretivity and the logarithmic lower tail

Fix $0 < \delta < 1$ and set

$$B_\delta = I - \delta U, \quad N_\delta = q(A) + 3\delta U B_\delta^{-1}.$$

The matrix U is symmetric with eigenvalues in $(0, 1)$, so $(1 - \delta)I \leq B_\delta \leq I$ and $UB_\delta^{-1} \geq 0$. A direct multiplication, in the displayed order, gives $\mathcal{M}_n = N_\delta B_\delta$. For $\Re T = (T + T^*)/2$, the contraction property of A gives

$$\Re q(A) = 2I + \Re A - \Re(A^2) \geq I + \Re A =: B, \quad \Re N_\delta \geq B. \quad (\text{III.69})$$

The diagonal sign matrix E changes B to $I - \Re A$, with diagonal one and off-diagonals $-w_i/2$. Since $1/\sqrt{3} \leq w_i < 1$, let $w_* = 1/\sqrt{3}$ and let B_0 have diagonal one and off-diagonals $-1/2$. The difference $EBE - w_*B_0$ has diagonal $1 - w_*$ and off-diagonals $-(w_i - w_*)/2$. Its absolute off-diagonal row sum is at most $1 - w_*$, so it is positive semidefinite, as follows from $2|xy| \leq |x|^2 + |y|^2$. Hence $EBE \geq w_*B_0$.

List eigenvalues and singular values in increasing order. The Dirichlet eigenvectors of B_0 are discrete sines, giving

$$\lambda_k(B) \geq w_* \left(1 - \cos \frac{k\pi}{n+1}\right) \geq 2w_* \left(\frac{k}{n+1}\right)^2 \quad (1 \leq k \leq n). \quad (\text{III.70})$$

The last bound uses $\sin y \geq 2y/\pi$ on $0 \leq y \leq \pi/2$. For any matrix N whose Hermitian part is positive, singular-value minmax gives $s_k(N) \geq \lambda_k(\Re N)$. Namely, on the span of the largest $n - k + 1$ eigenvectors of $\Re N$,

$$\|Nv\| \|v\| \geq \Re \langle v, Nv \rangle \geq \lambda_k(\Re N) \|v\|^2,$$

and the max-min characterization proves the claim. Also $s_k(NB_\delta) \geq (1 - \delta)s_k(N)$, by the minmax characterization and the lower norm bound for B_δ . Combining these facts gives

$$s_k(\mathcal{M}_n(\delta)) \geq c_\delta \left(\frac{k}{n+1}\right)^2, \quad c_\delta = \frac{2(1 - \delta)}{\sqrt{3}} > 0. \quad (\text{III.71})$$

This proves invertibility but provides much more than a bound for only the smallest singular value. If $N_n(\eta) = \#\{k : s_k(\mathcal{M}_n) < \eta\}$, then $N_n(\eta) \leq 2n\sqrt{\eta/c_\delta}$, and Tonelli's theorem yields

$$\frac{1}{n} \sum_{s_k < \eta} \log \frac{\eta}{s_k} = \int_0^\eta \frac{N_n(u)}{n} \frac{du}{u} \leq 4\sqrt{\frac{\eta}{c_\delta}}. \quad (\text{III.72})$$

Thus the negative logarithmic tail is uniformly integrable. A weak singular-value limit alone, even together with a polynomial lower bound for the very smallest value, would not justify the logarithmic determinant limit without this integrated estimate.

34.3 From mixed moments to the logarithmic determinant

The mixed trace limits (III.54)–(III.56) imply, for every nonnegative integer k ,

$$\frac{1}{n} \text{tr}[(\mathcal{M}_n^* \mathcal{M}_n)^k] \longrightarrow \int_0^1 \int_0^{2\pi} |m_\delta(x, \theta)|^{2k} \frac{d\theta}{2\pi} dx, \quad (\text{III.73})$$

where

$$m_\delta(x, \theta) = q(z) + \delta r(z) \mathbf{1}_{h(x, \theta) > 0}, \quad z = e^{i\theta}.$$

Uniform norm bounds and polynomial approximation show weak convergence of the empirical singular-value measures to the distribution of $|m_\delta|$. To pass to log, first use the continuous bounded function $\log \max(\eta, x)$ on the common compact support. Equation (III.72) bounds the difference

from $\log x$ uniformly in n by $4\sqrt{\eta/c_\delta}$. The limiting measure has the same bound, by bounded truncations and monotone convergence. Letting $\eta \downarrow 0$ proves

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log |\det \mathcal{M}_n(\delta)| = \int_0^1 \int_0^{2\pi} \log |m_\delta(x, \theta)| \frac{d\theta}{2\pi} dx. \quad (\text{III.74})$$

For real $t > 0$, $D_n(t) > 0$ by its negative-root factorization; thus (III.68) has no phase ambiguity. The elementary circle mean of $\log |q(e^{i\theta})|$ is $\log 2$: factor $q = (2-z)(1+z)$ and apply the power-series mean to the first factor and its boundary limit to the second. The logarithm at $z = -1$ is integrable. Equations (III.68) and (III.74) now give, for $3 < t < 4$,

$$\lim p_n(t) = \int_0^1 \int_0^{2\pi} \mathbf{1}_{h>0} \log \left| \frac{q(z) + (t-3)r(z)}{q(z)} \right| \frac{d\theta}{2\pi} dx. \quad (\text{III.75})$$

34.4 Exact evaluation of the phase space integral

For $0 \leq \theta \leq \pi$, the measure of $\{x \in [0, 1] : h(x, \theta) > 0\}$ is

$$Y(\theta) = \begin{cases} 1, & 0 \leq \theta \leq \pi/3, \\ 1 - X(\theta), & \pi/3 < \theta \leq \pi, \end{cases} \quad X(\theta) = \sqrt{\frac{1 - 2 \cos \theta}{2(1 - \cos \theta)}}.$$

This follows by factoring $h = x[(2 \cos \theta - 1) + 2(1 - \cos \theta)x^2]$. The substitution

$$\sin^2(\theta/2) = \frac{1 + 3 \sin^2 \phi}{4}, \quad 0 \leq \phi \leq \pi/2, \quad X(\theta) d\theta = \frac{6 \sin^2 \phi}{1 + 3 \sin^2 \phi} d\phi \quad (\text{III.76})$$

shows that $\pi^{-1} \int_{\pi/3}^{\pi} X(\theta) d\theta = 1/2$; hence the mean of Y is $1/2$.

For $3 < t < 4$, let $a > 1$ be determined by $a(a+1) = (t-1)/(4-t)$ and put $A_0 = a^2 + a + 1 = 3/(4-t)$. Then

$$q(z) + (t-3)r(z) = (4-t)(a+1-z)(a+z). \quad (\text{III.77})$$

Both roots are strictly outside the unit circle. We may therefore differentiate (III.75) on any compact subinterval of $(3, 4)$. Here are the integral details, including the branch sign. For real $c > 1$ or $c < -1$, define

$$L(c) = \frac{1}{\pi} \int_{\pi/3}^{\pi} X(\theta) \Re \frac{1}{c - e^{i\theta}} d\theta.$$

With $w = \sin^2 \phi$, the integrand after (III.76) is

$$\begin{aligned} & \frac{6w}{1+3w} \frac{c - 1/2 + 3w/2}{c^2 - c + 1 + 3cw} \\ &= -\frac{2}{(c-1)(3w+1)} + \frac{1}{c} + \frac{(c+1)(c^2 - c + 1)}{c(c-1)(c^2 - c + 1 + 3cw)}. \end{aligned} \quad (\text{III.78})$$

The elementary integral $\int_0^{\pi/2} (A + B \sin^2 \phi)^{-1} d\phi = \pi/[2\sqrt{A(A+B)}]$ for $A, A+B > 0$ follows by $y = \tan \phi$. Using it in (III.78) gives

$$L(c) = \frac{\text{sgn}(c+1)\sqrt{c^2 - c + 1} - 1}{2c(c-1)}. \quad (\text{III.79})$$

The full-circle mean of $\Re(c - z)^{-1}$ is $1/c$ for $|c| > 1$. Writing $F(t)$ for the right side of (III.75), (III.77) and (III.79) yield

$$\frac{dF}{da} = \frac{2a + 1 - \sqrt{A_0}}{a(a + 1)} - \frac{2a + 1}{2A_0}. \quad (\text{III.80})$$

The last term is the derivative of $(1/2) \log(4 - t)$. Since $dt/da = 3(2a + 1)/A_0^2$ and $\sqrt{A_0}/\sqrt{4A_0 - 3} = 1/\sqrt{t}$, simplification gives

$$F'(t) = \frac{\sqrt{t} + 3}{2\sqrt{t}(\sqrt{t} + 1)(\sqrt{t} + 2)}.$$

Finally $F(t) \rightarrow 0$ as $t \downarrow 3$, with a quantitative L^1 estimate at the approaching unit-circle root. Put $\eta = a - 1$ and $u = \pi - \theta$. Then

$$\left| \frac{a + e^{i\theta}}{1 + e^{i\theta}} \right|^2 = a + \frac{\eta^2}{4 \sin^2(u/2)}.$$

Using $\sin(u/2) \geq u/\pi$ bounds the integral of its nonnegative logarithm by $O(\log a)$ plus a constant multiple of $\int_0^\pi \log(1 + C\eta^2/u^2) du = O(\eta)$, as seen by $u = \eta v$ and integrability on $(0, \infty)$. The other root factor stays bounded away from zero and converges uniformly. Since Y is bounded, the pressure integral tends to zero. This fixes the integration constant without differentiation at the boundary. Integrating F' with this initial value gives

$$F(t) = 2 \log \frac{\sqrt{t} + 1}{\sqrt{3} + 1} - \log \frac{\sqrt{t} + 2}{\sqrt{3} + 2} = \log \frac{(\sqrt{t} + 1)^2}{2(\sqrt{t} + 2)}.$$

Differentiation directly at $t = 3$ under the original integral would miss a boundary contribution from the unit-circle root. The argument has differentiated only for $t > 3$ and will extend the answer analytically.

34.5 Analytic continuation from negative real roots

Parity gives $D_n = P_n P_{n+1}$; its degree is exactly n , all of its roots are strictly negative, and it is monic. Write those roots as $-\lambda_{n,j}$, $\lambda_{n,j} > 0$, with multiplicity. On $\Re t > 0$,

$$p_n(t) = \frac{1}{n} \sum_{j=1}^n [\text{Log}(t + \lambda_{n,j}) - \log(3 + \lambda_{n,j})], \quad |p'_n(t)| \leq \frac{1}{\Re t}. \quad (\text{III.81})$$

Since $p_n(3) = 0$, integrate this bound on the straight segment from 3 to t . It gives uniform boundedness on every compact subset of the right half-plane. The family is normal. Every subsequential analytic limit agrees on $(3, 4)$ with the function just evaluated, so the identity theorem identifies all subsequential limits. Thus convergence is locally uniform throughout the half-plane; Cauchy's integral formula then gives convergence of every derivative. For explicit branch control, if $s = \sqrt{t}$ then $\Re s > 0$ and

$$\frac{(s + 1)^2}{2(s + 2)} = \frac{1}{2} \left(s + \frac{1}{s + 2} \right)$$

has positive real part. Its principal logarithm is analytic, agrees with the real formula and vanishes at $t = 3$. This proves Theorem 34.1.

In particular, the first derivative at 3 proves the trace limit in (III.8). Combining it with (III.64) proves the calibrated individual mean limit. This proof has handled the singular Toeplitz factor

through the pole-cleared determinant and its integrated logarithmic tail. It does not use an unjustified bounded-symbol estimate on $R(A)$, and it does not assume the trace limit as an input. Below $t = 3$, a naive absolute-value continuation of (III.75) is not justified by this argument; the analytic continuation in (III.81) is the one being used.

35 Individual free energy and the conjectured linear coefficient

Let $c = \sqrt{3}$ and $R_n(s) = \log[Z_n(s)/Z_n(c)]$. Equation (II.24) bounds $|R_{n+1}(s) - R_n(s)|$ by the fixed constant $|\log(s/c)|$. The adjacent determinant definition also gives exactly

$$R_n(s) + R_{n+1}(s) = \log \frac{D_n(s^2)}{D_n(3)} + \log(s/c).$$

Together with Theorem 34.1, these identities yield

$$\frac{1}{n} \log \frac{Z_n(s)}{Z_n(\sqrt{3})} \longrightarrow \frac{1}{2} f(s^2) \quad (s > 0). \quad (\text{III.82})$$

This is an individual-size result on both parities, with a bounded adjacent correction before taking the limit.

Combining (III.82) with the bounded calibration expansion (II.35) proves, for every fixed $s > 0$,

$$\log Z_n(s) = \alpha n^2 + \beta(s)n + o(n), \quad \beta(s) = b + \log(s+1) - \frac{1}{2} \log(2(s+2)). \quad (\text{III.83})$$

At $s = 1$, $b = (3/4) \log 3 - (1/2) \log 2$, so

$$\boxed{\log a_n = \frac{n^2}{2} \log \frac{3\sqrt{3}}{4} + \frac{n}{4} \log 3 + o(n).} \quad (\text{III.84})$$

Thus the conjectured linear exponential coefficient is determined. The $o(n)$ error may dominate $\log n$; neither a logarithmic coefficient, a power of n , nor an amplitude can be inferred from (III.84). In particular, inserting $-\kappa \log n$ into its remainder would be a valid but uninformative rewriting, not a proof of the conjectured factor $n^{-5/72}$. The older fixed-slope upper and lower envelopes with bounded remainders remain compatible with this sharper slope, but their remainder strength does not transfer to the coincident limiting slope.

[write] Superseded, 5 October 2026. Part II proves the logarithmic term, $\log a_n = \alpha n^2 + \beta n - \kappa \log n + O(1)$ (Theorem 10.1), and Part I the full equivalent (Theorem 1.1). “The older fixed-slope upper and lower envelopes” are Part IV’s (IV.5).

For use in differentiation, the convergence in (III.82) is analytic at the level of P_n . Specifically,

$$\frac{1}{n} \log \frac{P_n(t)}{P_n(3)} \longrightarrow \frac{1}{2} f(t) \quad (\Re t > 0) \quad (\text{III.85})$$

locally uniformly with every derivative. The proof repeats (III.81): there are $\lfloor n/2 \rfloor$ negative roots, so the normalized derivative is bounded by $1/(2\Re t)$, and the real-axis limit is (III.82) after removing the $O(1/n)$ parity term. Normality, the identity theorem and Cauchy’s formula complete the argument.

There is also a direct half-plane statement in s . All n roots of Z_n lie on the imaginary axis, including the parity zero. The analytic branch $n^{-1} \text{Log}[Z_n(s)/Z_n(\sqrt{3})]$ has derivative bounded by $1/\Re s$. The same normal-family argument therefore extends (III.82), with every derivative, to all $\Re s > 0$, with limit

$$\text{Log}(s+1) - \frac{1}{2} \text{Log}(s+2) - \frac{1}{2} \log 2.$$

This avoids restricting the complex fugacity to the smaller sector where $\Re(s^2) > 0$.

36 The limiting root measure

Regard $[0, \infty]$ as a compact interval by adding the point at infinity. Let ν_n place mass $1/n$ at each positive magnitude $\lambda_{n,j}$ of a root of D_n , with multiplicity. Then

$$\nu_n \implies \nu = \frac{1}{2} \delta_\infty + \frac{3}{\pi \sqrt{\lambda}(\lambda+1)(\lambda+4)} d\lambda, \quad \lambda > 0. \quad (\text{III.86})$$

The same limit holds for the magnitudes of roots of P_n , normalized by their degree $\lfloor n/2 \rfloor$.

Here is a proof that accounts for the escaping mass. By compactness, any subsequence has a further weak limit. The kernel $(t+\lambda)^{-1}$, defined to be zero at infinity, is continuous on this compactification for $t > 0$. The derivative convergence in (III.66) therefore identifies its transform as $f'(t)$. To verify the candidate measure directly, substitute $\lambda = y^2$: its finite part becomes

$$\frac{6}{\pi(y^2+1)(y^2+4)} dy = \frac{2}{\pi} \left(\frac{1}{y^2+1} - \frac{1}{y^2+4} \right) dy.$$

Its total mass is $1/2$. With $s = \sqrt{t}$, the elementary integral $\int_0^\infty [(y^2+a^2)(y^2+b^2)]^{-1} dy = \pi/[2ab(a+b)]$ gives its Stieltjes transform

$$\frac{1}{s(s+1)} - \frac{1}{2s(s+2)} = \frac{s+3}{2s(s+1)(s+2)} = f'(t).$$

The remaining mass $1/2$ is placed at infinity.

For uniqueness, map λ to $u = (3+\lambda)^{-1} \in [0, 1/3]$, with $u = 0$ at infinity. Derivatives of the Stieltjes transform at $t = 3$ determine all positive moments of u , and total mass determines the zeroth moment. Uniform polynomial approximation identifies the measure. Every subsequential limit is therefore (III.86). For P_n , divide the derivative limit in (III.85) by $\lfloor n/2 \rfloor/n \rightarrow 1/2$ to obtain the same transform and repeat the argument.

The atom at infinity is essential. Every finite polynomial has finite roots, but half of their normalized mass escapes every bounded set. It is not correct to treat the displayed finite density, whose mass is $1/2$, as a probability density of total mass one. Nor is it valid to interchange the large-size limit with $s \rightarrow \infty$ without further estimates.

37 Means cumulants and Gaussian limits

For fixed $s > 0$, differentiation of (III.85), including the parity factor in $Z_n(s) = s^\varepsilon P_n(s^2)$, gives

$$\begin{aligned} \frac{M_n(s)}{n} &\longrightarrow m(s) = \frac{1}{2} - \frac{1}{(s+1)(s+2)}, \\ \frac{V_n^{\text{stat}}(s)}{n} &\longrightarrow v(s) = \frac{s(2s+3)}{(s+1)^2(s+2)^2} > 0. \end{aligned} \quad (\text{III.87})$$

Indeed the first limit is $tf'(t)$ at $t = s^2$, and the second is $2tf'(t) + 2t^2f''(t) = s dm(s)/ds$. More generally, if $\text{cum}_k(S)$ is the k th cumulant, then for every fixed $k \geq 1$,

$$\frac{\text{cum}_k(S)}{n} \longrightarrow (s\partial_s)^{k-1}m(s). \quad (\text{III.88})$$

These limits are uniform when s ranges over a fixed compact subset of $(0, \infty)$, by local analytic convergence. This concerns any fixed derivative order, not derivative orders increasing with n .

For uniform DSASMs, the first four cumulant limits are

$$\frac{1}{3}, \quad \frac{5}{36}, \quad -\frac{1}{27}, \quad -\frac{19}{216}. \quad (\text{III.89})$$

At the calibration,

$$m(\sqrt{3}) = 3 - \frac{3\sqrt{3}}{2}, \quad v(\sqrt{3}) = \frac{21}{2} - 6\sqrt{3}.$$

In particular $S/n \rightarrow m(s)$ in probability, since its variance is $O(n)$. These formulas describe the nonzero diagonal count, not the signed diagonal sum.

37.1 Concentration and the central limit theorem

The independent-Bernoulli representation already proved above gives, with $m_n = \lfloor n/2 \rfloor$,

$$S \stackrel{\text{law}}{=} \varepsilon + 2 \sum_{j=1}^{m_n} B_{n,j}, \quad p_{n,j} = \mathbb{P}(B_{n,j} = 1) = \frac{s^2}{s^2 + \gamma_{n,j}}.$$

It represents the distribution of the total; it does not assert independence of the actual diagonal indicators in a DSASM. The standard exponential bound for a variable in an interval of length one follows by convexity of its exponential and gives Hoeffding's inequality for their sum. After scaling by two,

$$\mathbb{P}_s(|S - M_n(s)| \geq x) \leq 2 \exp[-x^2/(2m_n)] \quad (n \geq 2). \quad (\text{III.90})$$

For any fixed $\eta > 0$, (III.87) implies exponentially high probability of $|S/n - m(s)| < \eta$. Precisely,

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P}_s(|S/n - m(s)| \geq \eta) \leq -\eta^2.$$

Let $\mu_n = M_n(s)$, $\sigma_n^2 = V_n^{\text{stat}}(s)$ and $T_n = (S - \varepsilon)/2$, with $v_n = \text{Var}(T_n) = \sigma_n^2/4$. Since $v_n \asymp n$, the elementary characteristic-function expansion for a centered Bernoulli variable $X = B - p$,

$$\log \mathbb{E} e^{itX} = -\frac{1}{2}p(1-p)t^2 + O(p(1-p)|t|^3),$$

is uniform in $p \in [0, 1]$ for small real t . Its uniform remainder follows by Taylor's theorem and $\mathbb{E}|X|^3 \leq p(1-p)$, followed by the expansion of the logarithm. Summing the independent terms at $t = z/\sqrt{v_n}$ leaves error $O(|z|^3/\sqrt{v_n})$ for fixed z . Therefore

$$\frac{S - \mu_n}{\sigma_n} \Longrightarrow N(0, 1). \quad (\text{III.91})$$

The variance scale can be replaced by $\sqrt{nv(s)}$ using (III.87). The center cannot yet be replaced by $nm(s)$: $\mu_n = nm(s) + o(n)$ does not imply the $o(\sqrt{n})$ error required for that substitution.

[write] Now possible, 5 October 2026. Part II proves $\mu_n = M_n(s) = nm(s) + O(1)$, by (II.121), and hence the explicitly centred CLT (II.122); Part I gives $M_n(s) - nm(s)$ a limit (Corollary 7.2). The local theorem (III.93) below may therefore also be centred at $nm(s)$: since $\mu_n - nm(s) = O(1)$ and $\sigma_n \asymp \sqrt{n}$, the argument of φ changes by $O(1/\sigma_n)$ uniformly in k , and φ is Lipschitz.

37.2 The parity lattice local central limit theorem

For $|t| \leq \pi$, independence gives

$$\begin{aligned} |\mathbb{E} e^{it(T_n - \mathbb{E} T_n)}| &= \prod_j \sqrt{1 - 4p_{n,j}(1 - p_{n,j}) \sin^2(t/2)} \\ &\leq e^{-2v_n \sin^2(t/2)} \leq e^{-2v_n t^2/\pi^2}. \end{aligned} \quad (\text{III.92})$$

After rescaling $t = z/\sqrt{v_n}$ and extending the rescaled function by zero outside $[-\pi\sqrt{v_n}, \pi\sqrt{v_n}]$, the right side is an integrable Gaussian majorant independent of n . Pointwise convergence from the preceding proof and dominated convergence give convergence in $L^1(\mathbb{R})$ to $e^{-z^2/2}$. Fourier inversion now yields, uniformly over all integers j ,

$$\sqrt{v_n} \mathbb{P}_s(T_n = j) - \frac{1}{\sqrt{2\pi}} e^{-(j - \mathbb{E} T_n)^2/(2v_n)} \rightarrow 0.$$

Writing $\varphi(x) = (2\pi)^{-1/2} e^{-x^2/2}$, this is exactly

$$\sup_{k \equiv n \pmod{2}} \left| \frac{\sigma_n}{2} \mathbb{P}_s(S = k) - \varphi\left(\frac{k - \mu_n}{\sigma_n}\right) \right| \rightarrow 0. \quad (\text{III.93})$$

The parity restriction and the factor $\sigma_n/2$ are necessary. Values off this lattice have probability zero. This local theorem still uses the actual mean and standard deviation; it does not supply an unproved explicit centering correction or a uniform formula in fugacities tending to zero or infinity.

38 Macroscopic large deviations

The individual pressure determines a full speed- n large-deviation principle for $X_n = S/n$ under each fixed fugacity $s > 0$. Its limiting logarithmic moment generating function is

$$\Lambda_s(\lambda) = \lim_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E}_s e^{\lambda S} = \log \frac{se^\lambda + 1}{s + 1} - \frac{1}{2} \log \frac{se^\lambda + 2}{s + 2} \quad (\lambda \in \mathbb{R}). \quad (\text{III.94})$$

It is smooth and strictly convex, with derivative $m(se^\lambda)$ and derivative range $(0, 1/2)$. For $0 < x < 1/2$, put

$$r(x) = \frac{\sqrt{1 + 8/(1 - 2x)} - 3}{2}.$$

Then the good rate function is

$$I_s(x) = x \log \frac{r(x)}{s} - \log \frac{r(x) + 1}{s + 1} + \frac{1}{2} \log \frac{r(x) + 2}{s + 2}, \quad 0 < x < \frac{1}{2}, \quad (\text{III.95})$$

with continuous endpoint values

$$\begin{aligned} I_s(0) &= \log(s + 1) - \frac{1}{2} \log(s + 2) + \frac{1}{2} \log 2, \\ I_s(1/2) &= \log(s + 1) - \frac{1}{2} \log(s + 2) - \frac{1}{2} \log s, \end{aligned} \quad (\text{III.96})$$

and $I_s(x) = +\infty$ outside $[0, 1/2]$. Precisely, for every open set $G \subset \mathbb{R}$ and closed set $F \subset \mathbb{R}$,

$$\begin{aligned} \liminf_n \frac{1}{n} \log \mathbb{P}_s(X_n \in G) &\geq - \inf_{x \in G} I_s(x), \\ \limsup_n \frac{1}{n} \log \mathbb{P}_s(X_n \in F) &\leq - \inf_{x \in F} I_s(x). \end{aligned} \quad (\text{III.97})$$

Here is a direct proof. The expression in (III.95) is the Legendre transform $\sup_\lambda(\lambda x - \Lambda_s(\lambda))$: solving $m(r) = x$ gives the displayed $r(x)$ and optimizer $\lambda = \log(r/s)$. Taking $\lambda \rightarrow \pm\infty$ gives (III.96) and infinity outside $[0, 1/2]$. These formulas show continuity on its finite domain and lower semicontinuity everywhere; compact sublevel sets follow.

For the lower bound at an interior point x , exponentially tilt by the optimizer. The exact tilted measure is simply fugacity $r(x)$. Under that measure $X_n \rightarrow x$ in probability by (III.87). On $|X_n - x| < \eta$, change of measure gives the lower exponent $\Lambda_s(\lambda) - \lambda x - |\lambda|\eta + o(1)$, since the tilted probability tends to one. Let $\eta \downarrow 0$ within a given open set. Endpoints are approached from $(0, 1/2)$ using continuity of the rate. If the open set has no point in $[0, 1/2]$, its lower bound is vacuous.

For the upper bound, an exponential Markov inequality at any real λ bounds a neighborhood of x by the affine exponent $\lambda x - \Lambda_s(\lambda)$, with an arbitrarily small local error. For any number strictly below $\inf_F I_s$, choose one such affine bound at every point of $F \cap [0, 1]$. This compact set has a finite subcover, and a union bound proves the desired upper bound. If the infimum is infinite, the same argument works for every finite number. The support $X_n \in [0, 1]$ handles points outside that compact set. This proves (III.97) without a separate general large-deviation theorem.

For each fixed $\delta > 0$, the upper tail $\mathbb{P}_s(X_n \geq 1/2 + \delta)$ is superexponentially small at speed n . No n^2 -scale rate is identified. The endpoint values in (III.96) govern neighborhoods in the large-deviation principle; they are not asymptotics for exact point probabilities, which can vanish because of parity. In particular this statement does not say that the event $X_n > 1/2$ without a fixed gap is superexponentially small.

39 Inverting the count threshold

From $Z_n(2) = a_{n+1}$ and the positive contribution of the identity matrix, $a_{n+1} > a_n$. For $T > 1$ let

$$N(T) = \min\{n \geq 1 : a_n \geq T\}, \quad n_0 = \sqrt{\frac{\log T}{\alpha}}, \quad \beta = \frac{1}{4} \log 3.$$

There exists a function $\eta(T) \downarrow 0$ (monotonicity can be imposed by taking a decreasing tail envelope) such that, for all sufficiently large T ,

$$\left\lceil n_0 - \frac{\beta}{2\alpha} - \eta(T) \right\rceil \leq N(T) \leq \left\lceil n_0 - \frac{\beta}{2\alpha} + \eta(T) \right\rceil. \quad (\text{III.98})$$

To prove this, write (III.84) as $\log a_n = \alpha n^2 + \beta n + n\varepsilon_n$, with $\varepsilon_n \rightarrow 0$. For each small fixed $\varepsilon > 0$, the functions $\alpha x^2 + (\beta \pm \varepsilon)x$ are increasing for large x and bound the sequence eventually. Their positive roots at level $\log T$ are

$$n_0 - \frac{\beta \pm \varepsilon}{2\alpha} + O(n_0^{-1}).$$

As in any monotone envelope inversion, the upper count envelope gives a lower ceiling for $N(T)$, and the lower count envelope gives an upper ceiling. A diagonal choice of $\varepsilon \downarrow 0$ and increasing thresholds proves (III.98).

The unrounded width tends to zero, so the bracket contains at most two consecutive integers eventually. The vanishing error must remain inside the ceilings; near an integer threshold it can change the answer. One cannot assert the real-valued estimate $N(T) = n_0 - \beta/(2\alpha) + o(1)$ for an integer-valued staircase. There is no effective numerical onset here, and no identified $\log n_0/n_0$

correction from the new $o(n)$ growth remainder. For the strict threshold $\min\{n : a_n > T\}$, replace each ceiling in (III.98) by $\lfloor \cdot \rfloor + 1$.

[write] **Superseded, 5 October 2026.** Part II's Corollary 10.2 identifies the $\log n_0/n_0$ correction ($\kappa \log r/(2ar)$ in the centre), and Part I's Corollary 7.1 adds the amplitude term with a vanishing error before the ceilings.

40 Remaining questions and reproducibility

The linear coefficient and fixed-fugacity laws resolve several questions left by one-point growth bounds. They leave specific subleading problems open.

1. Can the pressure error be controlled sharply enough to identify the logarithmic term at $s = 1$, then a positive amplitude in (III.5)? The logarithmic-tail argument here gives convergence, not a remainder expansion.
2. Can one prove $M_n(s) - nm(s) = o(\sqrt{n})$, or a finer expansion, uniformly on compact fugacity intervals? This would permit explicit limiting-mean centering in the Gaussian results.
3. At what scales does the escaping half of the root measure move to infinity? The compactified limit alone does not determine a joint large- n , large-fugacity regime or tails at speed n^2 .
4. Can one obtain coefficient asymptotics with relative errors and explicit prefactors away from the central window? The local central limit theorem gives uniform additive errors, and the large-deviation principle gives logarithmic probabilities, neither of which supplies such a prefactor.
5. Can quantitative constants and onsets be carried through the singular-value limit and inversion to give a certified finite-target threshold algorithm? The present inverse bracket is eventual and non-effective.

The full equivalent, possible parity-dependent corrections, and root simplicity remain outside the proved conclusions. Weak interlacing permits common or multiple roots and is all that was used. No claim of external peer review, formal proof verification, or global bibliographic novelty is made.

[write] **Status of the questions, 5 October 2026.** *Question 1 is answered:* the logarithmic term is $-\kappa \log n$ with bounded error (Part II, Theorem 10.1), and the positive amplitude exists (Part I, Theorem 1.1); its value is open (Section 9, item 1). *Question 2 is answered:* $M_n(s) - nm(s) = O(1)$ uniformly on compact fugacity intervals (Part II, (II.121)), and it converges (Part I, Corollary 7.2). *Questions 3–5 remain open.* For question 3, uncertified evidence from the intake dossier: the exact P_{60} has largest root about $7.3 \cdot 10^{32}$, and the fractions of its roots with magnitude below 1, 4, 10, 100 are 0.367, 0.467, 0.500, 0.533, against 0.352, 0.455, 0.485, 0.499 for the normalized finite part of ν , consistent with half the mass escaping. Of the last paragraph, the full equivalent is now proved (Part I); the parity-dependent corrections (Section 9, item 2) and *root simplicity* remain open: whether the roots of P_n are simple and the interlacing (II.21) strict is not known, and the exact P_n , $n \leq 31$, give no counterexample (dossier).

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Part IV

Diagonal fugacity roots and elementary growth envelopes

Bundle Report 120, archive A005163_Diagonal_Roots_Growth_Bounds_and_Inverse_Brackets_Source.zip, delivered as report120.tex (15 pages), dated 2 October 2026. Delivered title block: “REPORT 120 / Diagonal fugacity roots and DSASM growth bounds / A partial asymptotic result for A005163 / 2 October 2026 / Standalone proof and exact reproducibility companion”. Printed with [write] notes; Report 120’s Section k is Section $k + 40$ here, its statement $k.j$ is $(k + 40).j$, and its equation (k) is (IV. k). Its Sections 2, 4 and 5 repeat Part II’s Sections 11, 13 and 14, and its Section 3 repeats Part III’s Section 27; Sections 42–45 keep their titles, equation numbers, differing sentences and a pointer.

Reading conventions of Part IV	Elsewhere in the report
$D_n = \log(P_n(4)/P_n(1))$,	Parts I–III: $D_n(t)$ the companion polynomial; Part I:
$E_n = \log(P_n(3)/P_n(1))$, numbers, (IV.35)	$E_n(t)$
$\theta = \log 3 / \log 4$; ℓ , u maps on $J = [b - g, b]$;	Part I: $a = \pi / (3h)$; Part II: $F(x)$
$F = \ell \circ u$; $a = (1 - \theta) / \theta$	
$p < q$ the envelope slopes; $d = 5/36$	Part II: d_{ASM}
(logarithmic calibration)	
n_0 , $c(T)$ inverse scale and centre	Parts I, II: r , $z_C(T)$, $z(T)$

Abstract of this Part (as delivered). Let a_n count diagonally symmetric alternating sign matrices of order n , and let $Z_n(s)$ record their number of nonzero diagonal entries. We prove that $Z_n(s) = s^{n \bmod 2} P_n(s^2)$, where all zeros of P_n are strictly negative and real, with multiplicities allowed. The proof applies a degree-two case of the known Borcea–Brändén MOD operator to the published triangular six-vertex model. Finite determinant transformations give an exactly solvable calibration at $s = \sqrt{3}$. Combining that calibration with $Z_n(2) = a_{n+1}$, Jensen’s inequality and a quantitative two-step contraction yields, eventually,

$$\alpha n^2 + pn - \frac{5}{72} \log n - C \leq \log a_n \leq \alpha n^2 + qn - \frac{5}{72} \log n + C,$$

where $\alpha = \frac{1}{2} \log(3\sqrt{3}/4)$, $p \approx 0.2632463454$ and $q \approx 0.2960487269$ are precisely defined fixed-point constants. A threshold inverse has an eventual bracket of at most two consecutive integers; its error constants and onset are existential. This is a partial result. The positive gap $q - p$ leaves the true linear term, actual logarithmic power and amplitude undetermined, and the full equivalent in Behrend–Fischer–Koutschan’s Conjecture 11.1 remains open here. No bibliographic priority or external peer-review claim is made.

[write] **Status of Part IV, 5 October 2026.** The envelopes and the inverse of this Part are superseded: Part III identifies the linear coefficient $\beta = \frac{1}{4} \log 3 \in (p, q)$, Part II proves $\log a_n = \alpha n^2 + \beta n - \kappa \log n + O(1)$, and Part I proves the full equivalent of Conjecture 11.1’s leading term; so the sentence “The positive gap $q - p$ leaves the true linear term, actual logarithmic power and amplitude undetermined, and the full equivalent ... remains open here” is historical. The Part is kept for its method, which appears nowhere else in the report: Jensen and chord inequalities on the negative roots at the fugacities 1, $\sqrt{3}$, 2 (Section 46), a projected two-step contraction (Section 47) and exact interval certificates for p and q ; it uses only the exact prerequisites of Sections 11–14 and 27, no operator theory.

41 The model and the precise scope

An alternating sign matrix is a square matrix with entries in $\{-1, 0, 1\}$ whose nonzero entries alternate in sign along every row and column and sum to one on each such line. A diagonally symmetric alternating sign matrix (DSASM) satisfies $A = A^\top$. Define

$$a_n = |\text{DSASM}(n)|, \quad S(A) = |\{i : A_{ii} \neq 0\}|, \\ Q_n(s_1, \dots, s_n) = \sum_{A \in \text{DSASM}(n)} \prod_{i: A_{ii} \neq 0} s_i, \quad Z_n(s) = Q_n(s, \dots, s). \quad (\text{IV.1})$$

Here $n \geq 1$ is the matrix order, and $a_0 = 1$ can be added by the empty matrix convention. The count is OEIS A005163 [14]; in the notation of Behrend, Fischer and Koutschan [1], $Z_n(s) = X_n(1, s, 1)$. In particular, $Z_n(1) = a_n$. There is no factorial normalization or shift in the meaning of a_n .

Put

$$g = \log \frac{3\sqrt{3}}{4}, \quad \alpha = \frac{g}{2}, \quad b = \frac{g + \log 2}{2}, \quad \theta = \frac{\log 3}{\log 4}, \quad \kappa = \frac{5}{72}. \quad (\text{IV.2})$$

On $J = [b - g, b]$, define

$$\ell(x) = \frac{b}{\theta} - g - \frac{1 - \theta}{\theta}x, \\ u(x) = x - g + \frac{1}{2} \log \left(\frac{3e^{2(b-x)} - 1}{2} \right). \quad (\text{IV.3})$$

The logarithm is well-defined on J . Section 47 proves that $\ell \circ u$ has exactly one fixed point p in J . Set $q = u(p)$, so $p = \ell(q)$.

Theorem 41.1. *For every $n \geq 1$, Q_n is nonzero whenever every $\Re s_i > 0$. Writing $\varepsilon = n \bmod 2$, there is a polynomial with nonnegative integer coefficients such that*

$$Z_n(s) = s^\varepsilon P_n(s^2), \quad \deg P_n = \lfloor n/2 \rfloor, \quad P_n(0) > 0. \quad (\text{IV.4})$$

All zeros of P_n are strictly negative and real. They need not be simple for the purposes of this theorem.

There are constants $C > 0$ and n_ such that for all integers $n \geq n_*$,*

$$\alpha n^2 + pn - \kappa \log n - C \leq \log a_n \leq \alpha n^2 + qn - \kappa \log n + C. \quad (\text{IV.5})$$

The constants satisfy

$$p \approx 0.2632463454, \quad q \approx 0.2960487269, \quad p < q. \quad (\text{IV.6})$$

The corresponding threshold inverse is described in Theorem 48.1.

[write] **Superseded, 5 October 2026.** The linear coefficient is $\beta = \frac{1}{4} \log 3 = 0.2746530722\dots$, strictly between p and q (Part III, (III.84)); Part II proves $\log a_n = \alpha n^2 + \beta n - \kappa \log n + O(1)$ for all $n \geq 1$, and Part I the full equivalent. The structural statements are proved in Part II, Section 11 (this Part's Section 2 is printed there).

Equivalently, fixed positive constants c_-, c_+ give

$$c_- n^{-\kappa} e^{\alpha n^2 + pn} \leq a_n \leq c_+ n^{-\kappa} e^{\alpha n^2 + qn} \quad (n \geq n_*).$$

The $n^{-\kappa}$ factors belong to these two different exponential envelopes. Because $p < q$, their presence does not identify the power of n at the unknown true linear coefficient.

41.1 Established inputs and what is proved here

We use version 3 of the primary paper [1], dated 1 October 2026. Its Section 5.1 supplies the triangular six-vertex bijection; Equation (4.11) supplies the adjacent determinant; Proposition 5.1 supplies

$$Z_n(2) = a_{n+1} \quad (n \geq 1). \quad (\text{IV.7})$$

The quadratic growth rate $\log a_n \sim \alpha n^2$ was already established in its Theorem 11.2. It is not a new conclusion claimed by this report.

The stability-preserving MOD operator is established prior art in Borcea–Brändén [4, Section 8.4]. We give its elementary degree-at-most-two proof and apply it to the DSASM graph. The classical shifted determinant evaluation used for calibration is recorded in Krattenthaler [5, Theorem 34, Equation (3.24)]. All reductions from those inputs to (IV.5) are proved below. Finite exact checks accompany the proof but do not substitute for it.

The leading equivalent predicted in [1, Conjecture 11.1] is

$$a_n \sim C_{\text{DSASM}} e^{\alpha n^2 + (\log 3)n/4} n^{-5/72}, \quad C_{\text{DSASM}} > 0. \quad (\text{IV.8})$$

The source also conjectures parity-sensitive finer corrections. Neither (IV.8) nor those corrections are proved here. Even existence of a limit for $(\log a_n - \alpha n^2)/n$ remains unresolved by our argument. The conjectured value $(\log 3)/4 \approx 0.2746530722$ lies strictly between p and q . No equality of the two parity-dependent calibration amplitudes is asserted or used.

[write] **Historical, 5 October 2026.** Part III proves that $(\log a_n - \alpha n^2)/n \rightarrow (\log 3)/4$; Part I proves (IV.8) and the equality of the two parity calibration amplitudes, (27). The finer corrections remain unproved (Section 9).

42 An orientation polynomial and diagonal stability

[write] **Printed in Part II.** Report 120’s Section 2 is Part II’s Section 11 word for word, with these exceptions: it writes ε for ε_n ; its proof of the root structure ends “This proves the structural part of Theorem 41.1. For $n = 1$, $P_1 = 1$ and the root statement is vacuous.”; and its closing non-claim, printed below as delivered, replaces Part II’s last paragraph. Its equations (IV.9)–(IV.15) are (II.11)–(II.17) there; Lemma 2.1 of Report 120 is Lemma 11.1.

This argument does not prove simple roots, interlacing between sizes, or stability for arbitrary additional six-vertex weights or for a general r -refinement of $X_n(r, s, 1)$.

[write] **5 October 2026.** Interlacing between sizes is proved in Part II, Section 12; simple roots and the other two items remain unproved.

43 Two consequences for the diagonal statistic

[write] **Printed in Part III.** Report 120’s Section 3 is Part III’s Section 27 word for word (checked by diff); its equations (IV.16)–(IV.17) are (III.16)–(III.17) there.

44 The exact solvable calibration

[write] **Printed in Part II.** Report 120's Section 4 is Part II's Section 13 word for word; its equations (IV.18)–(IV.27) are (II.25)–(II.34) there.

45 Logarithmic calibration without matching parity amplitudes

[write] **Printed in Part II.** Report 120's Section 5 is Part II's Section 14 word for word, except that it writes $d = 5/36 = 2\kappa$ where Part II writes d_{ASM} (five occurrences). Its equations (IV.28)–(IV.32) are (II.35)–(II.39) there; (II.35) is the bounded calibration expansion used in Section 46.

46 Jensen and chord inequalities for the roots

For $n \geq 2$, use the factorization (III.16) and put

$$\xi_j = \log \frac{4 + \gamma_j}{1 + \gamma_j}, \quad \phi(x) = \log \frac{1 + 2e^x}{3}. \quad (\text{IV.33})$$

Then $0 < \xi_j < \log 4$ and

$$\phi(\xi_j) = \log \frac{3 + \gamma_j}{1 + \gamma_j}.$$

The function ϕ is increasing and strictly convex, since

$$\phi'(x) = \frac{2e^x}{1 + 2e^x} > 0, \quad \phi''(x) = \frac{2e^x}{(1 + 2e^x)^2} > 0.$$

Moreover, $\phi(0) = 0$ and $\phi(\log 4) = \log 3$, so the endpoint chord has slope θ . Jensen's inequality and that chord give the exact bounds

$$m\phi\left(\frac{D_n}{m}\right) \leq E_n \leq \theta D_n, \quad m = \lfloor n/2 \rfloor, \quad (\text{IV.34})$$

where, writing $L_n = \log a_n$,

$$\begin{aligned} D_n &= \log \frac{P_n(4)}{P_n(1)} = L_{n+1} - L_n - \varepsilon \log 2, \\ E_n &= \log \frac{P_n(3)}{P_n(1)} = W_n - L_n - \frac{\varepsilon}{2} \log 3. \end{aligned} \quad (\text{IV.35})$$

The equalities use (IV.7). The parity factors have been retained. In the chord inequality they cancel exactly, because $\theta \log 2 = (\log 3)/2$. The exceptional case $n = 1$ has $m = 0$ and is not included in Jensen's inequality; it has no asymptotic effect.

46.1 A bounded corrected residual

Normalize by

$$y_n = \frac{L_n - \alpha n^2 + \kappa \log n}{n}. \quad (\text{IV.36})$$

Positivity of the coefficients makes $Z_n(s)$ increasing on $s > 0$. Together with (IV.7) this gives, for $n \geq 2$,

$$Z_{n-1}(\sqrt{3}) \leq a_n \leq Z_n(\sqrt{3}). \quad (\text{IV.37})$$

By (II.35), the upper bound yields $y_n \leq b + O(1/n)$. For the lower bound, expand $\alpha(n-1)^2 + b(n-1) - \kappa \log(n-1)$; it yields $y_n \geq b - g - O(1/n)$. Thus

$$\text{dist}(y_n, J) = O(1/n), \quad J = [b - g, b]. \quad (\text{IV.38})$$

In particular the y_n are bounded. This does not assert that they converge or lie exactly inside J . Substituting (IV.36) and (II.35) into (IV.35) gives the uniform expansions

$$\begin{aligned} E_n &= n(b - y_n) + O(1), \\ D_n &= n(g + y_{n+1} - y_n) + O(1). \end{aligned} \quad (\text{IV.39})$$

For the second line, the additional terms are $\alpha + y_{n+1} - \kappa \log(1 + 1/n) - \varepsilon \log 2$, all uniformly bounded. Hence the chord half of (IV.34) implies

$$y_{n+1} \geq \ell(y_n) - O(1/n). \quad (\text{IV.40})$$

The inverse of ϕ is

$$\psi(v) = \log \frac{3e^v - 1}{2}, \quad v > -\log 3.$$

The Jensen half of (IV.34) gives $D_n \leq m\psi(E_n/m)$. Since $m = n/2 + O(1)$,

$$\frac{E_n}{m} = 2(b - y_n) + O(1/n).$$

By (IV.38), these arguments lie in a fixed compact neighborhood of $[0, 2g]$ contained in $(-\log 3, \infty)$. The derivative of ψ is bounded there. Therefore

$$D_n \leq \frac{n}{2}\psi(2(b - y_n)) + O(1),$$

and the second line of (IV.39) yields

$$y_{n+1} \leq u(y_n) + O(1/n). \quad (\text{IV.41})$$

All constants here are independent of parity; no limiting root distribution and no limiting value of y_n was assumed.

47 A projected two step contraction

Set $a = (1 - \theta)/\theta$. On J ,

$$\ell'(x) = -a, \quad u'(x) = -\frac{1}{3e^{2(b-x)} - 1}, \quad |u'(x)| \leq \frac{1}{2}. \quad (\text{IV.42})$$

Both maps decrease. Since $1/2 < \theta < 1$, we have $0 < a < 1$. At the endpoints,

$$\begin{aligned} \ell(b) &= u(b) = b - g, \\ \ell(b - g) &= b - g + ag, \\ u(b - g) &= b - 2g + \frac{1}{2} \log \frac{65}{32}. \end{aligned} \quad (\text{IV.43})$$

For the last identity, $e^{2g} = 27/16$. Its lower bound $u(b - g) \geq b - g$ follows from monotonicity and $u(b) = b - g$; its upper bound $u(b - g) \leq b$ follows from $65/32 < (27/16)^2$. The other endpoint values visibly belong to J . Thus both maps preserve J .

Define $F = \ell \circ u$ and $G = u \circ \ell$. They increase and are contractions of J with common Lipschitz bound

$$\rho = \frac{1 - \theta}{2\theta} < 1. \quad (\text{IV.44})$$

The contraction theorem gives a unique fixed point p of F . Then $q = u(p)$ satisfies $p = \ell(q)$ and $G(q) = q$, and is the unique fixed point of G .

47.1 Why projection is needed

It would not be correct to apply the derivative bound $|u'| \leq 1/2$ directly to all y_n , because (IV.38) only places them within $O(1/n)$ of J . Let

$$z_n = \max\{b - g, \min\{y_n, b\}\}$$

be their projection onto J . Then $|y_n - z_n| = O(1/n)$. The maps are smooth in a fixed neighborhood of J , so (IV.40)–(IV.41) imply, for a fixed C_0 ,

$$z_{n+1} \geq \ell(z_n) - \frac{C_0}{n}, \quad z_{n+1} \leq u(z_n) + \frac{C_0}{n} \quad (\text{IV.45})$$

for all sufficiently large n .

Both z_{n+1} and $u(z_n)$ lie in J . Since ℓ decreases and is a -Lipschitz, the second inequality in (IV.45) implies $\ell(z_{n+1}) \geq \ell(u(z_n)) - aC_0/n$. Apply the first inequality at $n + 1$. Similarly, both z_{n+1} and $\ell(z_n)$ lie in J , so the first inequality gives $u(z_{n+1}) \leq u(\ell(z_n)) + C_0/(2n)$; apply the second inequality at $n + 1$. Increasing the constant if necessary, we obtain

$$z_{n+2} \geq F(z_n) - \frac{C_1}{n}, \quad z_{n+2} \leq G(z_n) + \frac{C_1}{n}. \quad (\text{IV.46})$$

In particular, no derivative estimate is used at an uncontrolled point outside J .

47.2 Quantitative trapping with a bounded log remainder

Define $d_n^- = (p - z_n)_+$ and $d_n^+ = (z_n - q)_+$, where $x_+ = \max(x, 0)$. Since F and G increase, fix p and q respectively, and are ρ -Lipschitz, (IV.46) implies

$$d_{n+2}^- \leq \rho d_n^- + \frac{C_1}{n}, \quad d_{n+2}^+ \leq \rho d_n^+ + \frac{C_1}{n}. \quad (\text{IV.47})$$

Iterate either inequality separately on each parity. For a sufficiently large starting index N of that parity and $n = N + 2k$,

$$d_n^\pm \leq \rho^k d_N^\pm + C_1 \sum_{j=0}^{k-1} \frac{\rho^j}{n - 2 - 2j}.$$

For $j \leq k/2$, the denominator is bounded below by a fixed positive multiple of n , except for finitely many small cases absorbed into the constant. This part is $O(1/n)$ by the geometric sum. In the

remaining part $j > k/2$, use denominator at least N ; its total is $O(\rho^{k/2})$. The initial term is also exponentially small. Thus $d_n^\pm = O(1/n)$, uniformly on both parities.

It follows that $p - O(1/n) \leq z_n \leq q + O(1/n)$. Undoing the projection gives the same bounds for y_n , and multiplying (IV.36) by n proves (IV.5). The bounded remainder is therefore a consequence of quantitative trapping, not merely of a liminf–limsup argument. In particular,

$$\liminf_{n \rightarrow \infty} \frac{\log a_n - \alpha n^2}{n} \geq p, \quad \limsup_{n \rightarrow \infty} \frac{\log a_n - \alpha n^2}{n} \leq q.$$

47.3 Certified fixed point intervals

The exact rational interval computation in the companion proves

$$\begin{aligned} 0.2632463454286169293260401267680 &< p, \\ p &< 0.2632463454286169293260401267681, \end{aligned} \tag{IV.48}$$

$$\begin{aligned} 0.2960487268742950122628240495510507 &< q, \\ q &< 0.2960487268742950122628240495510786. \end{aligned} \tag{IV.49}$$

In particular $q > p$, and the inverse width obeys

$$0.12537 < \frac{q-p}{g} < 0.12538 < 1, \quad \frac{q-p}{g} \approx 0.1253798292. \tag{IV.50}$$

For completeness, a certificate requires only explicit rational tails. For $x \geq 1$, set $r = (x-1)/(x+1)$ and use

$$2 \sum_{k=0}^{M-1} \frac{r^{2k+1}}{2k+1} \leq \log x \leq 2 \sum_{k=0}^{M-1} \frac{r^{2k+1}}{2k+1} + \frac{2r^{2M+1}}{(2M+1)(1-r^2)}.$$

For $0 < x < 1$, use $\log x = -\log(1/x)$. For $0 \leq x < N+2$,

$$\sum_{k=0}^N \frac{x^k}{k!} \leq e^x \leq \sum_{k=0}^N \frac{x^k}{k!} + \frac{x^{N+1}}{(N+1)!} \frac{1}{1-x/(N+2)}.$$

Interval arithmetic with these tails bounds $F(p_-) - p_- > 0$ and $F(p_+) - p_+ < 0$ at the rational endpoints of (IV.48). The unique fixed point lies between them. Monotonicity of u then bounds q between $u(p_+)$ and $u(p_-)$, giving (IV.49)–(IV.50). No floating-point sign decision participates in the certificate.

48 The threshold inverse and its integer rounding

Since the identity matrix has $S = n > 0$, $Z_n(2) > Z_n(1)$, so (IV.7) implies $a_{n+1} > a_n$. The growth bound also shows $a_n \rightarrow \infty$. For $T > 1$ define

$$N(T) = \min\{n \geq 1 : a_n \geq T\}, \quad n_0 = \sqrt{\frac{\log T}{\alpha}}, \quad c(T) = n_0 + \frac{\kappa \log n_0}{2\alpha n_0}.$$

Theorem 48.1. *There are constants $K > 0$ and T_* such that, for all $T \geq T_*$,*

$$\left\lceil c(T) - \frac{q}{2\alpha} - \frac{K}{n_0} \right\rceil \leq N(T) \leq \left\lceil c(T) - \frac{p}{2\alpha} + \frac{K}{n_0} \right\rceil. \quad (\text{IV.51})$$

The difference between the two unrounded arguments tends to $(q - p)/g \approx 0.1253798292 < 1$. Consequently the bracket eventually contains at most two consecutive integers. No explicit numerical value of K or T_ is supplied by this report.*

Proof. Choose C for (IV.5) and define, for large real x ,

$$B_+(x) = \alpha x^2 + qx - \kappa \log x + C, \quad B_-(x) = \alpha x^2 + px - \kappa \log x - C.$$

Both functions are strictly increasing once x is large enough. Let $r_-(T)$ be the large solution of $B_+(x) = \log T$, and $r_+(T)$ the large solution of $B_-(x) = \log T$. For large T ,

$$\lceil r_-(T) \rceil \leq N(T) \leq \lceil r_+(T) \rceil. \quad (\text{IV.52})$$

Indeed, $\log T \leq L_{N(T)} \leq B_+(N(T))$ proves the lower bound. For the upper bound, the integer $\lceil r_+(T) \rceil$ is large enough for (IV.5), and its B_- value is at least $\log T$.

For a fixed real r and fixed constant c , the large root of $\alpha x^2 + rx - \kappa \log x + c = \log T$ satisfies

$$x = n_0 - \frac{r}{2\alpha} + \frac{\kappa \log n_0}{2\alpha n_0} + O(n_0^{-1}). \quad (\text{IV.53})$$

To check the stated error, substitute the three displayed terms into the equation. The $O(n_0)$ and $\log n_0$ terms cancel, leaving a bounded residual, including $c - r^2/(4\alpha)$ and an $o(1)$ term. The derivative of the left side is $2\alpha n_0 + O(1)$ in this neighborhood, so its root differs by $O(1/n_0)$. Apply this with $(r, c) = (q, C)$ and $(p, -C)$, enlarge a common K , and retain the errors inside the monotone ceilings in (IV.52). This proves (IV.51). Its unrounded width is $(q - p)/(2\alpha) + 2K/n_0$, which tends to (IV.50). $\square$

A width less than one before rounding need not produce a single integer. The two ceilings can differ by one, and errors cannot be discarded near an integer threshold. Thus (IV.51) is an eventual theoretical bracket, not a certified finite-target inversion algorithm or an $o(1)$ -accurate real approximation to an integer sequence. If the definition is instead $\min\{n : a_n > T\}$, replace both ceilings in (IV.52) and (IV.51) by $\lfloor \cdot \rfloor + 1$; this preserves the strict inequality at exact count thresholds.

[write] **Superseded, 5 October 2026.** Parts III, II and I give successively sharper brackets: (III.98), Corollary 10.2 and Corollary 7.1. None is effective.

49 Further questions and reproducibility

49.1 What would advance the full asymptotic problem

The present bounds leave a definite linear-scale gap. The following questions are narrower than asking directly for the full equivalent.

1. Can one prove existence of a single limit for $(\log a_n - \alpha n^2)/n$, and identify it as $(\log 3)/4$? The one-point Jensen and chord information used here does not force that convergence.

2. Can more information about the negative roots of P_n sharpen the relation between its values at 1, 3 and 4? A limiting root statistic or a quantitative improvement to Jensen could potentially reduce the interval $[p, q]$. Such a limit is not proved here.
3. Can one determine the large- n behavior of $D_n(1)/D_n(3)$ using the finite representation (II.32)? Its denominator is exactly known. Even an adjacent-product asymptotic requires additional parity control before it determines individual a_n amplitudes.
4. Can the derivative of the calibration determinant be analyzed through its inverse-matrix trace? The finite determinant permits an exact derivative formula, but no limiting trace evaluation is assumed or established in this report.
5. Can explicit uniform constants be carried through Stirling, the two-step trapping argument and scalar inversion to make the threshold bracket effective for a specified finite target?

Root simplicity and cross-size interlacing are not addressed by this report. Neither is needed for the bounds. Likewise, a common calibration amplitude is neither needed nor claimed.

[write] **Status of the questions, 5 October 2026.** *Question 1 is answered* by Part III, (III.84) (and with bounded error by Part II). *Question 2 is overtaken*: Part III proves the limiting root statistic (III.86), and the interval $[p, q]$ is no longer needed. *Question 3 is answered*: $\log(D_n(1)/D_n(3)) = n \log(2/3) + O(1)$ (Part II, (II.6)) and the remainder converges to $E(1)$ (Part I); the parity control it asks for is Part I's boundary limit (8). *Question 4 is answered*: the trace limit is (III.8) of Part III and, with error $o(1)$, (II.7) of Part II. *Question 5 remains open* (Section 9, item 4). Of the last paragraph: cross-size interlacing is proved in Part II, Section 12, and a common calibration amplitude in Part I, (27); root simplicity remains open (Section 40).

49.2 The reproducibility companion

The report source, deterministic PDF build, exact verification scripts, fixtures and closed inventories are supplied in the companion ZIP. The primary papers are linked in the references rather than copied into the archive. The archive contains no private working notes or source-paper PDFs.

The mathematical certificates use exact rational or integer arithmetic. The fixed-point intervals use explicit logarithm and exponential tails, not floating-point evidence. Small-size orientation and determinant computations check the model, indexing and formulas; they do not prove stability or an asymptotic statement for all sizes. The general proofs are the arguments of Sections 42 through 48.

The companion README describes the precise checked ranges and all commands. Every executable guard remains active under Python's optimization flag. Its negative tests deliberately change fixtures, schemas and manifests and require the intended failure diagnostic. The package inventory rejects missing, changed and unexpected files. Checksums detect modification; they are not authenticated signatures.

The PDF builder fixes dates and metadata, compiles in two independent temporary directories, rejects overfull boxes and unresolved references, and requires byte-identical PDFs. Archive creation uses sorted paths and fixed metadata. Fresh extraction, rebuilding, verification and repacking are tested in the same toolchain. This is same-toolchain reproducibility, not a claim that every TeX, font, Python or compression version produces identical bytes. Every page of the delivered PDF is rendered and visually checked.

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- [2] *Report 124: Logarithmic asymptotics of diagonally symmetric alternating sign matrices. Bounded determinant remainders and inverse thresholds*, 2 October 2026, 34 pp. Unchanged PDF and complete TeX source included as the earlier-input appendix to this bundle. Sections 2–4 and 6–14 are the precise inherited proof dependencies listed in Section 1. **[write]** Printed in full as Part II; its Sections 2–4 and 6–14 are Sections 11–13 and 15–23 here. The earlier-input copy is not shipped.
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